



Arab Academy for Science, Technology and Maritime Transport

College of Engineering and Technology

Mechatronics Department

B. Sc. Final Year Project

DESIGN AND IMPLEMENTATION OF AN ACTIVE ANKLE PROSTHESIS

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DECLARATION

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ABSTRACT

Below knee amputations are one of the most frequent limb removals. Because of armed disputes, medical conditions or other reasons. Amputees in the middle-east are in need of an active ankle that rehabilitates them into their society, since most amputees use passive ankles that could cause further medical problems. The goal of this project is to design a mechatronic smart ankle which mimics the function of the ankle joint in order to help amputees lead a normal life.

This report covers the mechanical design, control and electrical design and other important information that contributed to the design and implementation of the active ankle prosthesis.

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LIST OF ACRONYMS/ABBREVIATIONS

HS	Heel strike
FF	Flat foot
HO	Heel off
TO	Toe off
PO	Push off
CD	Controlled dorsiflexion
CP	Controlled plantarflexion
PP	Power plantarflexion
MENA	Middle east and north African region
Amp	Ankle mimicking prosthetic
BLDC	Brushless Direct Current
PI	Proportional Integral controller
Li-Po	Lithium Polymer
EHA	Electro-hydrostatic actuation

LIST OF UTILIZED STANDARDS

- For the mechanical design the metric standards were utilized
- DIN ISO 355/ DIN 720 in the bearings
- DIN 69051 Ballscrew and Ballscrew nut
- ISO 9001:2000 microchip standards for microcontrollers manufacturing
- IEEE JTAG 1149.1 standards for microcontrollers, Esp32
- NSK 30302 bearing 15x42x14.25mm ISO 10285 Taper roller bearing 30302.
- M8 Bolt-ISO 4017 - 2011 Hexagon head screws with full thread-Product grade A and B.
- M10 Bolt-ISO 4017 - 2011 Hexagon head screws with full thread-Product grade A and B.
- Nut M10-ISO 4017 - 2011 Hexagon head screws with full thread-Product grade A and B.
- Nut M8-ISO 4017 - 2011 Hexagon head screws with full thread-Product grade A and B.
- Round wire coil springs which are manufactured using Music Wire ASTM A228.
- BS EN 24035:1992, ISO 4035:1986 Hexagon thin nuts (chamfered).
- Aluminium 6061 UNS-A96061 ISO 209.
- Steel A36 ISO9001:2008.
- ISO 9443:1991, Heat-treatable and alloy steels — Surface quality classes for hot-rolled round bars and wire rods.
- Ball Screw M12 pitch 4mm-ISO 3408-4 was prepared by Technical Committee ISO/TC 39, Machine tools.

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LIST OF REALISTIC CONSTRAINTS

- Ankle joint angle range: Restricted by the natural movement of an ankle the angle range is from 10 to -20 (The angles are measured from the horizontal position of the ankle and the positive angle is when the ankle contracts upwards and the negative angle for downwards contraction).
- The weight of the active ankle must be light but strong enough to support the amputee and strong enough to withstand the dynamic and static loads.
- The active ankle should be comfortable for the user, known from the user's feedback.
- The energy consumed by the active ankle should be little considering that the amputee will need to use the active ankle for a day of walking and there will be batteries that need to be recharged at the end of the day. So batteries should be able to energize the active ankle for about 16 hours considering the user spends 16 hours outside and sleeps for 8 hours which is the largest possible time he could use the active ankle.
- The weight of the amputee must also be accounted for, since different weight classes require different torque at the ankle joint, so the system could be designed for a specific weight class or be adaptive.
- The height of the amputee using the active ankle should be accounted for. To make sure the ankle is compatible with the amputee's body without causing him imbalance or extra loading on his limbs due to imbalance.

- The speed of walking or the speed of the gait cycle must be considered. It could be set to a certain fixed speed or the system could be adaptable. To support the amputee through different speeds of walking.

1 INTRODUCTION

1.1 WHY DO AMPUTEES NEED ACTIVE PROSTHETIC ANKLES?

There are many people who lost their legs or parts of their legs in accidents or during armed disputes or other medical conditions like diabetic patients or chronic limb ischemia in atherosclerosis or infectious diseases. Below the knee amputations is an alarming health problem, causing high rates of physical, mental and considerable social impact on the amputee. Amputees in the MENA region struggle to afford prosthetic limbs in general since most manufacturers are outside of the region the prices of prosthetic limbs cost more because of customs and other taxes. Therefore, amputees in the Middle East struggle to lead a normal life and do regular activities like walking running, etc... By using passive ankle. Which could negatively affect the limb health and cause more damage to the limb and in general using the passive prosthesis is a painful process even in walking or participating in normal activities.

Furthermore, based on information from the national limb resource centre and world medical organization, two million amputees are living with below the knee amputations in the United States, and that number is expected increase by double by 2050, around the world there are nearly 185,000 people have amputation every year which means that are from 300 to 500 amputation surgeries each day, in countries like Afghanistan and Iraq there are nearly 1,558 military people lost their limbs, and the number of amputations caused by diabetes increased by 24% from 1988 to 2009. Overall below the knee amputees around the world and specifically in the MENA region struggle to lead a normal life and do regular activities like walking, running, etc... Which could cause them even more mental pain. So, we need to design a mechatronic active ankle prosthesis, to help people with disabilities lead a normal life and be productive in their society.

1.2 GOAL OF THE ACTIVE ANKLE PROTHESIS

The goal of the active ankle is to mimic the function of a real ankle, to make the amputee able to participate in day-to-day activities since using a passive ankle could make the simplest activities such as walking a tedious one and as mentioned before it could worsen the medical condition that the amputee already suffers from. So, to mimic the ankle joint there needs to be a smart system and this smart mechatronic system starts with detecting the phase in which the ankle joint is in the gait cycle and this is achieved by certain sensors that will be discussed in details in the control chapter. For now, it is crucial to explain what is the gait cycle and its different states as it will be used later in the report.

1.3 GAIT CYCLE

The definition of the gait cycle as a term it is the cycle which starts from when the heel strikes the ground till it strikes again only by one limb specifically one leg. The gait cycle starts from heel strike (HS), then foot flat (FF), then heel off (HO). In the first three states, which are ultimately the stance phase the spring is storing energy and the motor is even energizing the spring. From the heel off (HO) till the toe off (TO) states the leg is transitioning into the swing phase where the ankle is completely off the ground during which the energy from the motor returns the angle of the ankle to zero ready for taking the next step. The swing phase which starts from toe off (TO) to another heel strike (HS) is approximately 0.38 to 0.42 s. While the stance phase which starts from heel strike (HS) to heel off (HO) is approximately 0.59 to 0.67 s. The average duration of one gait cycle ranges from 0.98 to 1.07 s.

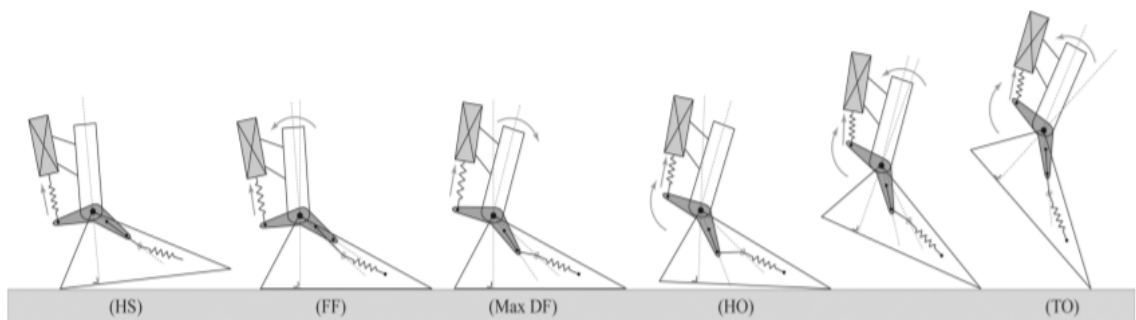


Figure 1-1: Gait cycle demonstration[1].

As shown in Figure 1-1 the gait cycle is clearly labeled by each state from HS to TO.

1.4 INFORMATION ABOUT THE ANKLE JOINT

For the ankle joint, positive angles represent dorsiflexion and negative angles represent plantar flexion as shown in Figure 1-2, the red point represents the fixed centre point of the joint. Also as shown in Figure 1-2: The angle of the joint is from 10 degrees in dorsiflexion and -20 degrees in plantarflexion.

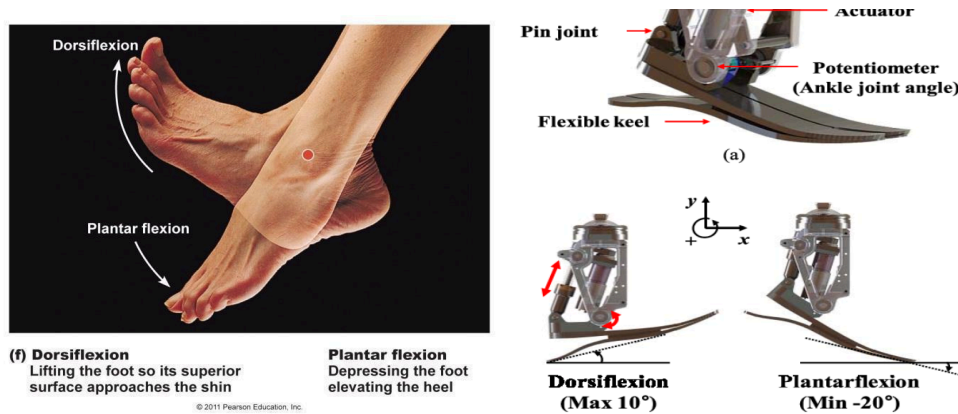


Figure 1-2 : Terminology & range of each ankle angle.[1]

1.5 METHODOLOGY

As mentioned, the ankle prosthesis needs to be smart in order to mimic the actual function of a real ankle and for that to happen it starts with detecting where is the foot in the gait cycle and after specifying the stance of the gait cycle a controlled torque should be outputted to assist the patient at push off (PO) or the heel off stance, then the motor returns the ankle to the original position which zero angle to be ready to take another step. The control scheme will be discussed in more details in its own chapter. As for the whole project a mechanical design needs to be made and by integrating the control components, we achieve the desired mechatronic system. In the next chapter the research effort made on this project will be shown in details and how it helped us build our own design for a smart ankle that is based on empirical data.

2 LITERATURE REVIEW AND BACKGROUND

2.1 PREFACE

This chapter goes over the details of the major research effort contributed to the active ankle project. By introducing the general classification of ankles, then going through the history of ankle prosthetics and then going through different types of active ankles.

2.2 CLASSIFICATION OF PROSTHETIC ANKLES

Prosthetic ankles could be classified into three main types, passive, passive energy storing and returning and bionic or active. The passive or conventional ankle provides no mimicking of the real ankle muscles which leads to more loading on other muscles to compensate for the ankle muscles. Using a passive energy storing and returning ankle is energy efficient but supplies less than needed torque to assist the amputee. Finally, an active prosthetic ankle uses a passive and an active actuator, the use of these actuators mimics the actual movement of an ankle which gives the patient more assistance in regular activities.

2.3 HISTORY OF PROSTHETIC ANKLES

The Seattle Foot shown in Figure 2-1, is the first energy-storing prosthetic component introduced in 1985, that attempted to substitute the human foot during the gait cycle. This type of passive prostheses can assist the amputee by storing and returning potential energy. It consists of three components: A Delrin keel, a Kevlar pad, and an exterior polyurethane foam matrix. Figure 2-2 shows the keels design back in the day.

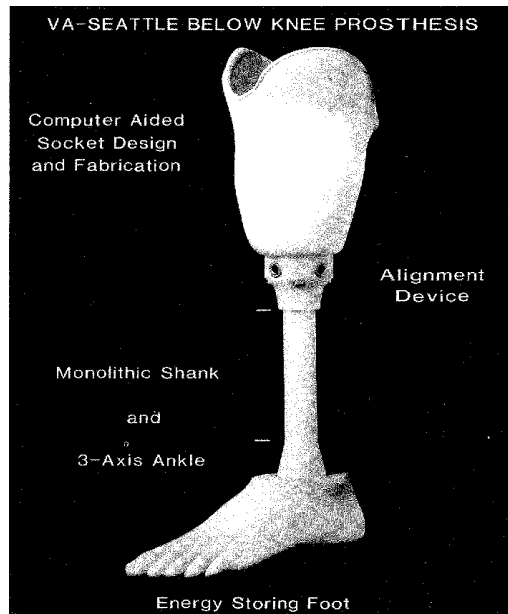


Figure 2-1 : The Seattle foot.[2]



Figure 2-2 : Old Keels design.[3]

The full control of a prosthesis ankle was introduced in 1997 by Ottobock, called C-leg. The control function of the prosthesis adapts to the individual gait pattern of each user. The prosthesis is operated by a microcontroller controlled hydraulic system that is flexible to different walking speeds. Simultaneously, the controller makes sure that the phase of the gait cycle is correct based on sensors. These sensors record the force on the knee joint every fraction of a second to clearly determine the stage the gait cycle. Two hydraulic valves that are adjusted by maxon RE-max 13 motors modify the characteristics of the hydraulic system. Two motors are used in each C-Leg. The key features of these DC motors are their compact design and high-power output.



Figure 2-3 : The proprio foot.[4]

In 2006, the Proprio Foot (Ossur) became the first bionic or microprocessor-controlled prosthetic foot. PROPRIO FOOT is a low-to-moderately active amputee's adaptable microprocessor-controlled ankle. Motorized ankle motion raises the forefoot in the swing phase, increasing toe clearance and minimizing the danger of trips and falls. It also adjusts the ankle angle in response to changes in topography, resulting in improved gait quality on stairwells and sloped surfaces. When compared to a traditional, passive, and elastic foot, active plantar flexion-dorsiflexion can reduce an amputee's metabolic energy required to transfer unit body weight to unit distance.

The SPARKy3 is the most current electrically powered ankle foot prosthesis created by Arizona State University's Human Machine Integration Lab (Spring Ankle with Regenerative Kinetics).shown in figure 2-4. Amputees in the military are generally energetic and youthful. They must execute physically hard dynamic tasks under extreme conditions as part of their job. Current state-of-the-art equipment, both commercially available and in development, do not meet their specific needs. SPARKy is the first and only gadget of its type, created to meet the technologically demanding needs of highly active Military amputees.

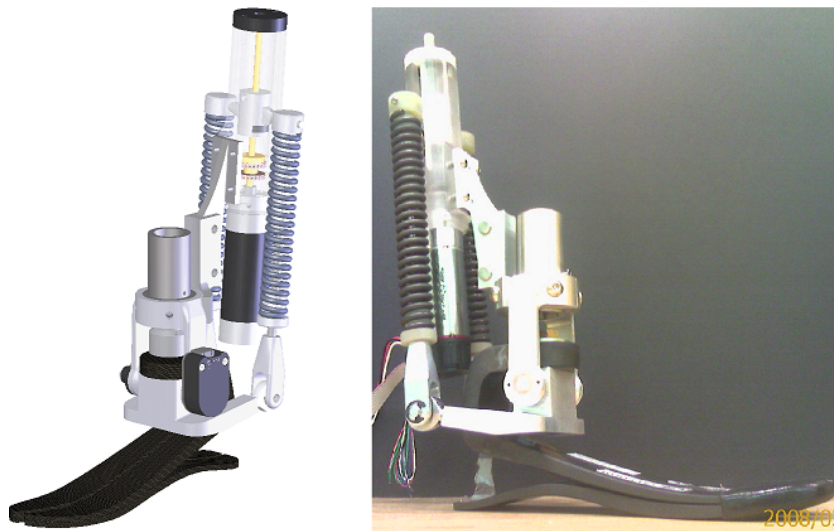


Figure 2-4 : Sparky3.[5]

2.4 DIFFERENT ACTIVE ANKLE PROSTHESIS

2.4.1 Ankle mimicking prosthetic foot (AMP) 2.0

Mechanical Design

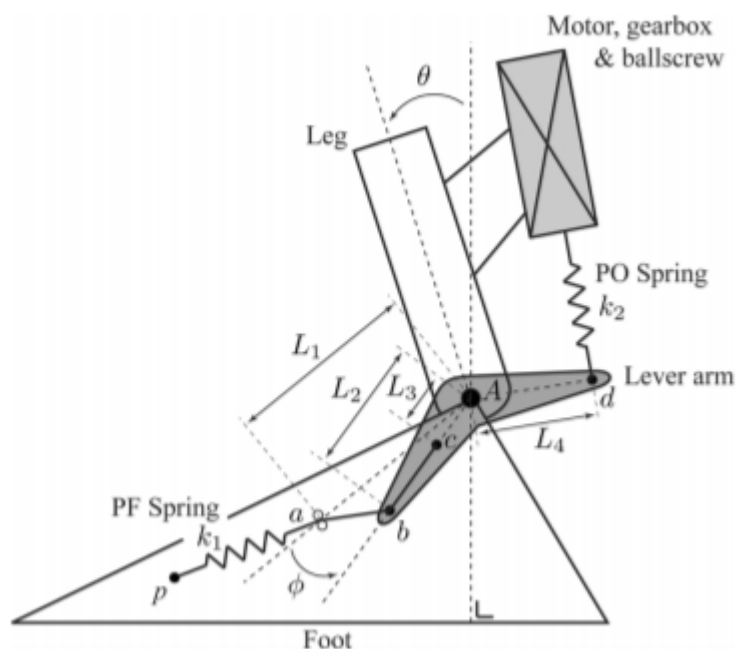


Figure 2-5 : Amp foot 2-D design.[6]

$L1$ Distance between ankle axis (A) and point a.

- $L2$ Distance between ankle axis (A) and point b.
- $L3$ Distance between ankle axis (A) and point c
- $L4$ Distance between ankle axis (A) and point d.
- θ Angle between foot and leg.
- ϕ Angle between foot and lever arm.
- $K1$ Plantar Flexion spring stiffness.
- $K2$ Push-off spring stiffness.

As shown in Figure 2-5 there are two springs in the amp 2.0 and a BLDC motor. These three actuators contribute to the ankle torque needed while taking a step or a stride which was discussed in the gait cycle. The idea of the ankle is to store energy when the person takes a step and return the ankle to angle zero which is the starting position for a step, the motor and ball screw mechanism apply more torque to the ankle to mimic the real muscles of the ankle.

The locking of the four-bar linkage shown in Figure 2-6 allows for a delayed injection of power into the PO spring which decreases the power needed from the actuator is reduced, which means reduction in size while still providing the torque needed for the motion.

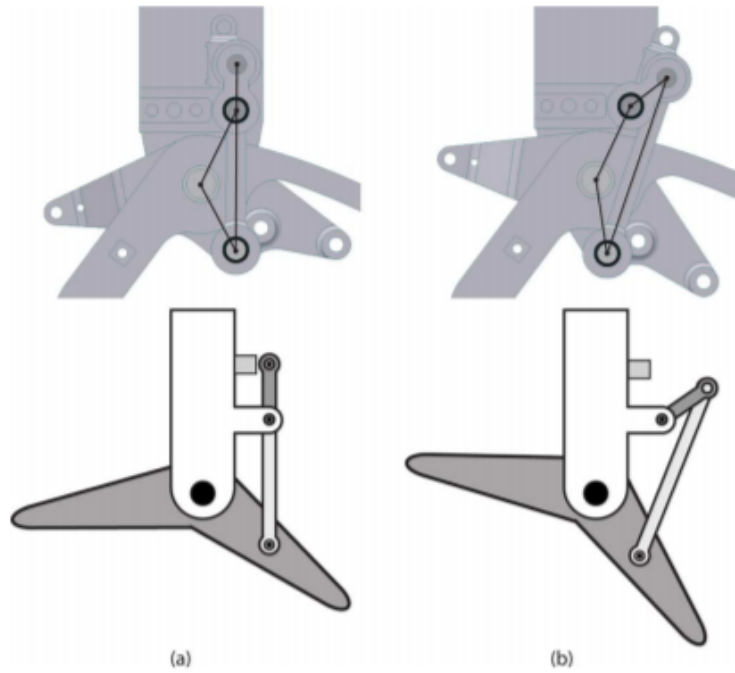


Figure 2-6 : Four bar linkage (a): locked state, (b): unlocked.[6]

Control of the AMP foot 2.0

Using two force sensing resistors at the heel and toes indicate contact with the ground therefore, indicating the state of the step. As shown in Table 2-1. Using a current sensor to control the motor power produced during the stance phase (which represents the first three phases of taking a step). While the motor returns to the original position during the swing phase by open loop velocity control.

Table 1: Full scale triggers./7/

Heel contact	Toe contact	Phase
1	0	HS to FF
1	1	FF to HO
0	1	HO to TO

0	0	Swing
---	---	-------

The torque required by the motor is set by the controller depending on the phase, torque required is represented by current which is sensed by the current controller and the error is fed to the PI controller to control the torque of the BLDC motor as shown in figure-5.

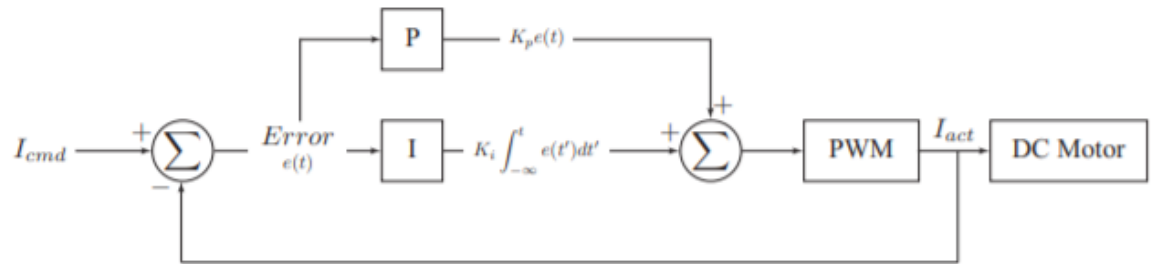


Figure 2-7 :PI control of the motor's torque./7/

Conclusions

Advantages:

- Simple control.
- Low energy consumption due to usage of two springs in the design. And using a four-bar linkage contributes to less power consumption.
- Lighter weight due to smartness in design and usage of 4 bar linkage.

Disadvantages:

- Simple control provides low level of smartness, therefore low level of assistance and comfortability for the amputee.
- Simple control assists the amputee if moving at a constant speed only.
- Simple control proved only applicable in preliminary research.

Due to these disadvantages the researchers came to the conclusion of using the sensors shown in Table-2 and a microcontroller (ATmega2560) to improve the control of the AMP foot 2.0. The algorithm of control quoted from the authors is as follows “*As soon as a first step has been detected, the walking mode is triggered. In this mode, the high level controller feeds a well-defined current command to the low level controller during the stance phase. While walking, the position of the motor is measured and compared with the desired end position. If the motor arrives too soon to its final destination, which means the subject walks slower than expected, the commanded current value is decreased for the next step. On the contrary, if the motor is not able to reach its end position before TO, which means that the amputee is walking faster than expected, the commanded current value is increased. If the subject suddenly stops walking, the controller detects that the phases of stance are not followed correctly and therefore the motor undergoes a reset. Before a new step is completely determined, the AMP-Foot 2.1 works passively. Using this system of flags, daily life scenarios like start or stop walking, opening doors, taking a step backward, increasing or decreasing the walking speed are made possible with a relatively simple control algorithm.*” [7]

Table 2: Sensors used in the amp 2.0 [7]

Sensor	Measurement
Current sensor	Current drawn by the DC motor (torque)
Load cell	Force acting on the spring
Linear potentiometer	Ballscrew position
Force Sensing Resistors (FSR)	Heel and toe contact
Magnetic encoder	Angle of the ankle joint

2.4.2 SPARKy3

Intentions and Goals

SPARKy 3 has adopted a novel method to giving the prosthetic ankle two degrees of flexibility in order to revolutionize powered transtibial prosthesis design. While the ordinary amputee may not require agility, many would want to be able to live a more active lifestyle without being restricted by their prosthetic ankle : Some people have communicated this desire to us directly. Military amputees may gain the most from such a gadget, as they must do skills like running, leaping, and passing their PT test in order to return to active duty service. Our long-term objective is to use two extremely strong, lightweight motors to allow an individual to return to active service with enhanced athletic abilities.

Pre-design and Sketching:

The initial design goals were to achieve just running and leaping, which necessitated the use of two EC Powermax 30 motors to boost power capacity. The installation of a second motor encouraged research into adding a third degree of freedom. The notion of employing two motors in tandem for powered running and regulating each individual motor to power the additional rotating degree-of-freedom was then developed. Another goal of the design was to relocate the springs above the keel but still inside the foot's envelope, allowing an unaltered shoe to be worn over the device. This would result in a lower-profile gadget with less machinery hidden below the leg.



Figure 2-8 : 3-D view of SPARKy3.[5]

Variable impedance control for the active ankle:

A series-elastic linear motor drives a commercial polypropylene ankle orthosis. The angle between the shank and the foot is measured using a rotary potentiometer. Capacitive force transducers on the bottom of the foot are used to determine ground reaction forces (GRFs). To detect heel striking occurrences, an extra footswitch is put on the shoe heel. The control method is implemented on a Linux computer using an analog input–output circuit board.

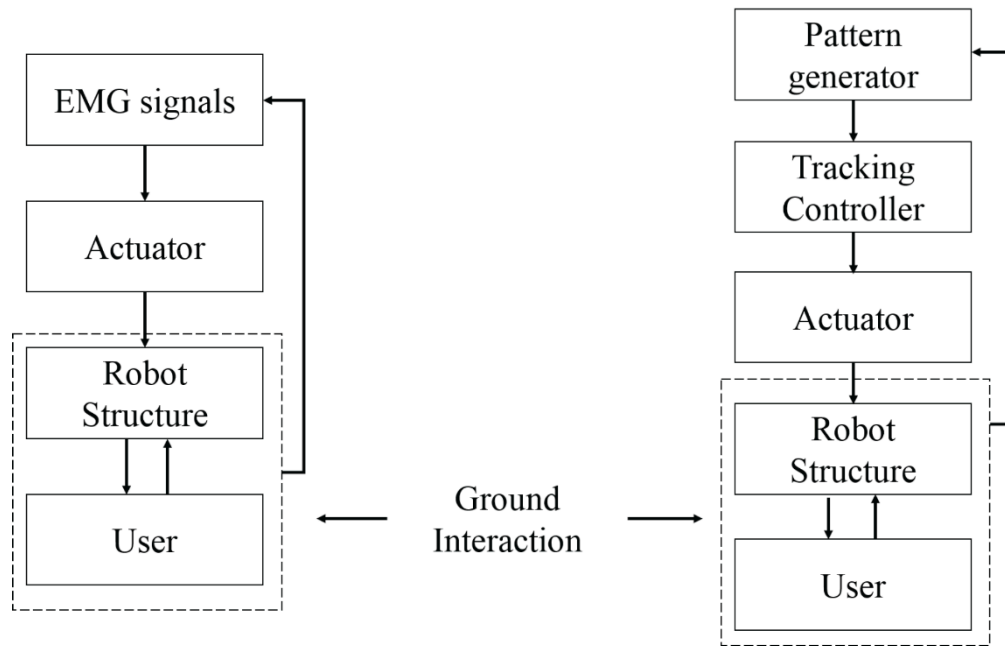


Figure 2-9 : Control Scheme for SPARKy3.[5]

2.4.3 Active Ankle Prosthesis by Electro-Hydrostatic Actuation

Goals of Electro-hydrostatic actuation

Ankle prosthesis attracts increasing attention, as an important human medical prosthesis serving as weight-bearing joints of the lower limb. Its stability and flexibility play an important role in various activities. It can withstand 5 times of the body weight. In pushing, the body goes forward and upward, 60% of the forces are produced by the ankle joints. According to different drive modes, ankle prosthesis can switch modes from passive, semi-active and active prosthesis with active actuation. Passive prostheses are widely used by amputees in the early stages to compensate in the body structure of amputees. However, these prostheses are inflexible so that amputees tend to be slow to walk. So we go to active prosthesis. To assist walking, particularly at high speed and climbing, several kinds of actuation have been used, including off-board electric motor with flexible tether, pneumatic actuation, a DC motor with a ball screw and electro

hydrostatic actuation. Motor-driven ankle prostheses have disadvantages such as the power density ratio is so small that the holistic installed power is large, output torque is not sufficient enough for performing daily activities. Hydraulically actuated ankle prostheses can provide necessary cushioning and active facilitating force is large enough, but its main drawbacks are overall excessive weight, poor energy endurance and so forth. Ankle joint prostheses have high requirements for peak power and output torque, while relatively low demands for average power in most work cycles. There is a contradiction between the driver's requirements for peak power and weight as well as endurance. Pure motor drives or hydraulic drives cannot solve these problems. The innovative solution of this study was to design an active ankle prosthesis based on the principle of electro hydrostatic actuation (EHA). EHA is a new actuator that uses the power transmission technology and the volumetric speed regulation principle to integrate the actuation system energy source and the output actuator. It has dual advantages of power transmission and hydraulic actuation. The output velocity Ratio and output force ratio of EHA are large. This actuator can continuously regulate speeds, has the ability of self-resistance overload and self-lubrication. It also has desirable characteristics such as strong safety, high efficiency, more reliability and easy maintainability, implement integration and miniaturization.

Components used

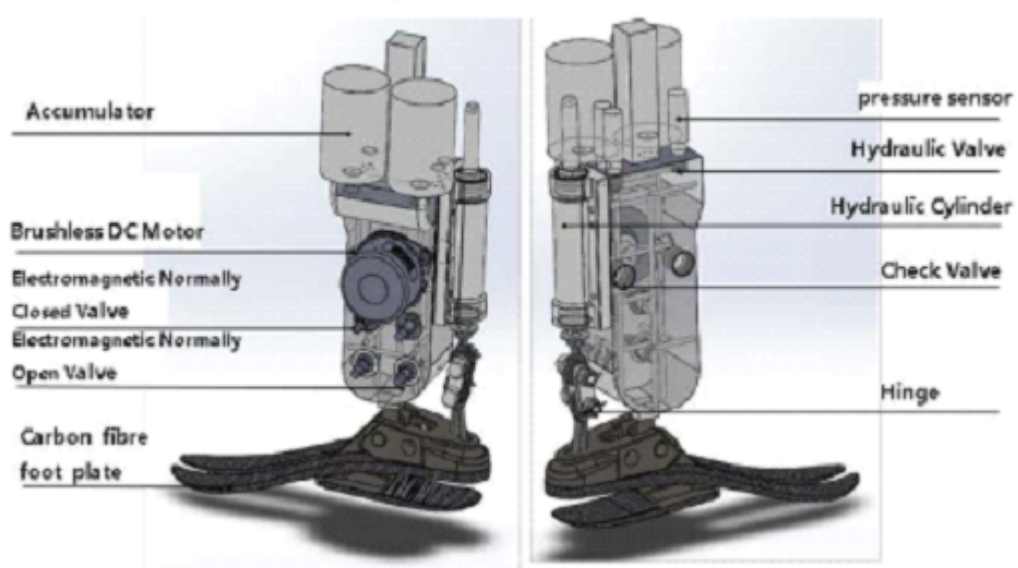


Figure 2-10 : Components used.[8]

It consists of more than ten parts, including BLDC motor and its drive, bidirectional pump, valves, double-outlet hydraulic cylinder, pressure sensors, accumulators, carbon fiber foot plate, gyroscope, displacement sensor, toe spring, microcontroller and Li-Po battery.

Hydraulic system Design

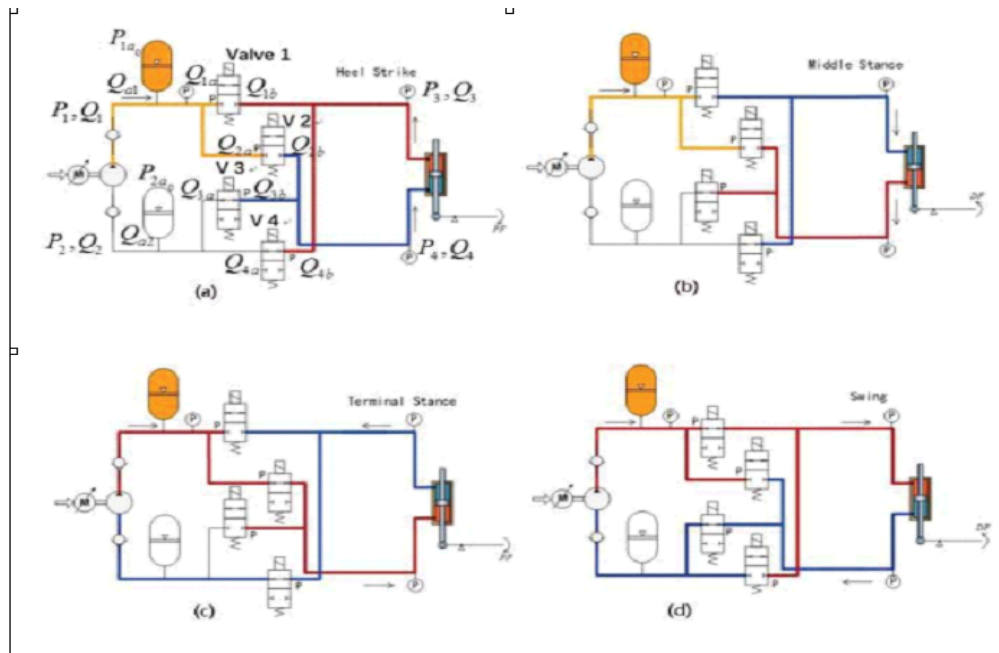


Figure 2-11 : Hydraulics system schematics diagram./8/

Large number of components which cause difficulty in controlling the system, but it's efficient and operates with high performance.

Controlling speed and angle of active ankle

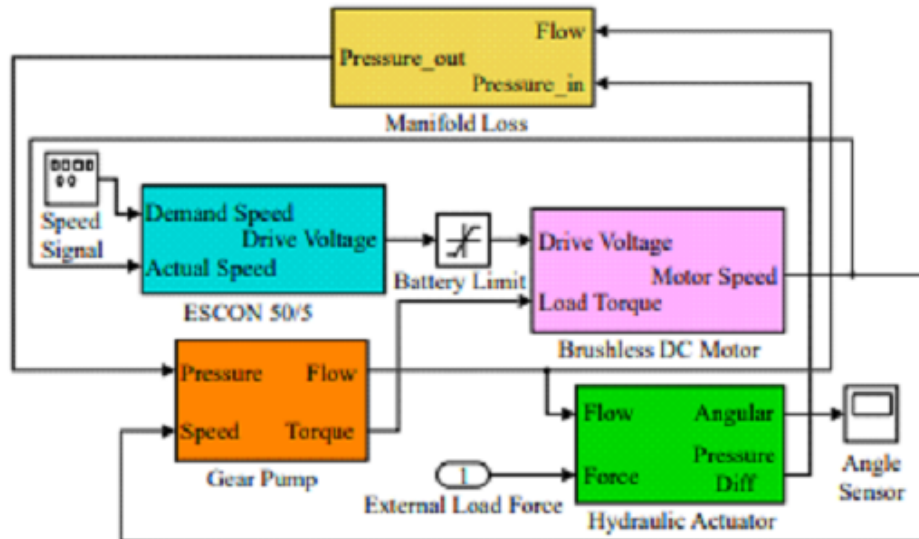


Figure 2-12 : Simulink model./8/

Controlling torque/force of active ankle

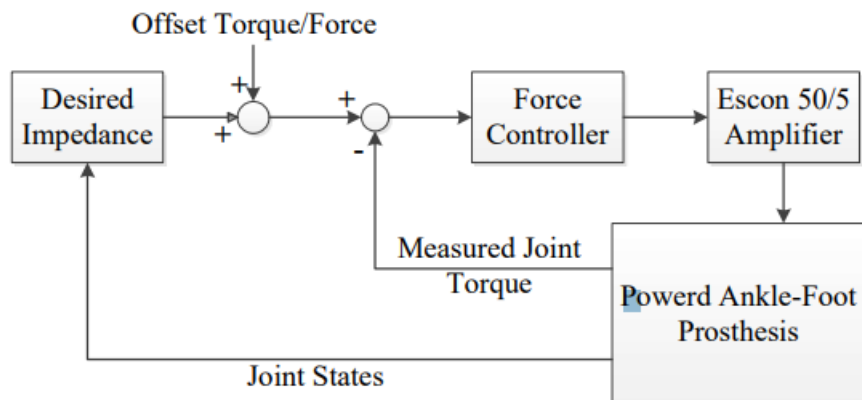


Figure 2-13 : Simulink model for torque control.[8]

Conclusions

Electro Hydraulic actuator EHA has the highest performance and efficiency in different modes of actuations. But it may be the most expensive as the large number of components in the system is one from its disadvantages. The other disadvantages are leakage of oil and complexity of controlling the system (not easy to be controlled as there are a large number of components to be controlled).

2.4.4 PKU-RoboTPro-II Prosthesis

To test the angle adjustment, a powered transtibial prosthesis (PKU-RoboTPro-II) was used. The PKU-RoboTPro-II utilised a DC motor (Maxon, RE-40, 150 W) to operate the ankle joint.

The average weight of the in the active ankle industry is 5 kg.

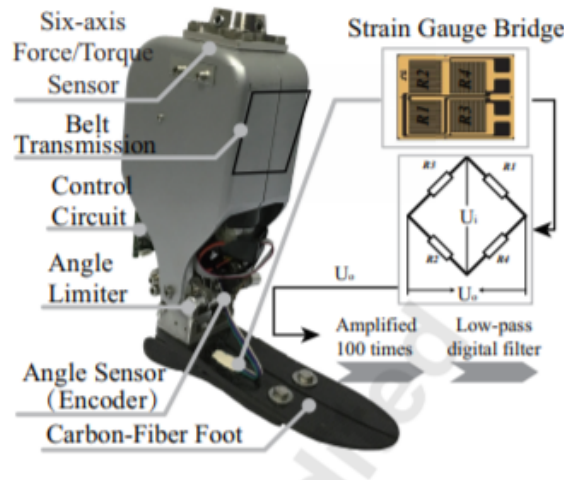


Figure 2-14 : PKU-RoboTPro-II prosthesis.[9]

Most sections of the PKU-RoboTPro-II were built of aluminum alloy to minimize overall weight; the total weight of the PKU-RoboTPro-II (without the rechargeable Li-ion battery) was 1.75kg. The ankle angle is defined as 0 when the shank is perpendicular to the plane of the carbon-fiber footplate. In this article, the positive angle indicates dorsiflexion and the negative angle represents plantar flexion. From 25 degrees dorsiflexion to 25 degrees plantar flexion, the joint angle was changed.

The following are the control system components utilized in the PKU-RoboTPro-II prosthesis: On the upper surface of the carbon fiber, a magnetic encoder was used to measure the ankle joint angle, a DC motor (Maxon, RE-40, 150 W) was used to actuate the ankle joint, a six-axis force/torque sensor (M3553E) was used to measure the interaction force between the amputee and the prosthesis, a control circuit (using the controller chip STM32F103), and an integrated strain gauge bridge with four strain gauges was used to measure. The original Strain Gauge Bridge signals were amplified 100 times and then processed via a low-pass digital filter with a cut-off frequency of 10 Hz.

Control Strategy

The gait cycle begins with one foot's heel strike and concludes with the same foot's following heel strike. Controlled flexion (CF) and powered plantar flexion are included in the stance phase. Controlled plantar flexion (CP) and controlled dorsiflexion (CD) are two types of controlled flexion (CD). Controlled plantar flexion (CP) begins with heel impact (the ankle joint begins to plantarflex) and finishes with the foot flat (figure B) (the ankle joint starts to dorsiflex). CD begins with the foot flat (figure B) and finishes when the ankle reaches the maximal dorsiflexion angle (figure C) (Figure A). After controller Dorsiflexion (CD), powered plantar flexion (PP) begins and finishes at toe off. The swing phase begins at toe off (pre-swing toe off figure E) and concludes at heel strike (terminal swing figure G). We develop a velocity control approach that establishes a precise motor velocity based on each subject's preferences.

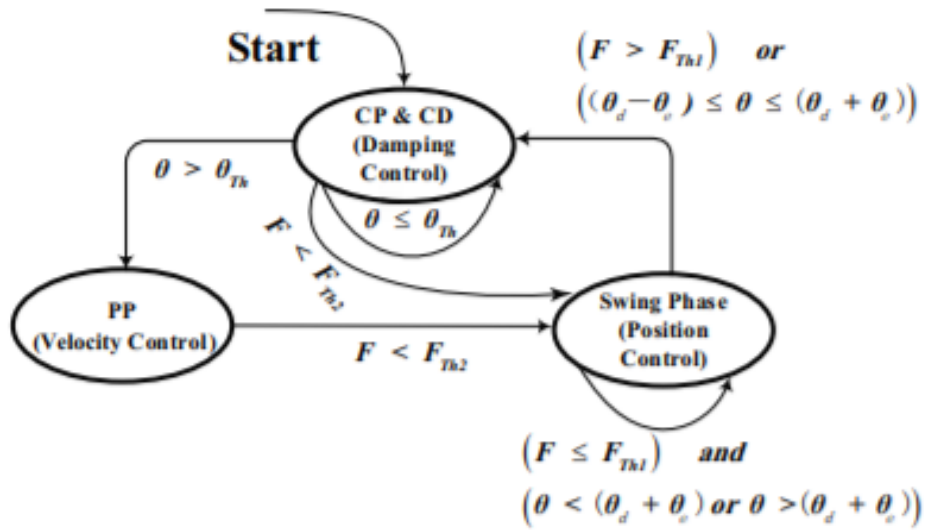


Figure 2-15 : Control logic.[9]

F represents the vertical pressure between an adaptor and a prosthesis.

F_{Th1} represents the gravitational pressure threshold at heel strike. (Refer to 200N)

F_{Th2} represents the gravitational pressure threshold at toe off. (Refer to 100N)

θ_e represents the allowable angle error. (Refer to 0.5 degree)

θ_{Th} represents a minimum angle to detect the beginning of push-off.

θ_d represents the desired ankle angle at the end of swing phase. θ_e represents the allowable angle of error.

θ represents the ankle joint angle.

For ramp ascent, level ground and ramp descent, the desired angle θ_d is 5, 0 and -5 degrees, respectively.

Angle Adjustment Necessity

Angle encoders positioned at the ankle rotation axis were utilized in previous techniques, however they were unable to monitor the changing angle owing to deformation. If the ankle angle measurement did not take the deformation of the carbon-fiber footplate into account, the ankle angle measurement would be incorrect, depending on the research. If the accurate ankle angle threshold (reference ankle angle) is used to determine push-off, the angle inaccuracy may affect walking performance and control effect evaluation. We discovered that deformation of the carbon-fiber footplate influenced the ankle angle, and we hypothesized that the deformation of the carbon-fiber footplate mostly resulted in angle error, with an error or minor discrepancy between the encoder angle and the deformation of the footplate. We needed to establish a link between the human weighted force (deformation of carbon-fiber-footplate) and the angle encoder sensor, therefore we conducted an experiment to discover the relationship between the strain gauge bridge's measured value and the altered ankle joint angle.

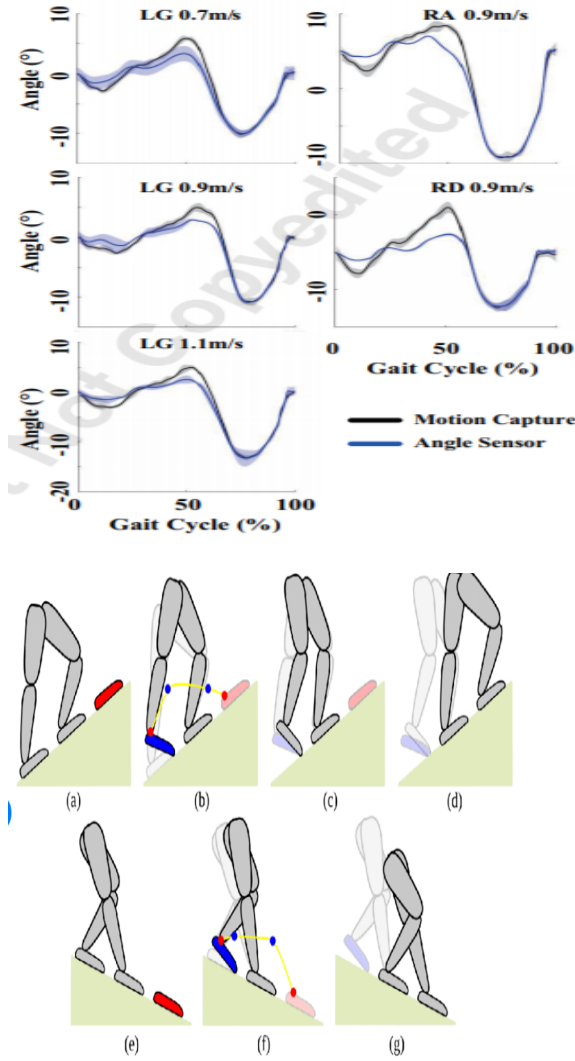


Figure 2-16 : The ankle angle using different measurements for different conditions.[9]

The black line represents motion capture data, while the blue line represents data from an ankle angle sensor built into the prosthesis' spinning shaft, s. A standard deviation is indicated by the darkened region. Level ground, ramp ascension, and ramp descent are all abbreviated as LG, RA, and RD, respectively. The ankle joint reference angle was calculated using the motion capture system's markers to record the motion trajectory. Amputees were put through walking tests to see if angle adjustment was possible. As shown in figure 2-16.

Angle Adjustment Theory

$$\theta = 90^\circ - \angle 2 = -90^\circ + \angle 1 + \angle 3 \quad (1)$$

- $\angle 2$ is the angle between the shank extension line and lower footplate surface

- $\angle 1$ is an angle directly measured by an angle sensor (encoder)
- $\angle 3$ is an angle between the extension line (upper rear footplate where an encoder was placed) and the lower footplate.

When 3 remains constant, such as throughout the swing phase, ankle angle may reflect the true angle. Where, as shown in Figure 2-17, 3 changes throughout the stance phase owing to footplate deformation. When you land on your back foot, 3 decreases, and the corresponding real decreases. On the other hand, when you land on your forefoot, 3 goes up, and the matching real goes up.

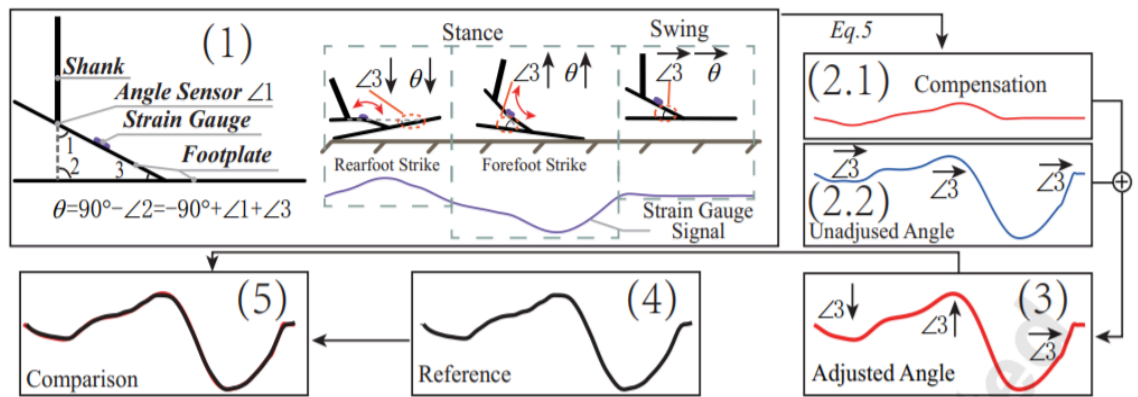


Figure 2-17: Angle adjustment flow diagram./9/

The strain gauge bridge can detect two-direction deformation during the stance phase. The strain gauge bridge bends downward as the backfoot strikes it. The strain gauge bridge, on the other hand, bends upward as the forefoot strikes. A 100-fold amplification was used to acquire the original signal, which was then processed via a low-pass digital filter with a 10 Hz cut-off frequency. A correction angle was produced by plugging the original signal into Eq. 1. An unadjusted ankle angle is only derived from an angle sensor reading plus a constant drift ($90 + 3$) when 3 is taken as a constant. In reality, throughout the stance phase, 3 was not a constant (see (1)). (3) The adjusted ankle angle is equal to the compensation angle (2.1) plus the unadjusted ankle angle (2.2). (4) The black line represents the motion capture system's measurement of the ankle angle. (5) When comparing an adjusted ankle angle to a measured angle from the motion capture system, there is a slight discrepancy. Figure 2-17 depicts the labels mentioned.

2.5 BIONIC RUNNING FOR UNILATERAL TRANSTIBIAL MILITARY AMPUTEES

Active prostheses serve to mimic the lower appendage Muscle-limb System in the sagittal plane with a combination of a linear spring and actuators. The linear springs store and regenerate the energy during the stand stage, while the actuator adequately reduces the equivalent length of the spring after the dorsal flexion of the stand stage supply the measure and amount of potential energy it stores and releases.

There is almost a threefold increment in energy needed during the gait cycle to transition to running from walking. Also, the step frequency almost double, requiring near to six times the output power from the muscle-tendon complex while running when contrasted with strolling.

The gait cycle frequency increase in transition from walking to running, but the acceleration of actuator components inside each walk cycle increments as a result of the changes in kinematics between strolling and running.

The impact of an increasing gait cycle frequency and decreased time for storing and releasing “discharging” vitality per cycle is a sensational rise in power required to overcome the inertial effects of the actuators and supply the needed rapid power surge at push-off. Robust finite state control frameworks adapt to changes within the gait by controlling invariant parameters inside distinctive parameters of the gait.

One execution divides the stance position into five regions based on the angular position of ankle, and controls either ankle stiffness or angular speed inside each zone. This strategy looks for to control angular speed when the ankle is plantarflexing and stiffness when the lower leg is dorsiflexing within the position stage.

The optimal angular speed and stiffness parameters change based on the position inside the gait cycle, which the control strategy addresses by characterizing particular parameters for three separate plantarflexion zones and two isolated dorsiflexion zones, accounting for five special parameters for a given stride cycle.

As the idea angular ankle angular speed and firmness profiles, also change beneath reasonable conditions, this control framework employs state logic to decide the

amputee action given the conditions, and maps the five ideal parameters to that certain activity.

Mechanical design:

The running prosthesis was basically an update of the SPARKy walking active ankle prosthesis. Refining the strolling prosthesis for running required component adjustment and modification resist the increased dynamic loads on the structural components and provide the expanded actuator power vital for running.

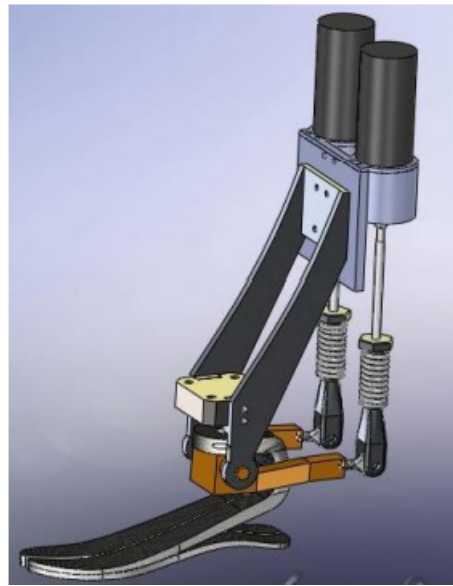


Figure 2-18 : 3-D View of the prosthesis.[10]

Finite component analysis of the components indicated three basic zones for update: the muscle interface. The robotic system interface at the beginning transmits inputs from the actuator to both springs, which resulting in critical internal moments and shear loads within the interface component. To relieve the large normal and shear stress at the contact each motor uses a single spring.

Increased moments applied on the joint at the ankle caused over the top shear loads on the bolted tibia-ankle interface adjoining to the ankle joint. To reduce the hazard of failure at the revolute joint, this interface is reinforcement to sustain these moments.

The 300-400W actuator control necessity ordered substitution of the Maxon RE40 based actuator system evaluated up to 150W. The group assessed three options with comparative control characteristics: a single Maxon RE75 brushed engine actuator, dual Maxon RE40 actuator system and a single EC40 brushless motor., the weight and

inertial characteristics of the components, and the clearance given for the lower appendage socket. The RE75 permitted for the best motor control algorithm, but provided less than ideal socket clearance due to the large diameter of the motor casing. In addition, the RE75's inertia was double to fourfold of the inertia of the other two options, which would have constrained its efficiency. Moreover, the single actuator system would have produced excessive moments in the muscle-tendon interface component. The double actuator framework needed additional control requirements to synchronize the two motors, but moderated chance of muscle-tendon component failure. Also, in spite of the fact that inertial power requirements were less than the RE75 system, they were double the EC40 brushless motor actuator control prerequisites. In any case, the EC40 brushless motor framework control had some instabilities during actuator direction inversion, and tested strategies demonstrated questionable at that specific transition in the gait cycle. Thus, the RE40 double framework demonstrated the leading compromise between motor control necessities, efficiency, and compatibility with the amputee.

Motors Selected:

1. Maxon RE 40 motor:

RE 40 Ø40 mm, Graphite Brushes, 150 Watt

2. EC 40 brushless motor:

EC 40 Ø40 mm, brushless, 120 Watt, with Hall sensors

Control system:

A PC104 microprocessor is used with a Sensory DAQ unit attached to PC104. Utilizing control code generated from MATLAB, it drives two custom brushed motor position controllers.

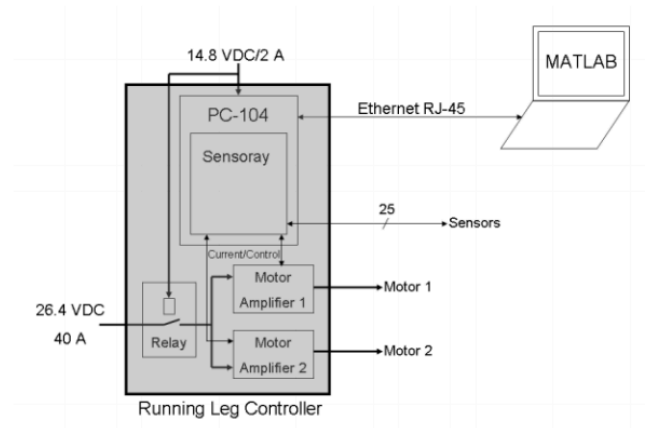


Figure 2-19 : Control schematics.[10]

Portable electronics are powered by a 26.4V battery pack. An Ethernet connected to the MATLAB PC is used to generate the control code as well as collect the data measured from the robotic ankle.

PC104 has several modules and it depends on the system parameters.

Conclusion

This prototype of active ankle prosthesis is efficient and more powerful for transition this ankle from walking process to running process. This ankle could operate at speed of 3.6 m/sec. comparing with sparky1 prototype that could operate at 2.1m/sec. its disadvantages are complexity in synchronizing the two motor to operate at the same time without any delay.

2.6 CHAPTER CONCLUSIONS

As different types of active ankles were mentioned the choice of mechanical design and components became clear. As for mechanical design most of the ankles have the same general design, so our active ankle must follow this general design, which will be mentioned later in details in the mechanical design section. As for the control

components we gained an insight of the most used components and their need in the system and more details will be given in the control chapter.

3 MECHANICAL DESIGN AND CALCULATIONS

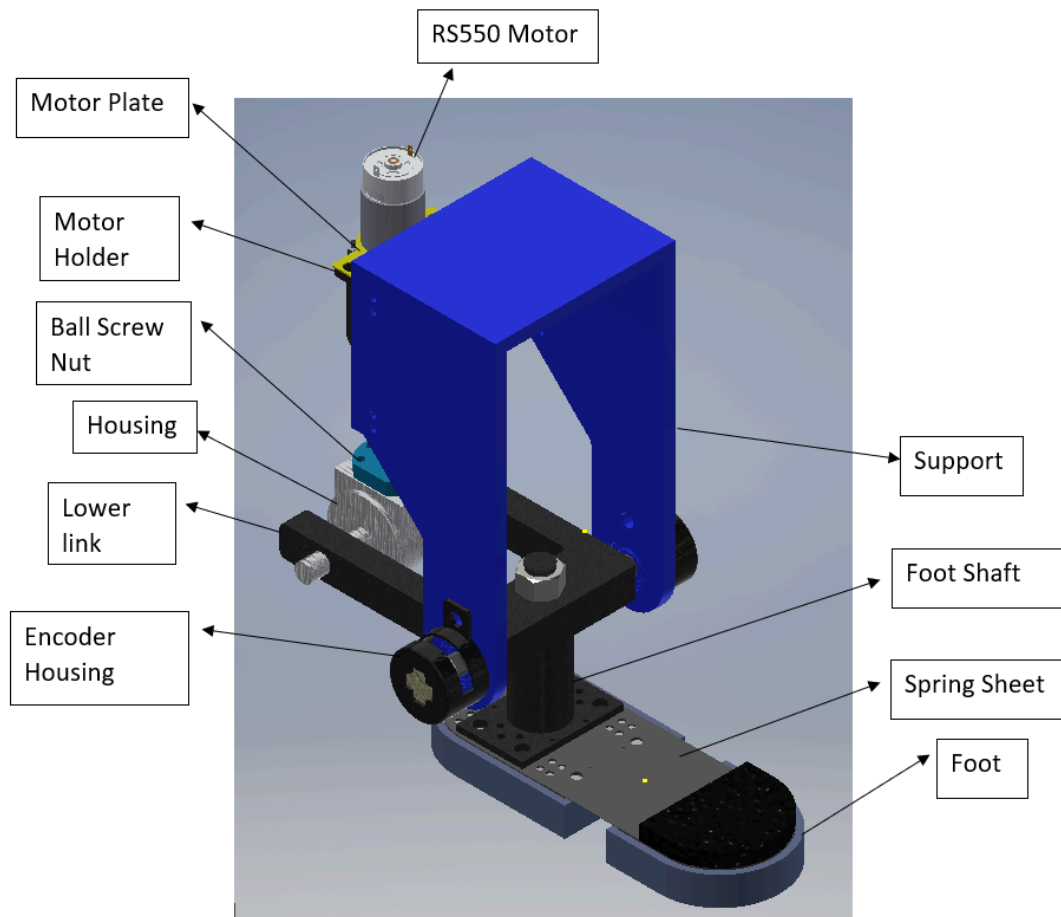


Figure 3-1: Assembly of the active ankle.

3.1 CALCULATIONS

The distance between centre of rotation and force action is equal to 137 mm. as shown in the figure:

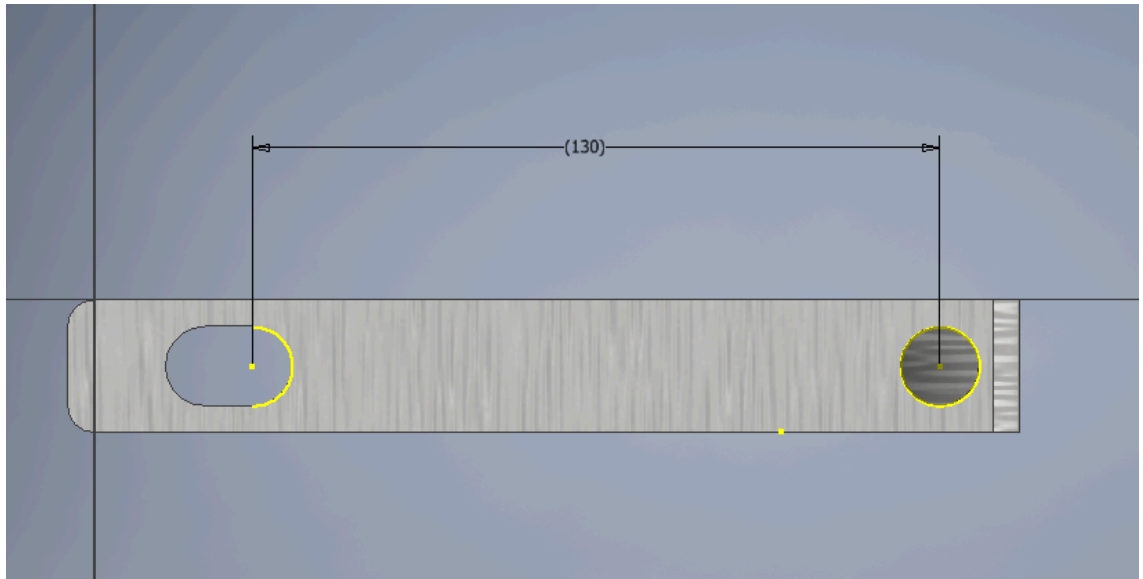


Figure 3-2: Distance between center of rotation and force action in lower link.

Distance travelled in plantar flexion phase:

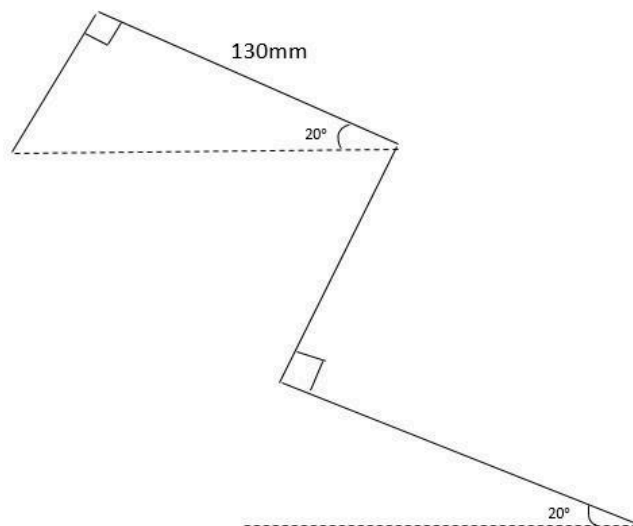


Figure 3-3: Ankle position in plantarflexion phase.

$$\frac{stroke}{130} = \tan(20)$$

$$stroke = 130 \times \tan(20) = 48 \text{ mm}$$

Distance travelled in dorsiflexion phase:

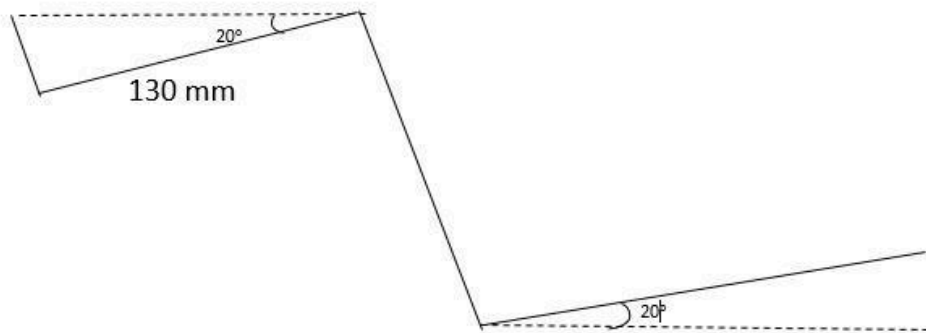


Figure 3-4: ankle position in dorsiflexion phase.

$$\frac{stroke}{130} = \tan(20)$$

$$stroke = 130 \times \tan(20) = 48 \text{ mm}$$

Ball screw: (SFU 1605)

Ball Screw with Nut SFU1605.

- Diameter (d): 16mm.
- Root Diameter (d_r) = 13 mm.
- Lead (l): 5 mm/rev.
- Length: 400 mm.

Speed for each phase of the joint inclination (in 0.6 sec):

$$\text{number of revolutions in plantar flexion to transfer } 48 \text{ mm} = \frac{\text{stroke}}{\text{lead}} = \frac{48}{5}$$

$$\text{ball screw should rotate } 9.6 \text{ revolution in } 0.6 \text{ sec} = \frac{\text{revolution}}{\text{time}} = \frac{9.6 \times 60}{0.6} = 960 \text{ rpm.}$$

Speed for each phase of the joint inclination (in 0.4 sec):

$$\text{number of revolutions in dorsiflexion} = \frac{\text{stroke}}{\text{lead}} = \frac{48}{5} = 9.6 \text{ rev.}$$

$$\text{ball screw should rotate } 9.6 \text{ revolution in } 0.4 \text{ sec} = \frac{\text{revolution}}{\text{time}} = \frac{9.6 \times 60}{0.4} = 1440 \text{ rpm.}$$

Torque of Ball Screw:

$T_{\text{Joint}} = 120 \text{ N.m}$, the mechanical joint torque exerted by normal ankle

$$T_{\text{Joint}} = F \cdot x$$

$$F = \frac{T_{\text{Joint}}}{x} = \frac{120}{0.13} = 924 \text{ N} \approx 1500 \text{ N}, \text{ we approximated the force to } 1500 \text{ N for Safety purposes.}$$

$x = \text{Ball Screw Stroke}$

$$T_{screw} = \frac{F \times l}{2\pi \times \xi} = \frac{1500 \times 5 \times 10^{-3}}{2\pi \times 0.9} = 1.32 \text{ N.m}$$

T_{screw} : torque acting on ball screw

F: force acting on ball screw (force calculated from mechanical torque joint).

l: lead of ball screw

ξ : efficiency of ball screw

$$\tau_{torsional} = \frac{2 \times T_{screw}}{\pi \times r_{root}^3}$$

$$\tau_{torsional} = \frac{2 \times 1.32}{\pi \times (\frac{13}{2} \times 10^{-3})^3} \cong 3.059 \text{ MPa}$$

$$\sigma_{axial} = \frac{F_{axial}}{\pi \times r_{root}^2}$$

$$F_{axial} = F_{max} = 1500 \text{ N.}$$

$$\sigma_{axial} = \frac{1500}{\pi \times (\frac{13}{2} \times 10^{-3})^2} = 11.3 \text{ MPa}$$

$$\sigma_{eq} = \sqrt{(\sigma_{axial})^2 + 3 \times (\tau_{torsional})^2}$$

$$\sigma_{eq} = \sqrt{(11.3)^2 + 3 \times (3.059)^2} = 12.48 \text{ MPa}$$

Safety factor:

Material of ball screw is steel alloy which has a permissible strength equal to 147 MPa.

$$N = \frac{\sigma_{strength}}{\sigma_{eq}} = \frac{147}{12.48} = 11.7 \text{ (safe)}$$

Buckling of ball screw:

$$P = \frac{\lambda \pi^2 EI}{l_b^2}$$

P : Buckling load [N]

l_b : Distance between mounting positions [m] ($l_b = 2 \times l$)

E : Elastic modulus [Pa] (200 GPa)

I : Second moment of inertia [m⁴]

λ : Support factor (Fixed – supported: $\lambda = 2$)

$$I = \frac{\pi}{64} \times d_{root}^4$$

$$I = \frac{\pi}{64} \times (13 \times 10^{-3})^4 = 1.401 \times 10^{-9} \text{ m}^4$$

$L = 130 \text{ mm}$.

$$P = \frac{2 \times \pi^2 \times 200 \times 10^9 \times 1.401 \times 10^{-9}}{(0.5 \times 130 \times 10^{-3})^2} = 1.309 \text{ kN (safe)}$$

Since maximum load is equal to 1500 N. which is less than 1.309 kN. Thus, the ball screw is safe.

Housing Calculations:

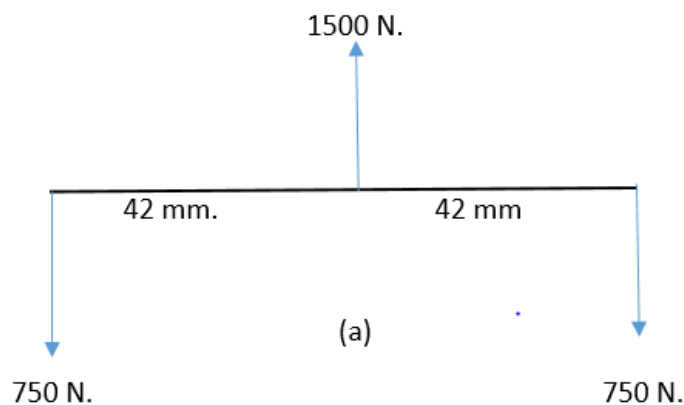


Figure 3-5 : Force diagram of the housing.

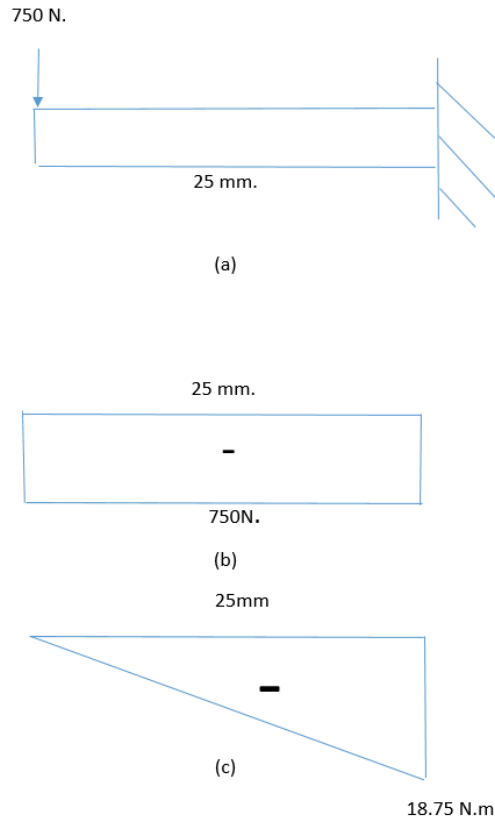


Figure 3-6 : (a) schematic of bolt 1. (b) shear force diagram. (c) bending moment diagram.

Maximum bending moment = 18.75 N.m.

$$\tau_{shear} = \frac{F}{A} = \frac{750}{\frac{\pi}{4} \times (15 \times 10^{-3})^2} = 4.244 \text{ MPa}$$

$$\sigma_{bending} = \frac{M.y}{I}$$

$$M = 18.75 \text{ N.m}$$

$$y = \frac{15}{2} = 7.5 \text{ mm}$$

$$I = \frac{\pi}{64} (d_r)^4 = \frac{\pi}{64} (15 \times 10^{-3})^4 = 2.485 \times 10^{-9} \text{ m}^4$$

$$\sigma_{bending} = \frac{18.75 \times 7.5 \times 10^{-3}}{2.485 \times 10^{-9}} = 56.59 \text{ MPa.}$$

$$\sigma_1 = \left(\frac{\sigma_z + \sigma_x}{2} \right) + \frac{\sqrt{(\sigma_x - \sigma_z)^2 + 4 \times \tau_{xy}^2}}{2}$$

$$\sigma_1 = \left(\frac{56.59}{2} \right) + \frac{\sqrt{(56.59)^2 + 4 \times (4.244)^2}}{2} \cong 57 \text{ MPa}$$

Select Aluminium 7075 with strength equal to 480 MPa.

Safety factor:

$$N = \frac{\sigma_{strength}}{\sigma_1} = \frac{480}{57} = 8.42 \text{ (safe)}$$

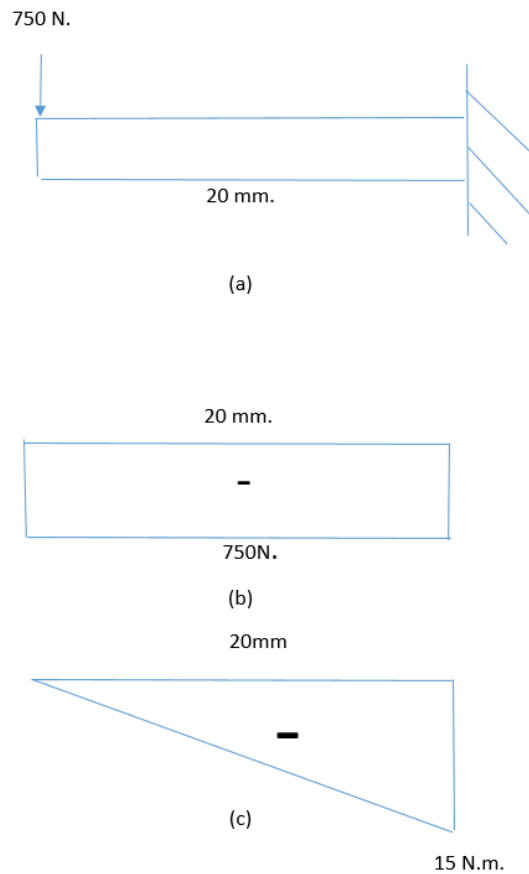


Figure 3-7 : (a) schematic for bolt 2. (b) shear force diagram. (c) bending moment diagram.

Maximum shear force = 1500 N.

Maximum bending moment = 15 N.m.

$$\tau_{shear} = \frac{F}{A} = \frac{750}{\frac{\pi}{4} \times (15 \times 10^{-3})^2} = 4.244 \text{ MPa}$$

$$\sigma_{bending} = \frac{M.y}{I}$$

$$M = 15 \text{ N.m}$$

$$y = \frac{15}{2} = 7.5 \text{ mm}$$

$$I = \frac{\pi}{64} (d_r)^4 = \frac{\pi}{64} (15 \times 10^{-3})^4 = 2.485 \times 10^{-9} m^4$$

$$\sigma_{bending} = \frac{15 \times 7.5 \times 10^{-3}}{2.485 \times 10^{-9}} = 45.27 MPa.$$

$$\sigma_1 = \left(\frac{\sigma_z + \sigma_x}{2} \right) + \frac{\sqrt{(\sigma_x - \sigma_z)^2 + 4\tau_{xy}^2}}{2}$$

$$\sigma_1 = \left(\frac{45.27}{2} \right) + \frac{\sqrt{(45.27)^2 + 4 \times (4.244)^2}}{2} \cong 45.67 MPa$$

Select Aluminium 7075 with strength equal to 480 MPa.

Safety factor:

$$N = \frac{\sigma_{strength}}{\sigma_1} = \frac{480}{45.67} = 10.51 \text{ (safe)}$$

Actual Motor Selected:



Figure 3-8 : RS550 Motor.

Specifications

Table 3-1: RS-550S Motor specifications

Manufacturer ID	HRS-550S-18V
Nominal voltage:	18V DC
Operating voltage:	3-24V
No load speed (min.):	22,000 rpm
No load current:	0.9A
Stall current*:	62A
Max. efficiency current:	8A
Max. efficiency power output:	103W
Max. efficiency torque:	600g-cm
Stall torque*:	5145g-cm

3. RS550 Motor with Motor Plate:

General objective and settings:

Design Objective	Single Point
------------------	--------------

Study Type	Static Analysis
Last Modification Date	26-May-21, 10:42 PM
Detect and Eliminate Rigid Body Modes	No
Separate Stresses Across Contact Surfaces	No
Motion Loads Analysis	No

Mesh settings:

Avg. Element Size (fraction of model diameter)	0.1
Min. Element Size (fraction of avg. size)	0.2
Grading Factor	1.5
Max. Turn Angle	60 deg
Create Curved Mesh Elements	No
Use part based measure for Assembly mesh	Yes

Material(s)

Name	Steel	
General	Mass Density	7.85 g/cm ³
	Yield Strength	207 MPa
	Ultimate Tensile Strength	345 MPa
Stress	Young's Modulus	210 GPa

	Poisson's Ratio	0.3 ul
	Shear Modulus	80.7692 GPa
Part Name(s)	User Library-Motor-RS-550 Motor Plate 3 Motor holder2	

Name	Aluminum 6061	
General	Mass Density	2.7 g/cm ³
	Yield Strength	275 MPa
	Ultimate Tensile Strength	310 MPa
Stress	Young's Modulus	68.9 GPa
	Poisson's Ratio	0.33 ul
	Shear Modulus	25.9023 GPa
Part Name(s)	RS550 Gearbox	

Name	ABS Plastic	
General	Mass Density	1.06 g/cm ³
	Yield Strength	20 MPa
	Ultimate Tensile Strength	29.6 MPa
Stress	Young's Modulus	2.24 GPa

	Poisson's Ratio	0.38 ul
	Shear Modulus	0.811594 GPa
Part Name(s)	Gearbox Plastic piece	

Name	Steel, Mild	
General	Mass Density	7.85 g/cm ³
	Yield Strength	207 MPa
	Ultimate Tensile Strength	345 MPa
Stress	Young's Modulus	220 GPa
	Poisson's Ratio	0.275 ul
	Shear Modulus	86.2745 GPa
Part Name(s)	AS 1110 - Metric M8 x 25 AS 1110 - Metric M8 x 25 AS 1474 - Metric M8 AS 1474 - Metric M8	

Operating conditions

Force:1

Load Type	Force
Magnitude	1500.000 N
Vector X	0.000 N

Vector Y	0.000 N
Vector Z	1500.000 N

Selected Face(s)

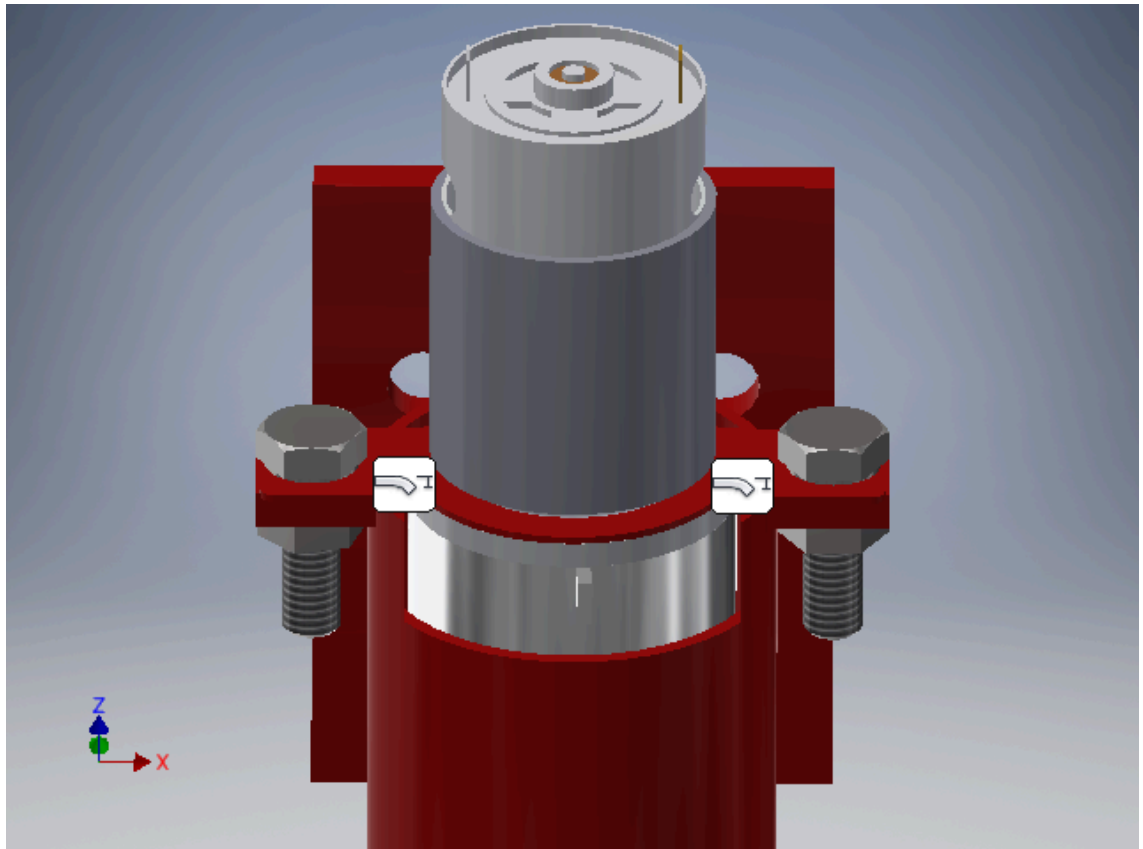


Figure 3-9: Motor with motor holder.

Results

Reaction Force and Moment on Constraints

Constraint Name	Reaction Force		Reaction Moment	
	Magnitude	Component (X,Y,Z)	Magnitude	Component (X,Y,Z)
Fixed Constraint:1	1500 N	0 N	50.8823 N m	50.8823 N m

		0 N		0 N m
		-1500 N		0 N m

Result Summary

Name	Minimum	Maximum
Volume	174826 mm ³	
Mass	0.943599 kg	
Von Mises Stress	0.0000910351 MPa	111.924 MPa
1st Principal Stress	-11.511 MPa	120.252 MPa
3rd Principal Stress	-85.2662 MPa	15.4122 MPa
Displacement	0 mm	0.0214286 mm
Safety Factor	1.84948 ul	15 ul
Stress XX	-61.3066 MPa	115.947 MPa
Stress XY	-34.8269 MPa	33.7423 MPa
Stress XZ	-14.3895 MPa	16.75 MPa
Stress YY	-33.1683 MPa	39.2404 MPa
Stress YZ	-18.2932 MPa	9.0065 MPa
Stress ZZ	-28.1992 MPa	19.0773 MPa
X Displacement	-0.00734433 mm	0.00743681 mm
Y Displacement	-0.00451677 mm	0.0185818 mm

Z Displacement	-0.000146538 mm	0.0168226 mm
Equivalent Strain	0.000000000370972 ul	0.000937093 ul
1st Principal Strain	-0.0000091557 ul	0.000545518 ul
3rd Principal Strain	-0.00108282 ul	0.000000757001 ul
Strain XX	-0.000227533 ul	0.00051887 ul
Strain XY	-0.000215595 ul	0.000208881 ul
Strain XZ	-0.000467407 ul	0.00064112 ul
Strain YY	-0.0000966384 ul	0.000165359 ul
Strain YZ	-0.000261311 ul	0.000173856 ul
Strain ZZ	-0.000749664 ul	0.000157017 ul
Contact Pressure	0 MPa	40.8912 MPa
Contact Pressure X	-31.5279 MPa	24.7899 MPa
Contact Pressure Y	-16.0015 MPa	19.6177 MPa
Contact Pressure Z	-30.3345 MPa	21.9605 MPa

Figures

Von Mises Stress

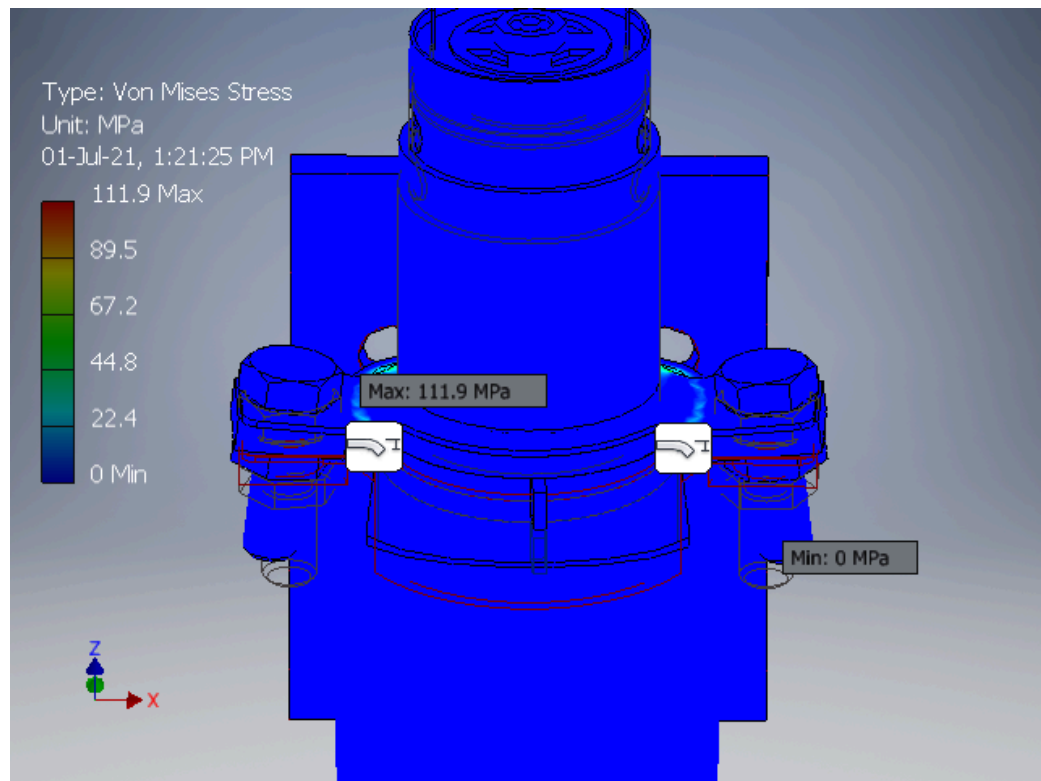


Figure 3-10: max. von mises stress for motor holder.

Displacement

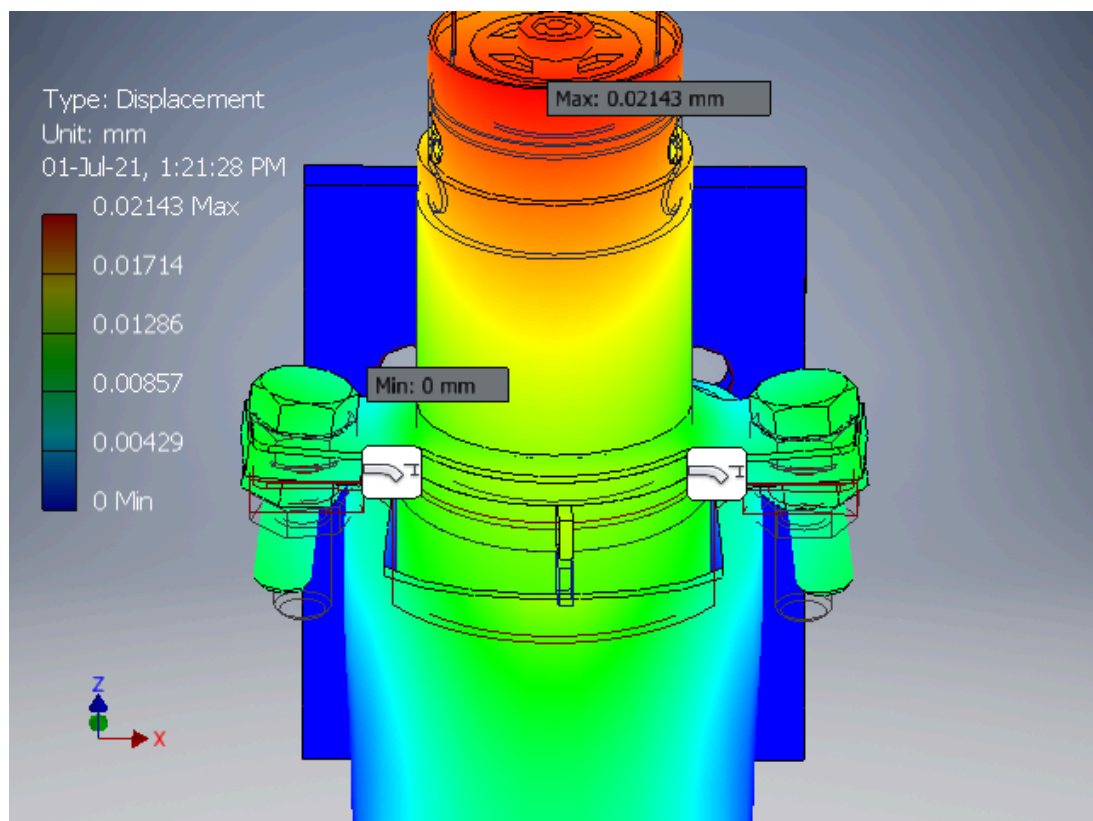


Figure 3-11: max. displacement for motor holder

Safety Factor

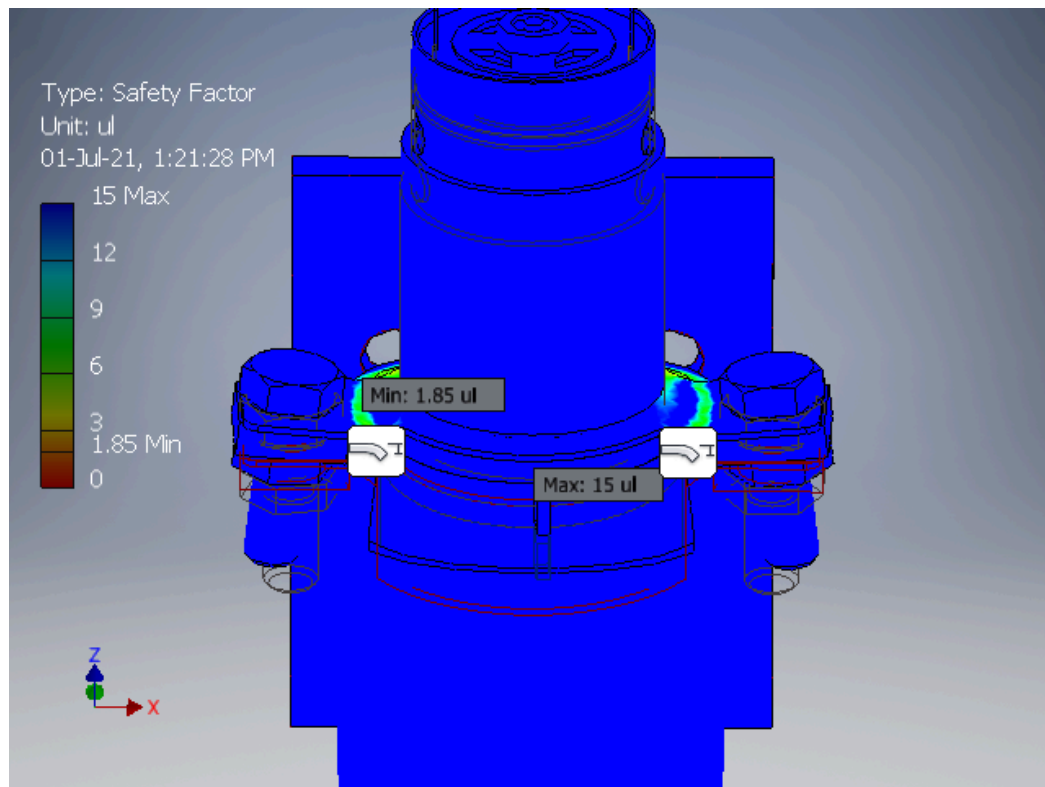


Figure 3-12: max. and min. safety factor.

Equivalent Strain

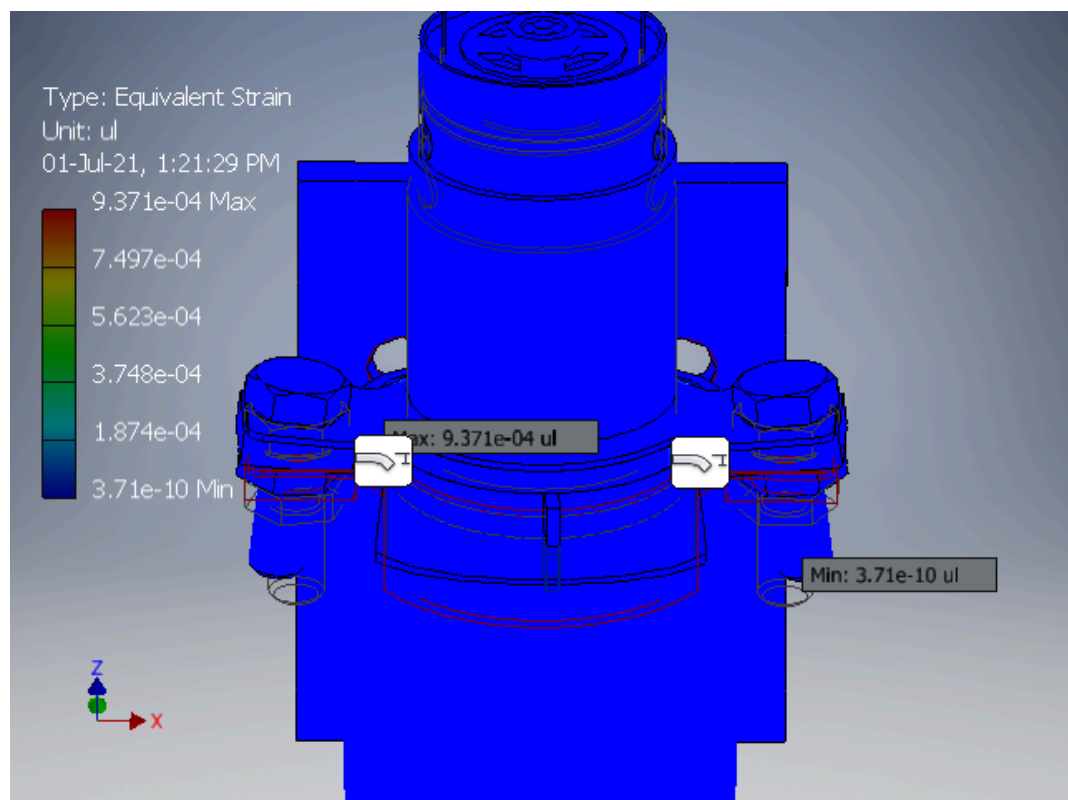


Figure 3-13: max. equivalent strain for motor holder.

Contact Pressure

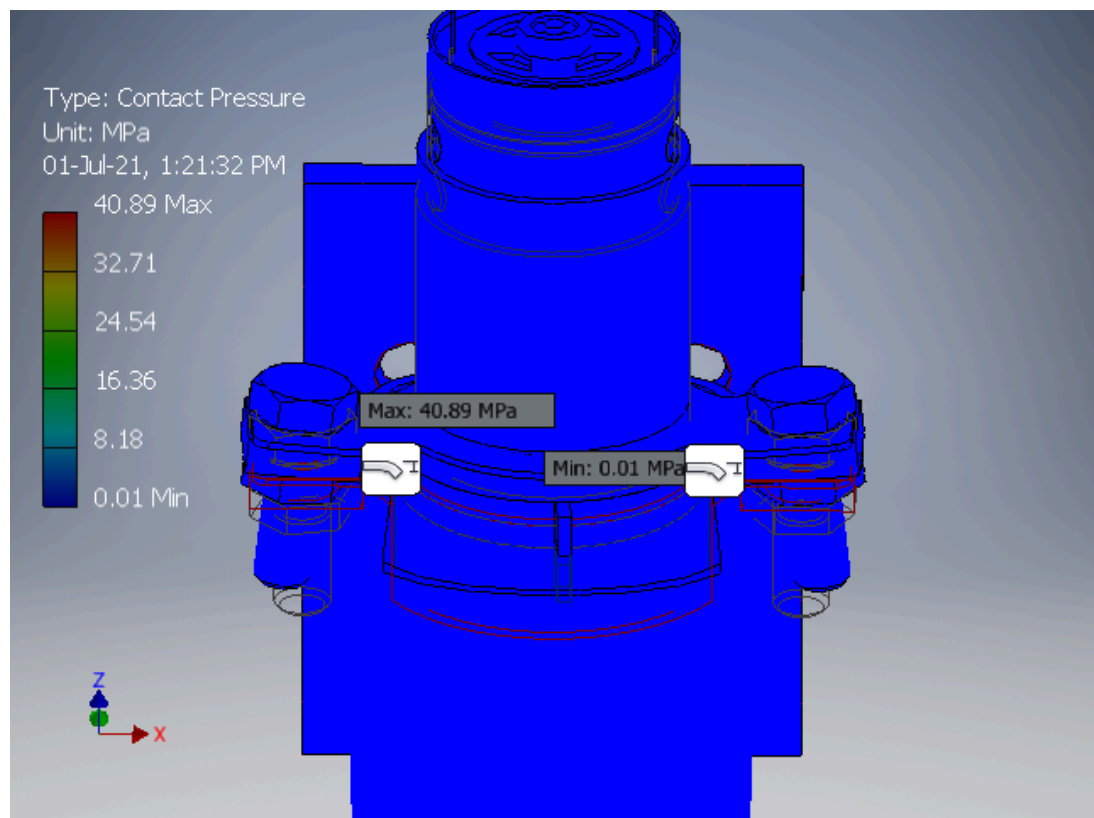


Figure 3-14: max. contact pressure for motor holder

4 MECHANICAL COMPONENTS AND MANUFACTURING

4.1 MECHANICAL COMPONENTS

4.1.1 BALL SCREW (SFU1605)

The main component to transform the rotational motion of the ball screw using our RS550 motor into linear motion to move the nut housing up and down, so we can perform the plantar flexion and dorsiflexion with their desired angles successfully. We chose this particular type of screw due to higher efficiency than lead screw and that it can handle higher loads. The specifications of this screw can be found in figure 4.1.

Size Table of SFU Ball Screws

Technical drawings of SFU ball screws showing end views with dimensions (d, l, Da, D, A, B, L, W, X, H, O, n, Ca, Coa, K) and a side view showing the screw and nut assembly. The unit is mm.

单位 (Unit):mm

1: 导程 Lead Da: 球径 Ball Dia. n: 珠圈数 Number of Circuits K: 刚性 Stiffness (Kgt/μm)
Ca: 动额定负荷 Basic Dynamic Rating Load (Kgf) Coa: 静额定负荷 Basic Static Rating Load (Kgf)

滚珠螺杆、螺帽之基准数据 Dimensions

型号 Model No.	d	l	Da	D	A	B	L	W	X	H	O	n	Ca	Coa	K	
★ SFU01204-4	12	4	2.5	24	40	10	40	32	4.5	30		4	593	1129	12.5	
★ SFU01604-4	4	2.381	28	48	10	40	38	5.5	40	M6	4	629	1270	35		
★ SFU01605-4	16	5	3.175	28	48	10	50	38	5.5	40	M6	4	780	1790	20	
★ SFU01610-3	10	3.175	28	48	10	57	38	5.5	40	M6	3	721	1249	15		
★ SFU02004-4	4	2.381	36	58	10	42	47	6.6	44	M6	4	699	1617	41		
★ SFU02005-4	5	3.175	36	58	10	51	47	6.6	44	M6	4	1130	2380	25		
★ SFU02504-4	4	2.381	40	62	10	42	51	6.6	48	M6	4	777	2052	48		
★ SFU02505-4	5	3.175	40	62	10	51	51	6.6	48	M6	4	1280	3110	35		
★ SFU02506-4	25	6	3.969	40	62	10	54	51	6.6	48	M6	4	1528	3284	40	
★ SFU02508-4	8	4.762	40	62	10	63	51	6.6	48	M6	4	1941	3863	38		
★ SFU02510-4	10	4.762	40	62	12	85	51	6.6	48	M6	4	1944	3877	33		
★ SFU03204-4	4	2.381	50	80	12	44	65	9	62	M6	4	871	2661	56		
★ SFU03205-4	5	3.175	50	80	12	52	65	9	62	M6	4	1450	4150	40		
★ SFU03206-4	32	6	3.969	50	80	12	57	65	9	62	M6	4	1720	4298	47	
★ SFU03208-4	8	4.762	50	80	12	65	65	9	62	M6	4	2189	5079	44		
★ SFU03210-4	10	6.350	50	80	12	90	65	9	62	M6	4	3390	7170	79		
★ SFU04005-4	5	3.175	63	93	14	55	78	9	70	M8	4	1610	5330	49		
★ SFU04006-4	6	3.969	63	93	14	60	78	9	70	M6	4	1911	5458	55		
★ SFU04008-4	8	4.762	63	93	14	67	78	9	70	M6	4	2435	6469	52		
★ SFU04010-4	10	6.350	63	93	14	93	78	9	70	M8	4	3910	9520	50		
★ SFU05010-4	10	6.350	75	110	16	93	93	11	85	M8	4	4450	12500	65		
★ SFU05020-4	20	7.114	75	110	16	138	93	11	85	M8	4	4644	14327	59.5		
★ SFU06310-4	10	6.350	90	125	18	98	108	11	95	M8	4	5070	16000	80		
★ SFU06320-4	20	9.525	95	135	20	149	115	13.5	100	M8	4	7573	23860	84.1		
★ SFU08010-4	10	6.350	105	145	20	98	125	13.5	110	M8	4	5620	21300	90		
★ SFU08020-4	20	9.525	125	165	25	154	145	13.5	130	M8	4	8485	30895	84.1		
★ SFU10020-4	100	20	9.525	150	202	30	180	170	17.5	155	M8	4	9420	39183	110.1	

Figure 4-1: Ball Screw (SFU1605).

4.1.2 Bearing (SKF 6002)



Figure 4-2: Deep Groove Bearing

SKF 6002 is a deep groove bearing that is used to provide an angular motion to the support and also the lower link by rotating of its inner race for “lower link” and outer race for “support”. It has an inner diameter of (15 mm.), outer diameter of (32 mm.) and width (9 mm.). It can withstand dynamic load up to ($C = 5.85 \text{ kN}$) and static load up to ($C_0 = 2.85 \text{ kN}$). It is available in local market used in various applications and many industries.

4.2 MANUFACTURED COMPONENTS

Support Back

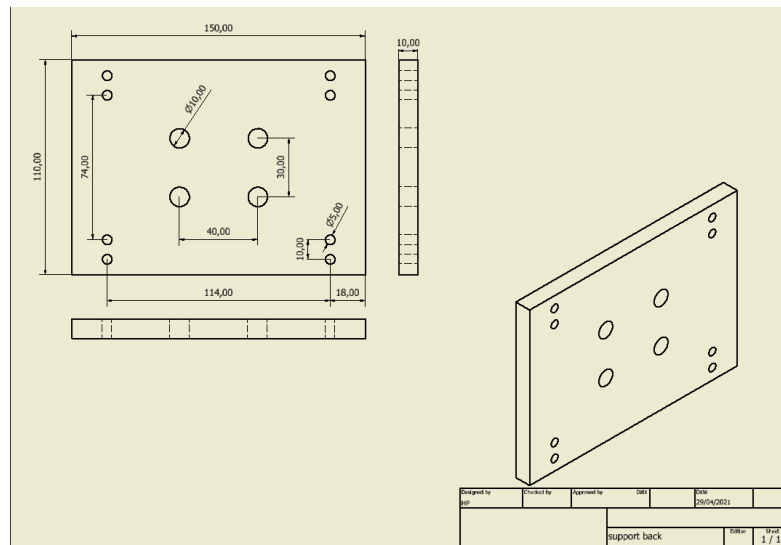


Figure 4-3: 2D sketch of support back.

The function of this part is to connect the whole support which carries the weight of the patient using it by the motor connected to the ball screw to initiate the movement of the whole system. We used the raw material Aluminium 7075 because it was fit for our project regarding the yield strength and weight which are very crucial factors in designing the system. It was cut using the Laser cutting machine for lower cost.

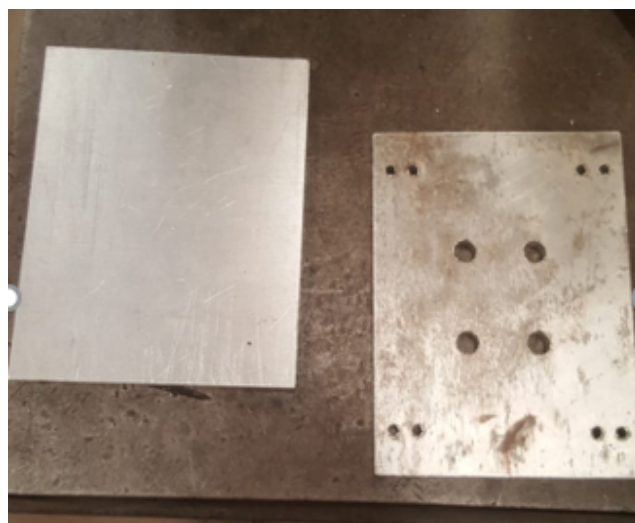


Figure 4-4: Final shape of support back.

Support Side

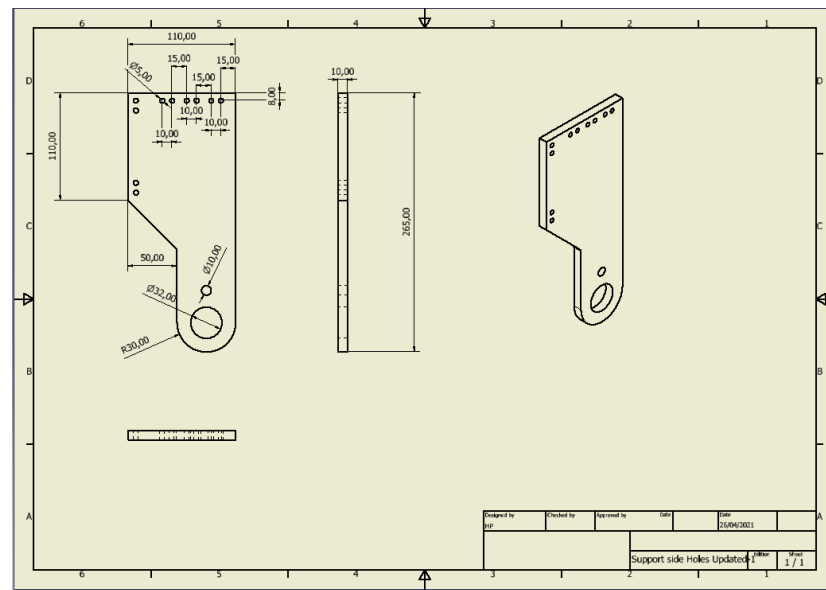


Figure 4-5: 2D sketch of support side.

The support side is the part responsible for the system body rotation about the bearing seated on the lower link shaft. It is designed to endure the weight of the patient and the rest of the system. It was made using laser cutting machine for precision and lower cost from the raw material Aluminium 7075.



Figure 4-6: Final shape of support side.

Foot Shaft

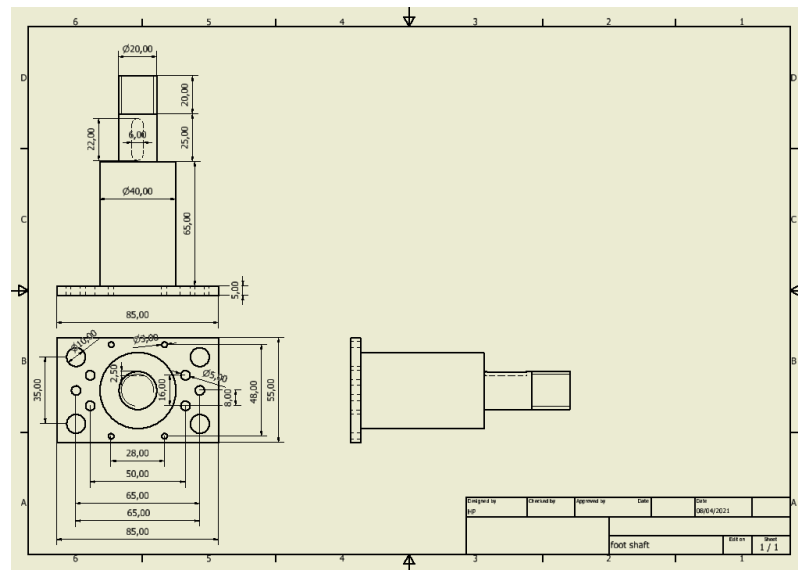


Figure 4-7: 2D sketch of foot shaft.

Foot Shaft is the part that technically carries the whole system starting from the lower link to the support and motor holder. It was manufactured using the turning machine to reach the exact diameters as shown in figure 4-7.



Figure 4-8: Final shape of foot shaft.

Heel and fore foot:

Heel and fore foot are made from steel, we purchased a steel sheet with dimension 23x108x1.5 mm from scrap and get onto laser cutting machine to cut it to desired shape. The cad file illustrate the desired dimensions needed to be achieved.

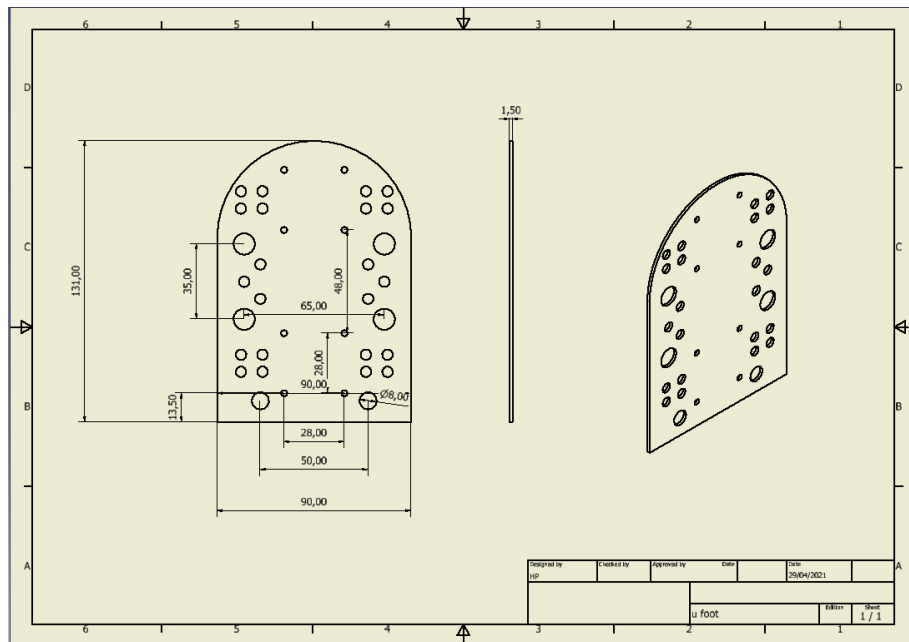


Figure 4-9: 2D sketch of heel.

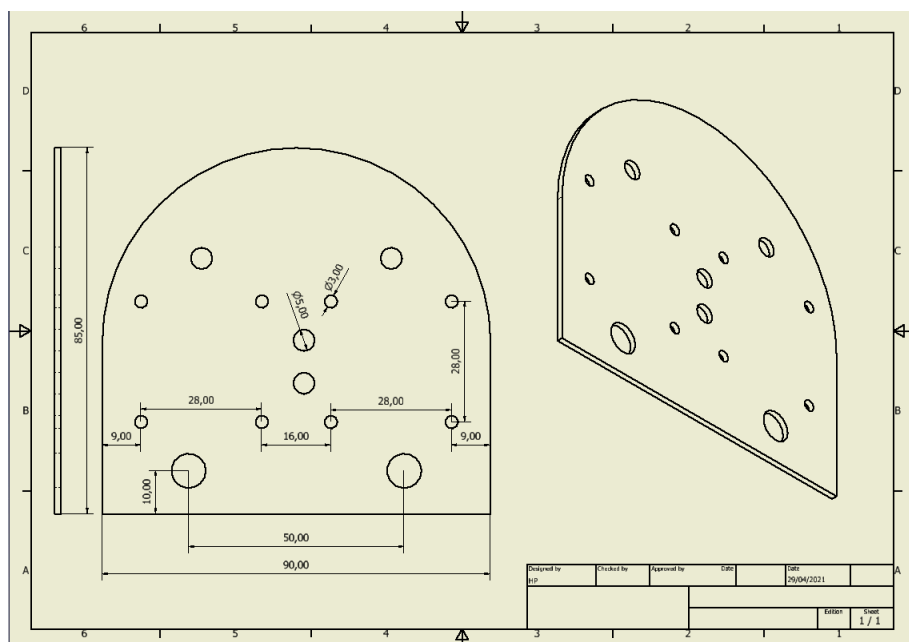


Figure 4-10: 2D sketch of forefoot.

And the desired shape after finishing the cutting process is to be:

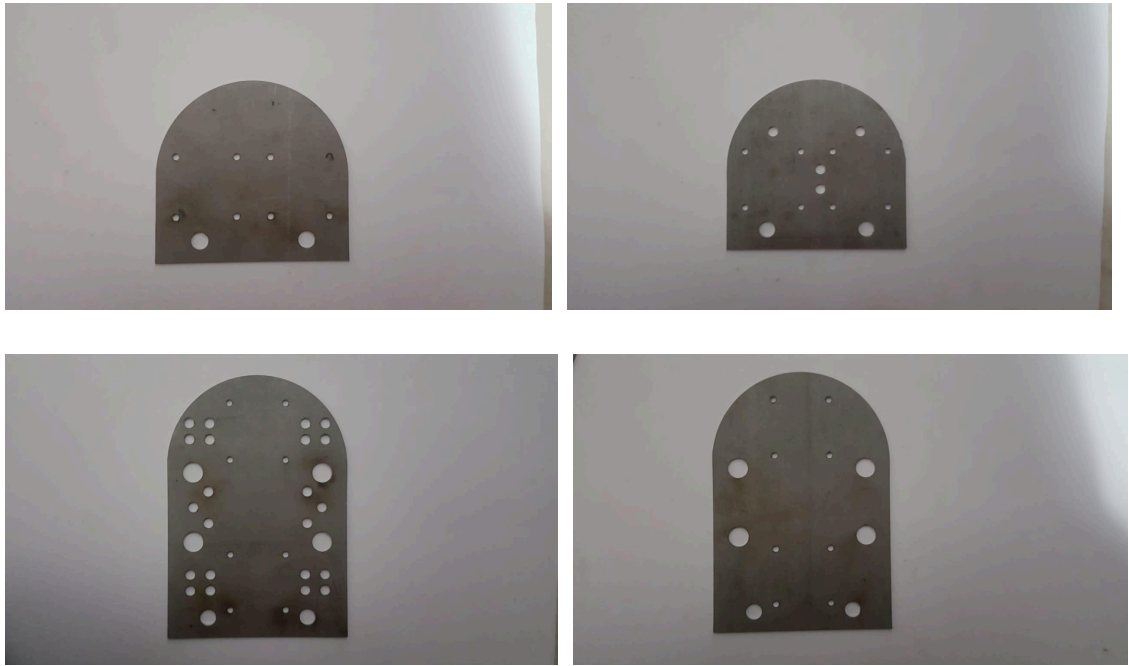


Figure 4-11: Final shape of heel and forefoot.

Spring sheet:

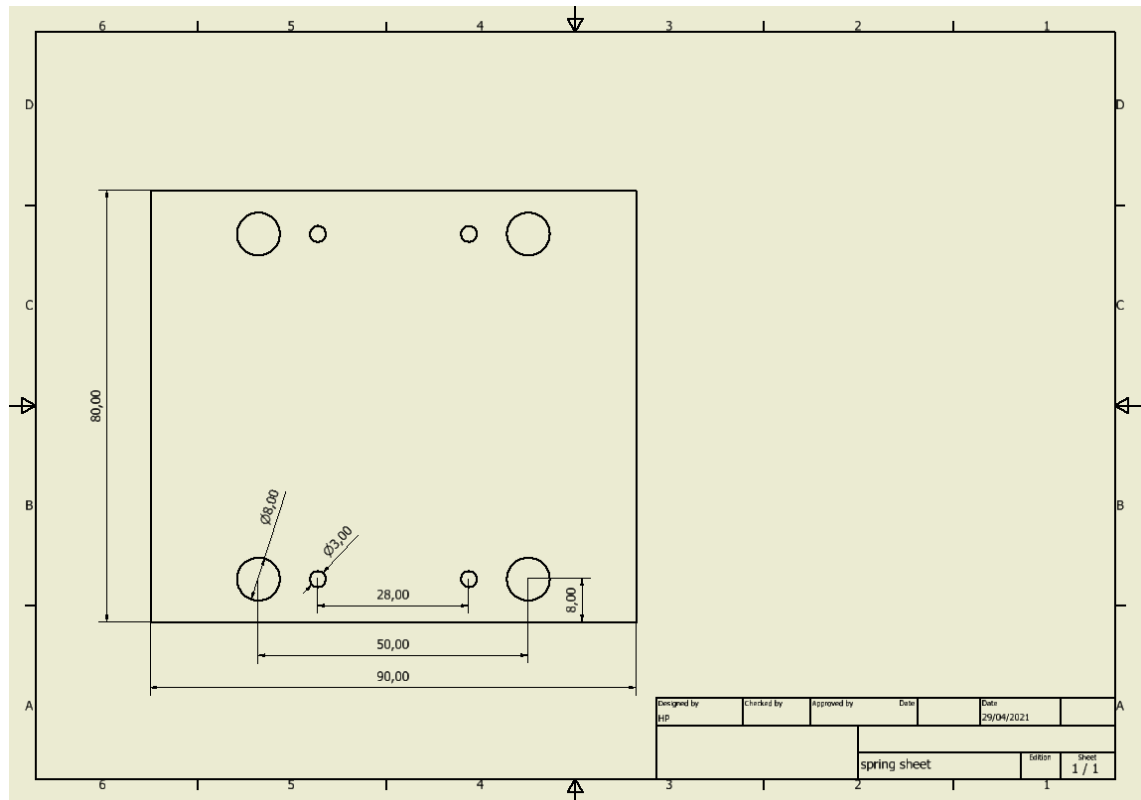


Figure 4-12: 2D sketch of spring sheet.

It is a spring steel that has an elastic property and provide the telt of foot during walking. It provides a connection between the fore foot and heel. It's cut by laser cutting to machine to achieve its desired dimension and shape



Figure 4-13: Final shape of spring sheet.

Lower Link:

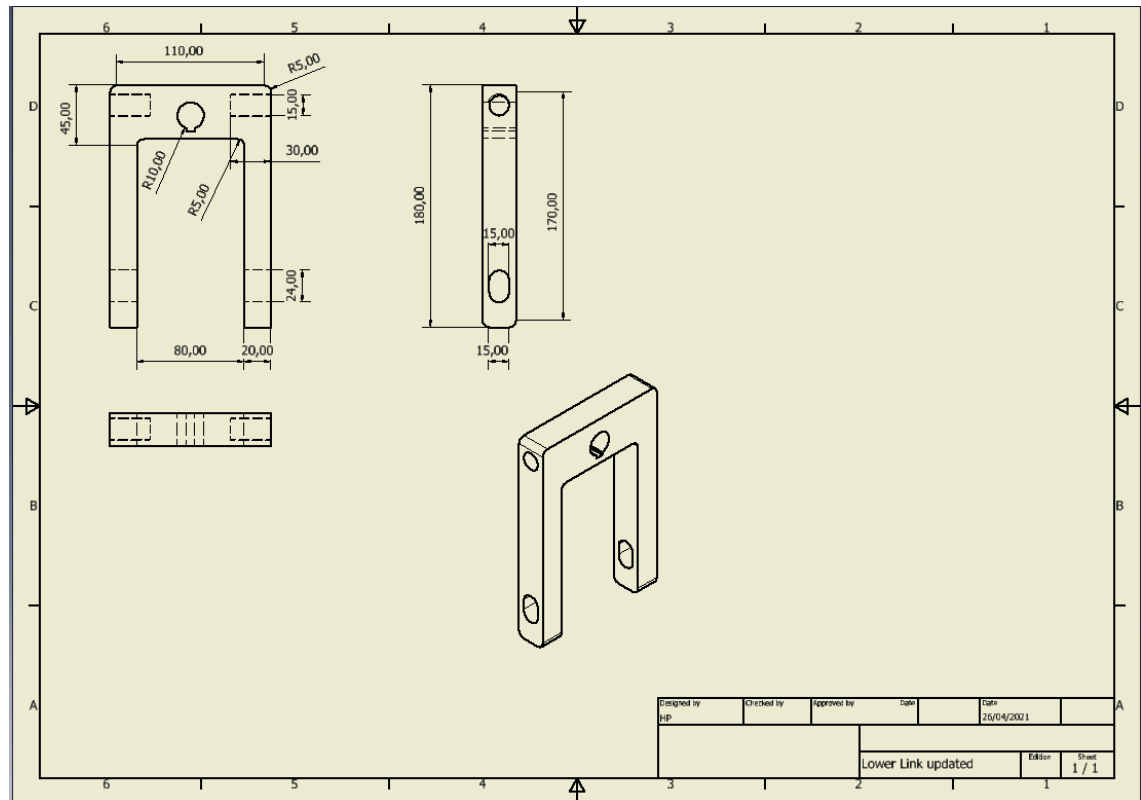


Figure 4-14: 2D sketch of lower link.

Lower link is the part responsible for linking the nut housing and the foot to mimic the move of the normal foot. It has been manufactured using the plasma cutter machine for being cost effective. We used a 7075 Aluminium for material.

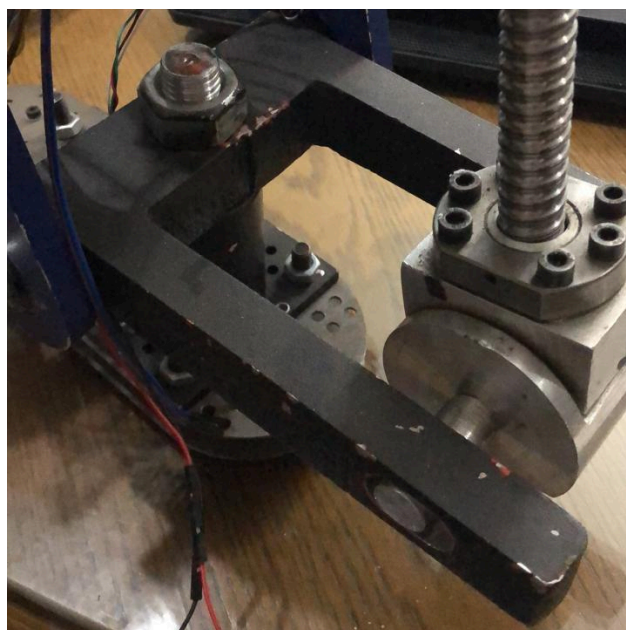


Figure 4-15: Final shape or lower link.

Motor Holder:

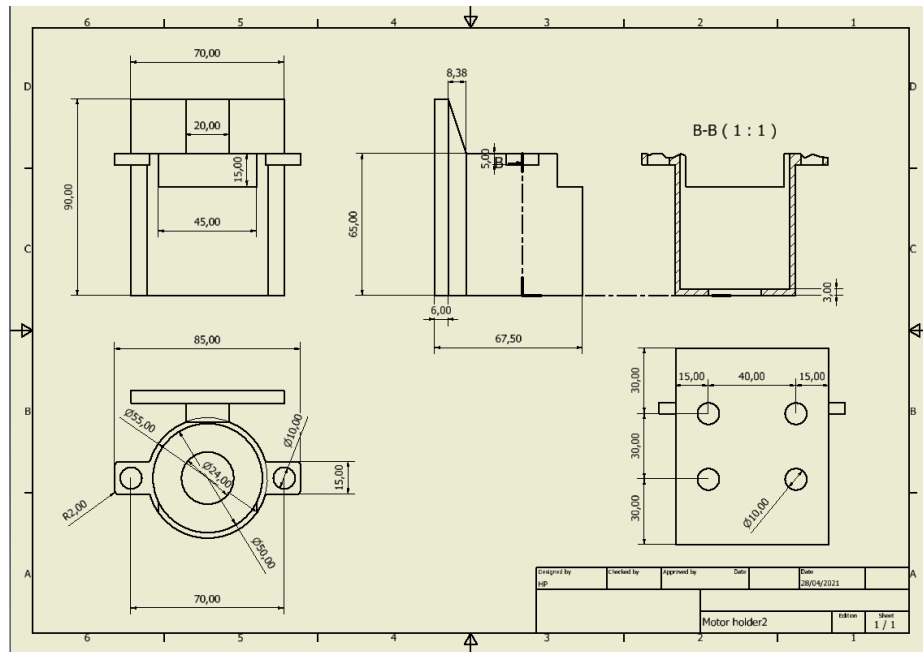


Figure 4-16: 2D sketch for motor holder.

Motor holder is the part that holds the motor to fixate the position of the ball screw to lift and lower the housing of the nut. It has been manufactured by dividing it into three parts. The first one is a 6 mm plate with the dimensions shown in the figure. The second one was a cube and the last one was a cylinder done using the turning machine. Its material is steel.



Figure 4-17: Final shape of motor holder.

Motor Plate:

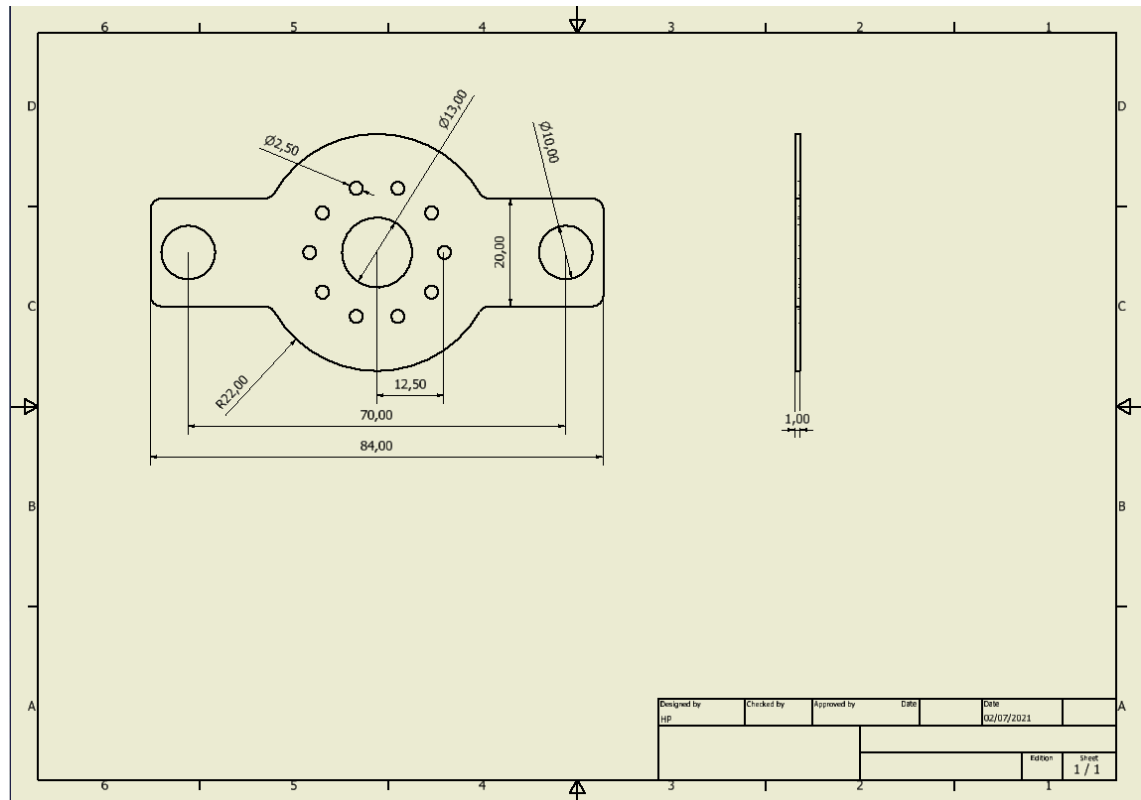


Figure 4-18: 2D sketch for motor plate.

Motor plate is used to link between the motor and the gearbox. They prevent the motor from moving upward and downward. In addition, they prevent the motor from rotating around itself. It has been manufactured using laser cutting machine and we used a steel material so it can hold high forces.



Figure 4-19: Final shape of motor plate.

5 ELECTRICAL AND CONTROL DESIGN

5.1 INTRODUCTION TO EMBEDDED COMPONENTS AND ELECTRICAL CIRCUIT DESIGN

In any control system, there is an interconnection between the constituent components, these individual components may be electrical, mechanical, or hydraulic pneumatic in nature. The embedded electronic system in Figure 4-1 of our prosthesis ankle include a commercial inexpensive low cost microcontroller ESP32 with a dual core 32 bit chip that contains programmable electronic components that perform communication as Wi-Fi and bluetooth module and control functions which are chosen to be the main embedded board due to its high frequency clock 16MHz, containing 34 general purpose input-output (GPIO), the using of inexpensive brushed DC motor (RS-550S) contain a commutator which is a rotary electric switch to provide a reverse direction between external circuit and rotor, also it can be varied in speed by changing the voltage, speed linear to the applied voltage or the drawn current and its advantage is the efficient for small loads and doesn't need to a complex control circuits, the only disadvantage is the noise of motor. The dc motor is driven by the h-bridge driver Cytron (MD13S) which used to provide bi-directional control to the brushed dc motor, This motor driver which able to produce high current to DC motor with 13A continuous and 30A peak current for 10 seconds, can drive motor with voltage ranges from 6V to 30V, a pwm frequency up to 10KHz. An encoder is an electromechanical feedback device that provides information about position, speed and direction, there are two types of encoders linear and rotary, our application needed to rotary encoder where there are many different types of rotary encoders which divide into an incremental, absolute encoder and magnetic encoder, we don't use the incremental rotary encoder because it need to a set reference point which indicate us about the real angle position of ankle joint which need to always perform the zero of encoder return , if the light transduces doesn't detect one pulse due to an error or noise the count value of encoder can not be corrected and the positional data will remain incorrect, also the results will be lost without electrical supply, the absolute rotary encoder has a disadvantage when it is related with output of

binary code absolute occurs when the output of the encoder changes its value, if 4-bit binary encoder when it changes from 7 (binary 0111) to 8 (binary 1000) the state of its output bits, the transition between 7 bit to 8 bit can get an inaccurate angle position due not all the bits changes its binary number, these problem occurs due to two reasons the the variations in slew rates of the photo-transistors, any small alignment errors in the relative positions of the photo-transistors will indicate us a wrong position. The best choice for our application is the magnetic encoder where which can be used for our prosthesis ankle which provides a good advantage of being robust in an environment contaminated with water, oil, and dust, also it is suitable for our prosthesis ankle which requires a sensor with small size and lightweight and high reliability and high accuracy, precision and resolution.

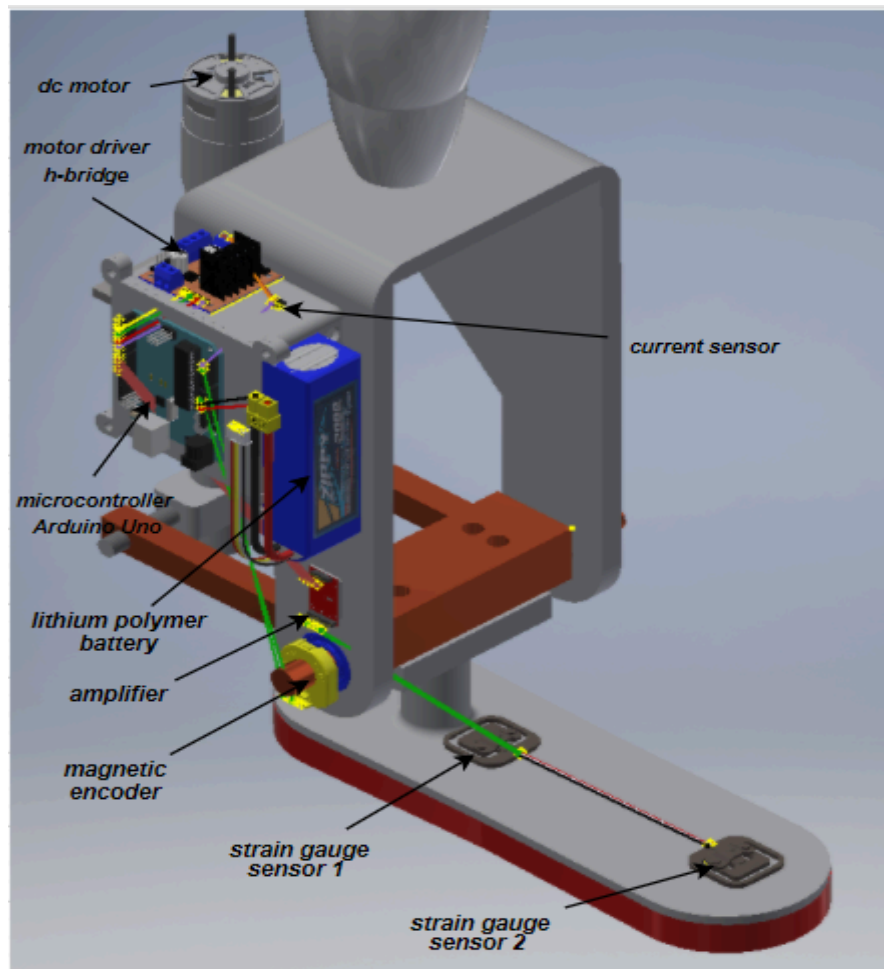


Figure 5-1: electrical control system

The magnetic encoder detects the rotational position information when occurs a change in magnetic field, converting the change of magnetic field into electrical signals, it consists of a permanent magnet and magnetic sensor where the permanent magnet is attached to the ankle joint and the magnetic sensor is fixed at a position which receives the magnetic field produced by the permanent magnet where the encoder detects the position, speed, and direction of ankle joint. The magnetic encoder consisting of a magnetic sensor the Hall element that detects the magnetic field strength which uses the phenomena of the hall effect to produce an output of a voltage proportional to the strength of the magnetic field, The use of a smart and contactless magnetic encoder (AS5600) due to its high reliability, durability, simple angular excursion programming and automatic low power modes 3.3v, providing 12-bit output resolution, an analog output, and PWM encoded output, the build-in hall sensor module determine the changes of magnetic field direction, where there is no limitation in rotation angle. Our prosthesis ankle needs to indicate the different phases of its gait cycle with the help of the strain gauge sensor implemented at foot will indicate the different phases. The strain gauge sensor, changing the resistance depending on the changing of the applied force on the load cell sensor. The chosen of four strain gauge sensor full-bridge circuit 50kg, we prefer the full-bridge circuit rather than half-bridge sensor because it provides higher measurement sensitivity to the external load applied on it and the output voltage is directly proportional to the force without needing to any approximation to the voltage value as it occurs in half bridge. The strain gauge sensor will be located on the foot, two installed at the heel and another two of load cell sensor installed at the toe to indicate the phase of the gait cycle, the usage of an amplifier signal conditioning module HX711 with a precision 24-bit designed for weighing scales and to interface directly with the strain gauge sensor, which amplifies the voltage signal from the strain gauge sensor and convert it from analog to digital value by AD converter and returning the amplified digital electrical signal to the microcontroller. The choice of lithium polymer batteries rather than lithium-ion batteries due to its benefits usage in our prosthesis ankle which providing more lightweight to decrease the weight of prosthesis ankle and also retains charging capacity, and more safety less chance of explosion.

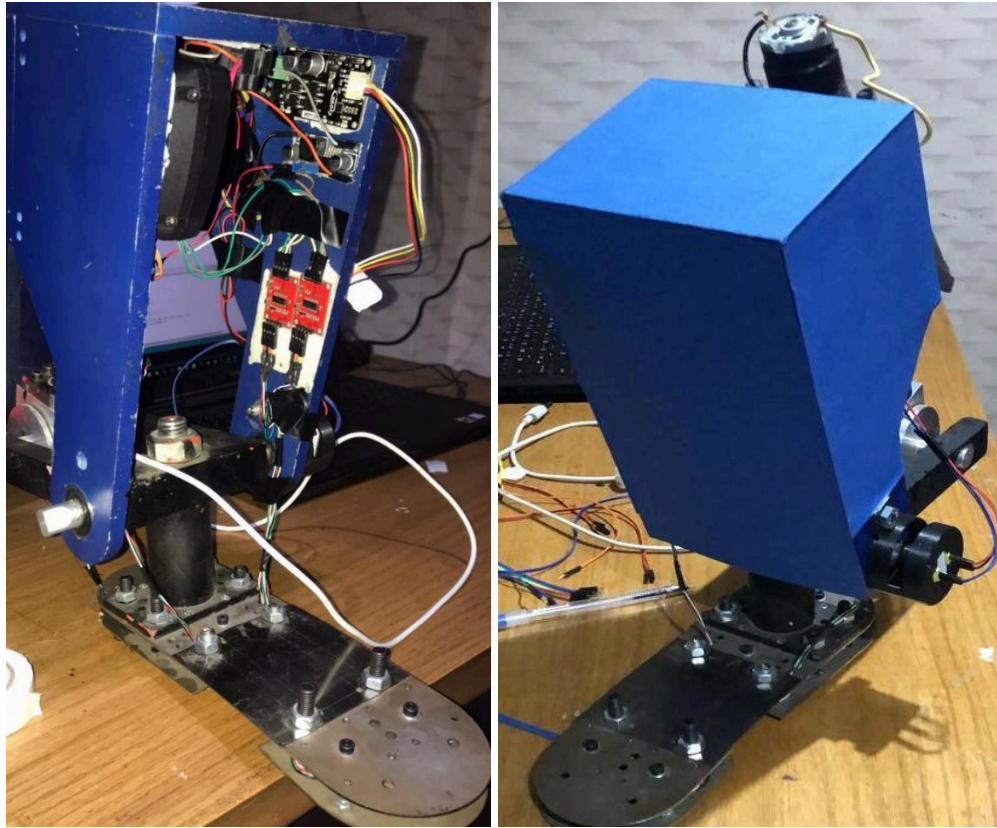


Figure 5-2: active ankle prosthesis

5.2 COMPONENTS SCHEMATICS

In Figure 5-2 shows the full circuitry of the electronics used in our project, the controller (ESP32), signal conditioning circuit (HX711) and magnetic encoder (AS5600) needing to supply voltage of 3.3V by using a buck converter to step down the battery from 18V to 3.3V with important notes that the supply for the electronics even the microcontroller (ESP32) is drawn from the main lithium-polymer battery (18V) which supplying our whole system. All logic level power is divided as following: AS5600 draws 100mA maximum, HX711 amplifier draws 1.6mA each and we are using four of them, microcontroller (ESP32) draws 80mA. This in total does not exceed 500 mA therefore the DC-DC buck step down converter (LM2596) in Figure 5-2 as 5V step down which supplies 2 Ampere is more enough than to power consumed by our electronic components and the microcontroller.

5.3.2 CONTROL SYSTEM BASED ON THE SEQUENCE OF THE GAIT CYCLE

There are many differences between the biomechanics of the human ankle in different daily activities. Our prosthesis ankle is designed for level ground walking and designed to be one degree of freedom at the ankle joint. The understanding of the gait cycle as shown in figure 5-3 helping inbuilt our control system, the gait cycle which is divided into four different phases, the controlled plantarflexion (CP) which starts at initial contact of heel and ends with the foot flat, and the prosthesis will absorb the impact force, controlled dorsiflexion (CD) which starts at foot flat and ends with maximum dorsiflexion of the second leg to provide the ability of the first leg to reach the next state of gait cycle, the powered plantarflexion or push-off phase (PP) which starts at maximum dorsiflexion (foot flat) and ends with toe-off (TO), the prosthesis foot will release the energy stored in the previous phases and will apply a suitable torque to push off the human body and during the swing phase, the prosthesis ankle will returning to the initial state of gait cycle. Due to the different phases of gait cycles, which will be determined by the strain gauge sensors, we need to get a relation between the percentage of gait cycle and ankle angle. We design a control system based on the state machine (high-level controller) and PID controller (low-level controller).

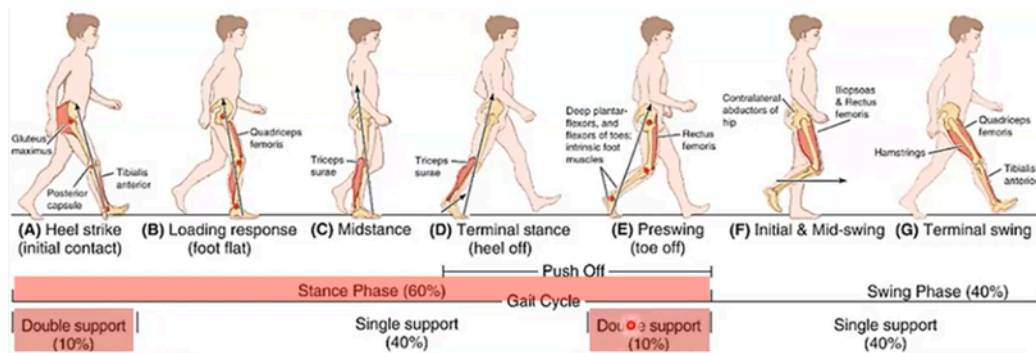


Figure 5-4: sequence of gait cycle

5.3.3 CONTROL STRATEGY WITH FINITE STATE MACHINE

To estimate the performance of prosthesis ankle in the controlled plantarflexion (CP) and the powered plantarflexion (PP) phases, the control system is designed based on two different layers, the state machine which is the high-level controller and PID

controller is the low-level controller. The block diagram shown in Figure 5-4 describes the sequence of the control system.

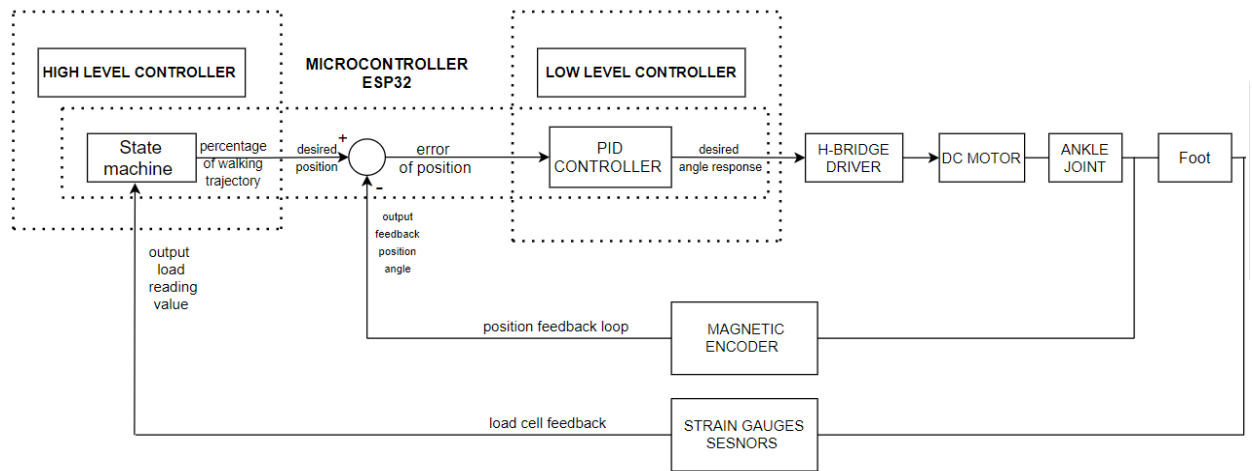


Figure 5-5: block diagram of control system

The state machine is a model that behaves the system response due to external and internal events, it consists of a finite number of states or conditions. The main function of the state machine is to collect the data from the four strain gauge sensors and construct different conditions which describe and divide our gait cycle to a percentage as shown in figure 5-5

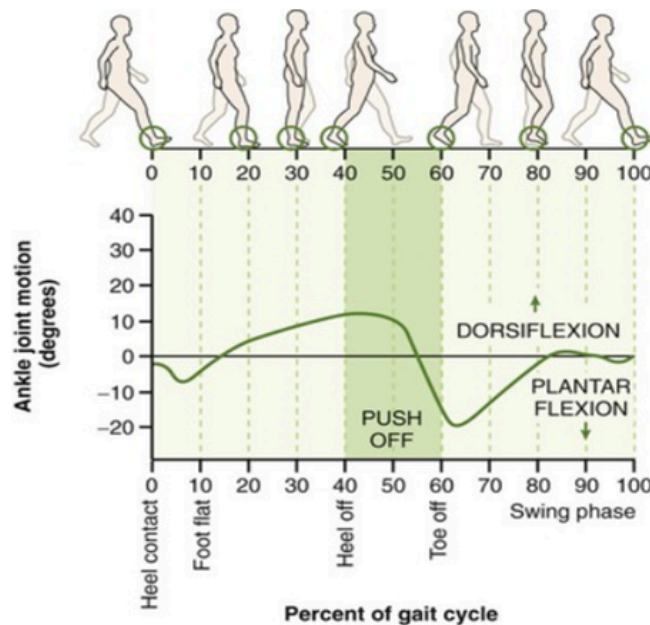


Figure 5-6: percentage of gait cycle with ankle angles

This conditions is determined by using the data collected from load cell in the form of the applied load (force) on each load cell which inserted in the foot, two load cells at the heel with acceptable maximum load up to 100kg and two load cells at the toe with acceptable maximum load up to 100kg and to get also angles from the absolute magnetic encoder installed on the prosthesis ankle joint to measure the different ankle angles in each phase, by making a relation between the conditions of load cell which determines the current state of gait cycle and the changing of ankle angles, the result is to perform a sequence walking trajectory as shown in figure 5-6. The output of the high-level controller is the desired percentage of gait cycle based on conditions with angle profile.

5.3.4 CONTROL SYSTEM BASED ON THE SEQUENCE OF THE GAIT CYCLE (WALKING TRAJECTORY SEQUENCE)

After the state is defined the controller starts the walking trajectory. For example the state machine defines that we are in the HS to FF state so the trajectory is set from 0% to 20% of the gait cycle, in the FF to HO state so the trajectory is set from 20% to 40% of the gait cycle, in the HO to TO state so the trajectory is set from 40% to 60% of the gait cycle, in swing phase to heel strike state so the trajectory is set from 60% to 100% of the gait cycle.

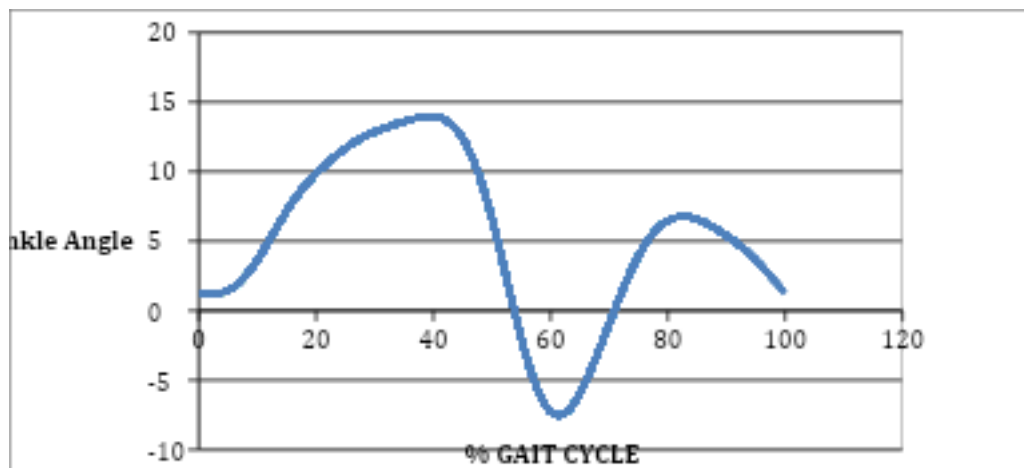


Figure 5-7: walking trajectory target response

Simply put the state machine determines the state of the foot and therefore determines the next phase of the gait cycle. Four different states determined by toe and heel load cell sensors contact. As shown in Table 5-1.

Table 5-1 : State Machine

Heel Contact	Toe Contact	State	%Range Taken from Gait Cycle
1	0	HS to FF	0% to 20%
1	1	FF to HO	20% to 40%
0	1	HO to TO	40% to 60%
0	0	Swing to HS	60% to 100%

5.3.5 PID CONTROLLER

PID stands to proportional, integral and derivative, the importance of using PID which to improve the steady-state accuracy by decreasing the steady-state error, which will result in improving the stability of the system and it provides a reduction in unwanted offsets or deviations produced by the system, at the proportional the error is multiplied by a gain (K_p), at the integral the error is multiplied by (K_i) then the error is integrated, in the derivative error is multiplied by (K_d) and then is differentiated as shown in figure 5-7, The proportional controller used in two states when the deviation should not be large or there is no large deviation between the input and output, and the deviation is not sudden, the proportional gain is directly proportional to the error where high the value of K_p gain will increase the overshoot which results in risk of instability. By decreasing the proportional gain (K_p) the overshoot will decrease but it will have a poor rise time and the steady-state error increases, by increasing the K_p gain the overshoot will increase and steady-state error will decrease so there are no compromises between the overshoot and steady-state error so we need to apply the integral controller which work on the past (memory) or the history errors by collecting all previous error and integrate them, it integrates the error until it eliminates the steady-state offset, the effective of

integral gain K_i is no longer applied when the error is so small, derivative gain (K_d) increasing the stability of the system, improving transient response by reducing the overshoot by making a damping to the system.

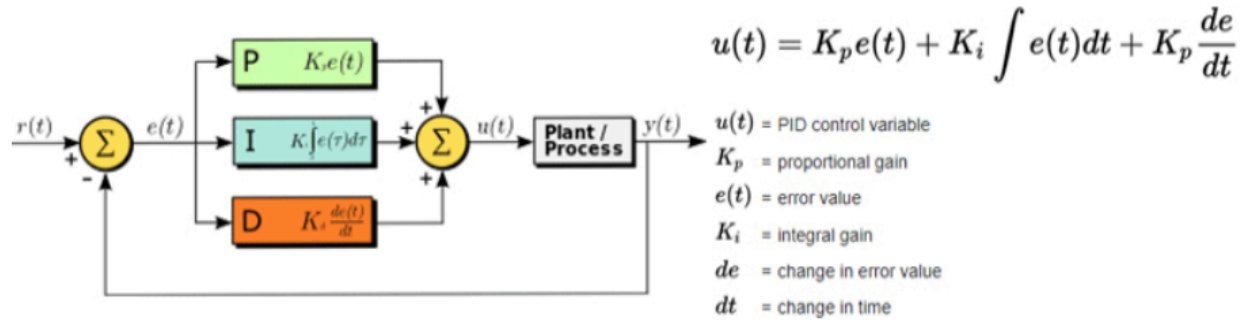


Figure 5-8: block diagram of PID controller

5.3.6 USING POSITION PID CONTROLLER IN CONTROLLING ANGLE OF ANKLE JOINT

By applying a position PID controller in our system to control the angles of ankle joint by using the rotary magnetic encoder to achieve the walking trajectory target by a very good system response with the position PID control. Simply our prosthesis control system takes readings from the load cell sensors identifies toe and heel contacts then the state machine defines the state based on the state a trajectory for the motor is set with ankle joint angles as in figure 5-8, Then our position PID controlled mechanism tries to reach this set of walking trajectory to successfully mimic the real ankle joint movement through the gait cycle, figure 5-8 shows when toe contact as the load cell sensor inserted at toe is high and the heel contact as the load cell sensor inserted at heel is high.

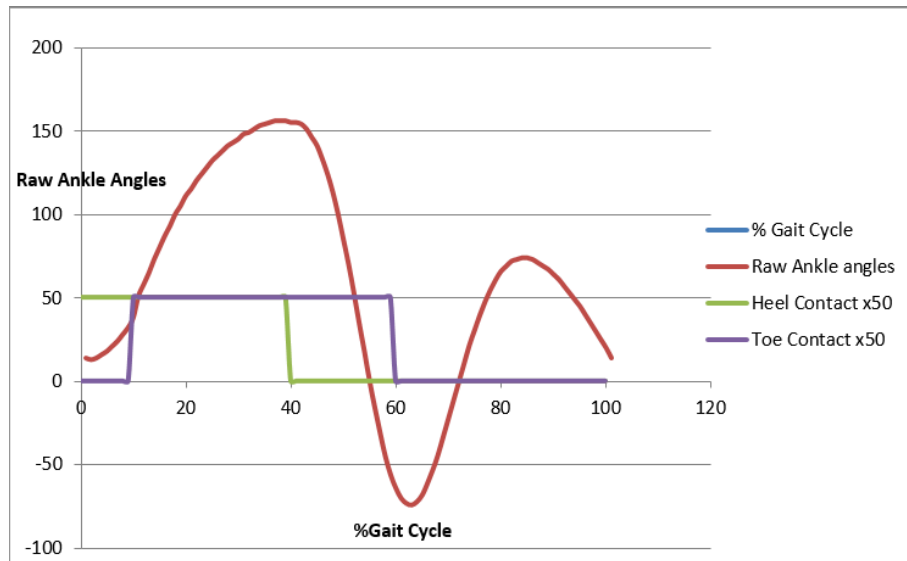


Figure 5-9: heel and toe with raw angles

5.3.7 OUR CONTROL SYSTEM USING RTOS

In any control system the program or the code is running in series between the tasks on one processing core where each processing core can only run one program at one time, we use (RTOS) when we have a multitasking objects and tasks need to be prioritized and used when there is a high number of interrupts, interrupt is when something or condition happens in the system and we need to stop the running program , by using RTOS, it switches between the tasks in a very small time, where it can run a multiple tasks in parallel way at the same time.

5.3.8 COMPONENTS SELECTED

Table 5-2-2: components

Mfr.No	Description	Qty.	Unit price
ESP32	Microcontroller	1	EGP 300

Load cell	Strain gauge sensor 50kg-full bridge	8	EGP 4250.0
AS5600	Magnetic encoder (12bit resolution)	3	EGP 270.0
HX711	Interface analog to digital converter (24bit)	5	EGP 310.00
Cytron MDC13S	h-bridge motor driver	1	EGP450.00
RS-550S	DC motor	1	EGP85.00
Switch	On and off power	1	EGP75.00
LLV-81D51EC33D6EF0866 5A4C4FA09330679	Ball screw	1	EGP800.00
C5X5.D19.L25	coupler	1	EGP150

6 CONCLUSIONS AND FUTURE WORK

In this report we came to the realization of the active ankle prosthesis needed to mimic the human ankle and why previous manufacturers of different ankles used certain components. Through our mechanical design that was completed by us based on the information gained and referenced from other ankle designs and the project were manufactured. Also the control strategy and control schematics are also completed and each component was tested separately. We could adjust our active ankle to add a custom bracket designed for a lower limb amputee and test our system more based on the feedback and comfort of the amputee. Moreover we could add an IoT application for our system that detects the steps of the amputee to measure his overall activity, to start the active ankle through wireless control and also to send the amputee information about the battery of the system and notify him when it needs to be recharged.

7 APPENDIX A: (INVENTOR CALCULATIONS)

1. Mid stance phase:



Analyzed File:	Support Updated 5mm.iam
Autodesk Inventor Version:	2017 (Build 210142000, 142)
Creation Date:	29/04/2021, 20:31
Study Author:	HP
Summary:	

7.1.1 Project Info (iProperties)

Summary

Author	HP
--------	----

Project

Part Number	Support Updated 5mm
Designer	HP
Cost	£0.00
Date Created	26/04/2021

Status

Design Status	WorkInProgress
---------------	----------------

Physical

Mass	4.24829 kg
Area	469805 mm ²
Volume	1607740 mm ³
Center of Gravity	x=152.549 mm y=76.4406 mm z=-14.6847 mm

Note: Physical values could be different from Physical values used by FEA reported below.

7.2 STATIC ANALYSIS:1

General objective and settings:

Design Objective	Single Point
Study Type	Static Analysis
Last Modification Date	29/04/2021, 20:31
Detect and Eliminate Rigid Body Modes	No
Separate Stresses Across Contact Surfaces	No
Motion Loads Analysis	No

Mesh settings:

Avg. Element Size (fraction of model diameter)	0.1
Min. Element Size (fraction of avg. size)	0.2
Grading Factor	1.5
Max. Turn Angle	60 deg
Create Curved Mesh Elements	No
Use part based measure for Assembly mesh	Yes

7.2.1 Material(s)

Name	Aluminum 7075	
General	Mass Density	2.81 g/cm ³
	Yield Strength	145 MPa
	Ultimate Tensile Strength	276 MPa
Stress	Young's Modulus	71.7 GPa
	Poisson's Ratio	0.33 ul
	Shear Modulus	26.9549 GPa
Part Name(s)	Support side Holes Updated.ipt Support side Holes Updated.ipt Grab Plate3 (5mm).ipt support back.ipt foot shaft.ipt	

	Lower Link updated.ipt Lower link bolt.ipt Lower link bolt.ipt
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Name	Steel	
General	Mass Density	7.85 g/cm ³
	Yield Strength	207 MPa
	Ultimate Tensile Strength	345 MPa
Stress	Young's Modulus	210 GPa
	Poisson's Ratio	0.3 ul
	Shear Modulus	80.7692 GPa
Part Name(s)	L shape 3.ipt L shape 3.ipt L shape 3.ipt L shape 3.ipt L shape 3.ipt L shape 3.ipt L shape 3.ipt L shape 3.ipt L shape 3.ipt L shape 3.ipt L shape 3.ipt L shape 3.ipt User Library-15mm 6002 BEARING.ipt User Library-15mm 6002 BEARING.ipt	

Name	Stainless Steel	
General	Mass Density	8 g/cm ³
	Yield Strength	250 MPa

	Ultimate Tensile Strength	540 MPa
Stress	Young's Modulus	193 GPa
	Poisson's Ratio	0.3 ul
	Shear Modulus	74.2308 GPa
Part Name(s)	u foot.ipt fore foot.ipt spring sheet.ipt u foot 2.ipt fore foot 2.ipt	

Name	Aluminum 7075	
General	Mass Density	2.81 g/cm ³
	Yield Strength	480 MPa
	Ultimate Tensile Strength	550 MPa
Stress	Young's Modulus	68.9 GPa
	Poisson's Ratio	0.33 ul
	Shear Modulus	25.9023 GPa
Part Name(s)	load cell.ipt load cell.ipt load cell.ipt load cell.ipt	

Name	Rubber, Silicone
------	------------------

General	Mass Density	1.25 g/cm ³
	Yield Strength	10.34 MPa
	Ultimate Tensile Strength	6.5 MPa
Stress	Young's Modulus	0.003 GPa
	Poisson's Ratio	0.49 ul
	Shear Modulus	0.00100671 GPa
Part Name(s)	lower fore foot.ipt lower heel.ipt	

Name	ABS Plastic	
General	Mass Density	1.06 g/cm ³
	Yield Strength	20 MPa
	Ultimate Tensile Strength	29.6 MPa
Stress	Young's Modulus	2.24 GPa
	Poisson's Ratio	0.38 ul
	Shear Modulus	0.811594 GPa
Part Name(s)	Hx711 box.ipt	

Name	Steel, Mild
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General	Mass Density	7.85 g/cm ³
	Yield Strength	207 MPa
	Ultimate Tensile Strength	345 MPa
Stress	Young's Modulus	220 GPa
	Poisson's Ratio	0.275 ul
	Shear Modulus	86.2745 GPa
Part Name(s)	ISO 2491 A A 6 x 4 x 22	

7.2.2 Operating conditions

Force:1

Load Type	Force
Magnitude	1000.000 N
Vector X	0.000 N
Vector Y	-1000.000 N
Vector Z	-0.000 N

Selected Face(s)

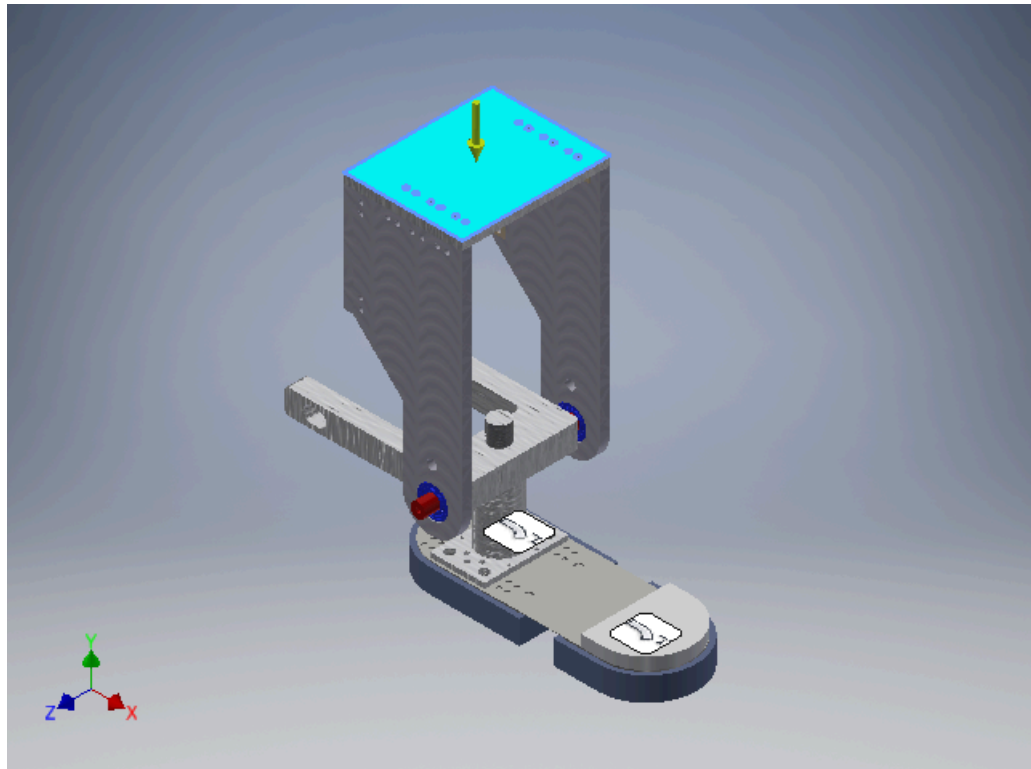


Figure 7-1 : Assembly design for mid stance phase.

Results

Reaction Force and Moment on Constraints

Constraint Name	Reaction Force		Reaction Moment	
	Magnitude	Component (X,Y,Z)	Magnitude	Component (X,Y,Z)
Fixed Constraint:1	1000 N	0 N	110.526 N m	-0.417908 N m
		1000 N		0 N m
		0 N		-110.525 N m

Figures

Von Mises Stress

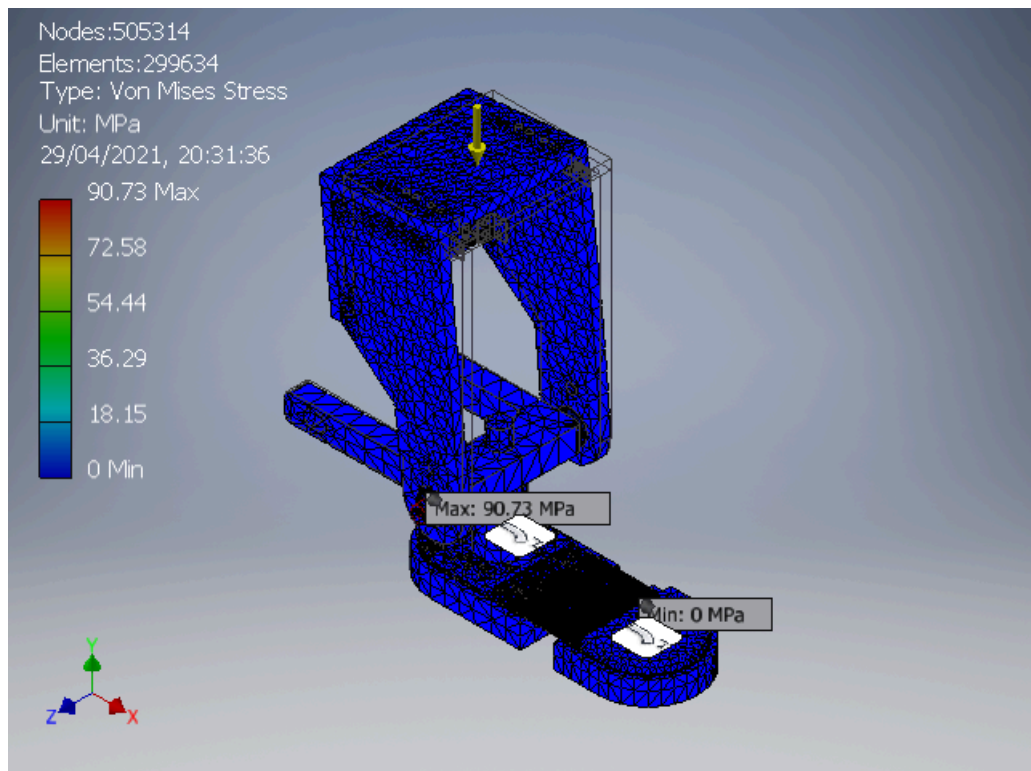


Figure 7-2 : : Von Mises stress for assembly in mid stance phase.

Displacement

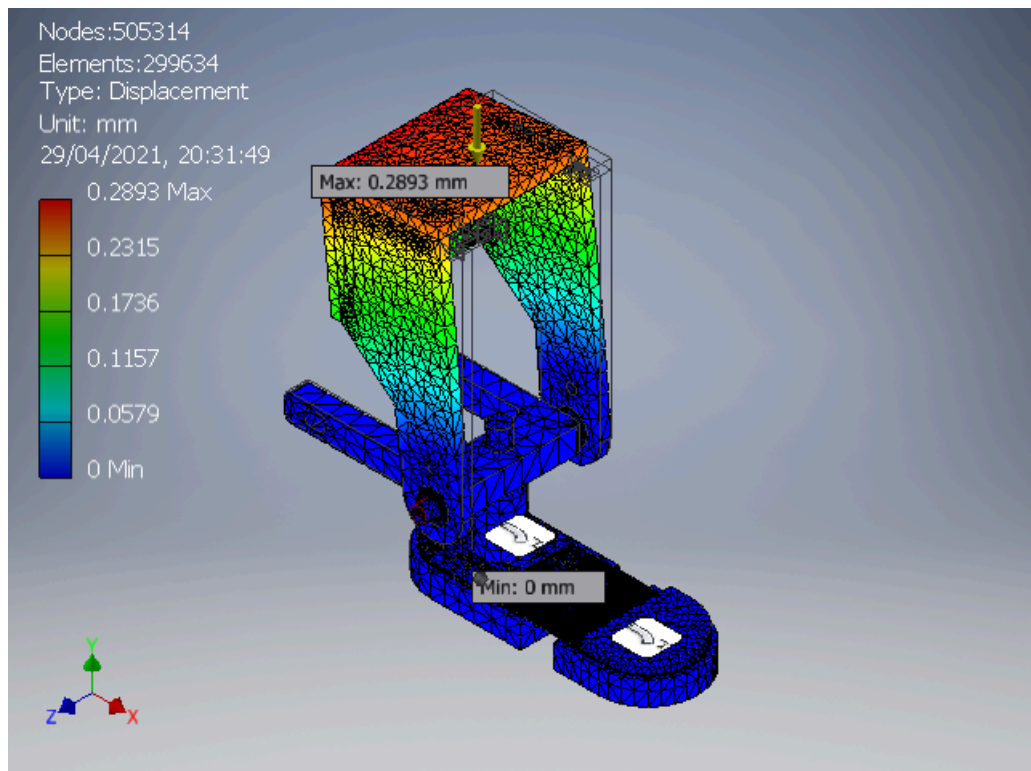


Figure 7-3 : max. displacement for assembly in mid stance phase.

Safety Factor

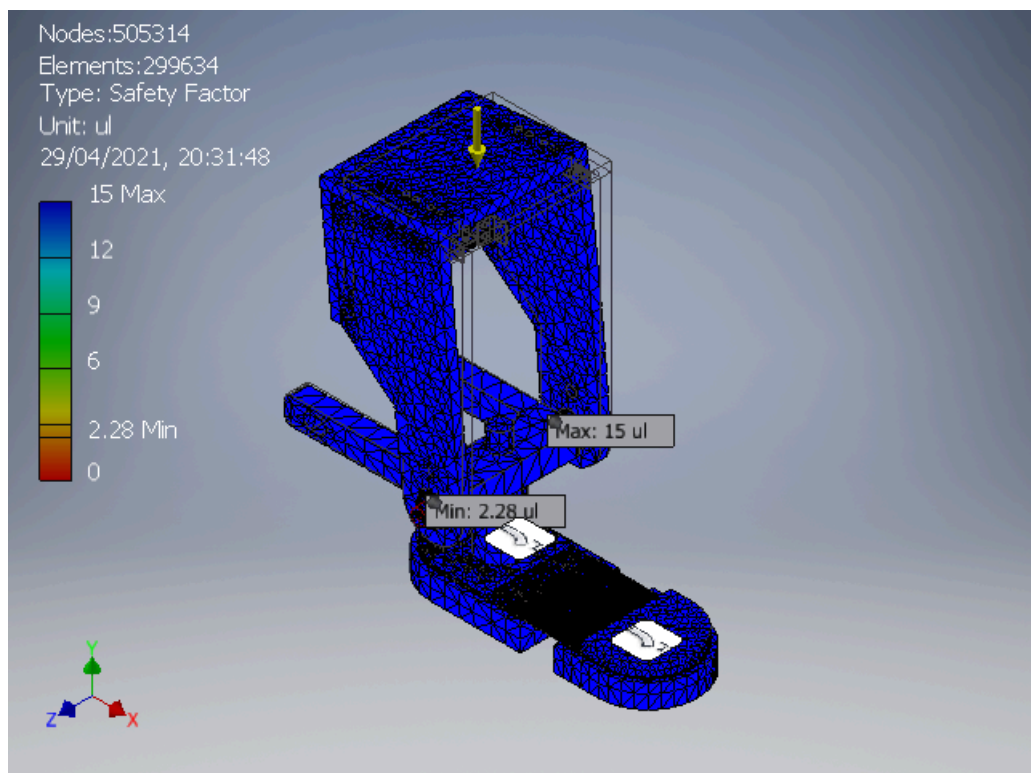


Figure 7-4: max. and min. safety factor for assembly in mid stance phase.

Equivalent Strain

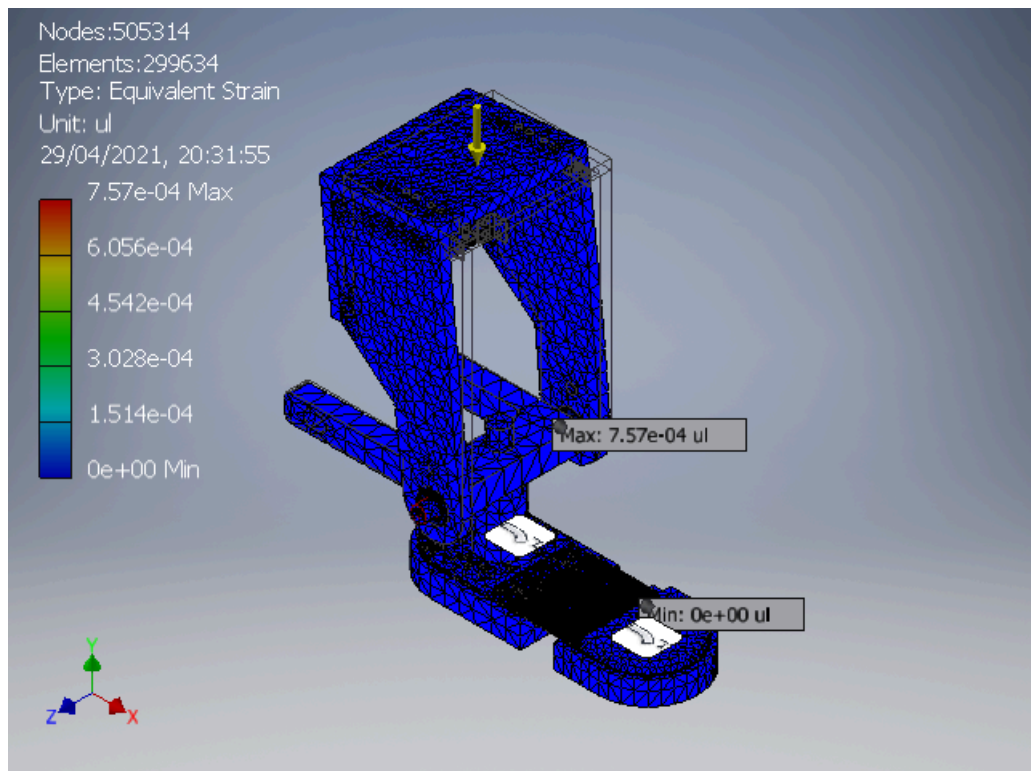


Figure 7-5: max. and min. strain for assembly in mid stance phase.

Contact Pressure

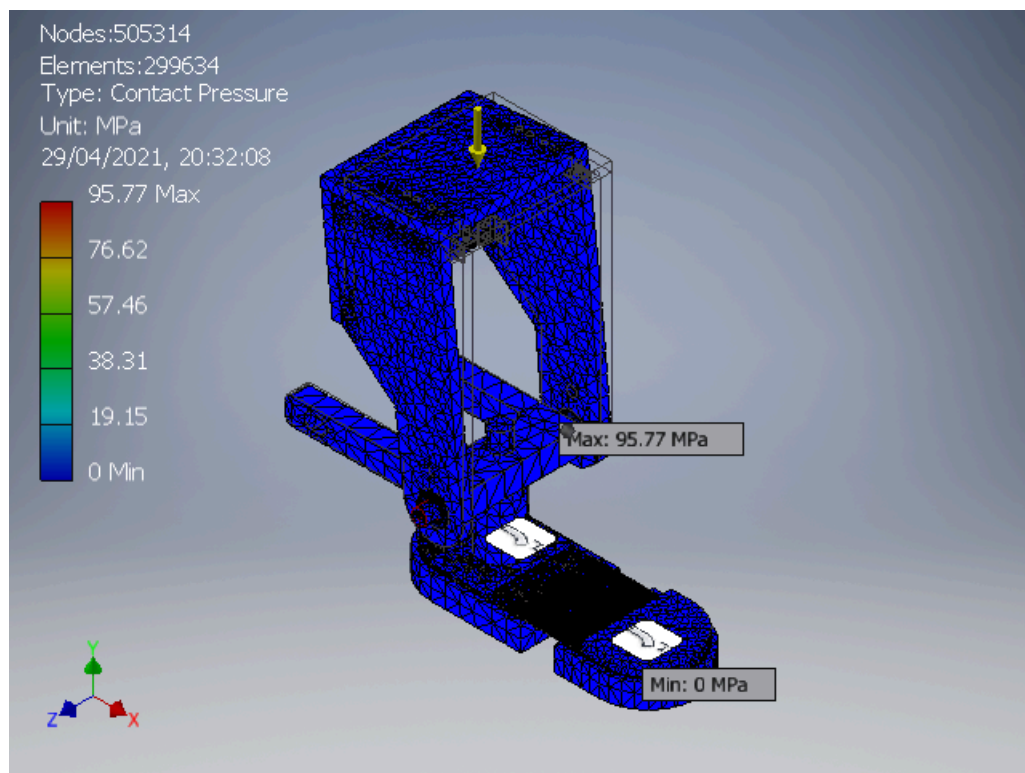


Figure 7-6 : Contact pressure.

2. Heel strike phase:

General objective and settings:

Design Objective	Single Point
Study Type	Static Analysis
Last Modification Date	29/04/2021, 20:39
Detect and Eliminate Rigid Body Modes	No
Separate Stresses Across Contact Surfaces	No
Motion Loads Analysis	No

Mesh settings:

Avg. Element Size (fraction of model diameter)	0.1
Min. Element Size (fraction of avg. size)	0.2
Grading Factor	1.5
Max. Turn Angle	60 deg
Create Curved Mesh Elements	No
Use part based measure for Assembly mesh	Yes

7.2.3 Material(s)

Name	Aluminum 7075	
General	Mass Density	2.81 g/cm ³

	Yield Strength	480 MPa
	Ultimate Tensile Strength	550 MPa
Stress	Young's Modulus	71.7 GPa
	Poisson's Ratio	0.33 ul
	Shear Modulus	26.9549 GPa
Part Name(s)	Support side Holes Updated.ipt Support side Holes Updated.ipt Grab Plate3 (5mm).ipt support back.ipt foot shaft.ipt Lower Link updated.ipt Lower link bolt.ipt Lower link bolt.ipt	

Name	Steel	
General	Mass Density	7.85 g/cm^3
	Yield Strength	207 MPa
	Ultimate Tensile Strength	345 MPa
Stress	Young's Modulus	210 GPa
	Poisson's Ratio	0.3 ul
	Shear Modulus	80.7692 GPa
Part Name(s)	L shape 3.ipt L shape 3.ipt L shape 3.ipt L shape 3.ipt	

	L shape 3.ipt L shape 3.ipt L shape 3.ipt L shape 3.ipt L shape 3.ipt L shape 3.ipt User Library-15mm 6002 BEARING.ipt User Library-15mm 6002 BEARING.ipt
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Name	Stainless Steel	
General	Mass Density	8 g/cm ³
	Yield Strength	250 MPa
	Ultimate Tensile Strength	540 MPa
Stress	Young's Modulus	193 GPa
	Poisson's Ratio	0.3 ul
	Shear Modulus	74.2308 GPa
Part Name(s)	u foot.ipt fore foot.ipt spring sheet.ipt u foot 2.ipt fore foot 2.ipt	

Name	Aluminum 6061	
General	Mass Density	2.7 g/cm ³
	Yield Strength	275 MPa

	Ultimate Tensile Strength	310 MPa
Stress	Young's Modulus	68.9 GPa
	Poisson's Ratio	0.33 ul
	Shear Modulus	25.9023 GPa
Part Name(s)	load cell.ipt load cell.ipt load cell.ipt load cell.ipt	

Name	Rubber, Silicone	
General	Mass Density	1.25 g/cm ³
	Yield Strength	10.34 MPa
	Ultimate Tensile Strength	6.5 MPa
Stress	Young's Modulus	0.003 GPa
	Poisson's Ratio	0.49 ul
	Shear Modulus	0.00100671 GPa
Part Name(s)	lower fore foot.ipt lower heel.ipt	

Name	ABS Plastic	
General	Mass Density	1.06 g/cm ³

	Yield Strength	20 MPa
	Ultimate Tensile Strength	29.6 MPa
Stress	Young's Modulus	2.24 GPa
	Poisson's Ratio	0.38 ul
	Shear Modulus	0.811594 GPa
Part Name(s)	Hx711 box.ipt	

Name	Steel, Mild	
General	Mass Density	7.85 g/cm^3
	Yield Strength	207 MPa
	Ultimate Tensile Strength	345 MPa
Stress	Young's Modulus	220 GPa
	Poisson's Ratio	0.275 ul
	Shear Modulus	86.2745 GPa
Part Name(s)	ISO 2491 A A 6 x 4 x 22	

7.2.4 Operating conditions

Force:1

Load Type	Force
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Magnitude	1000.000 N
Vector X	258.819 N
Vector Y	-965.926 N
Vector Z	-0.000 N

Selected Face(s)

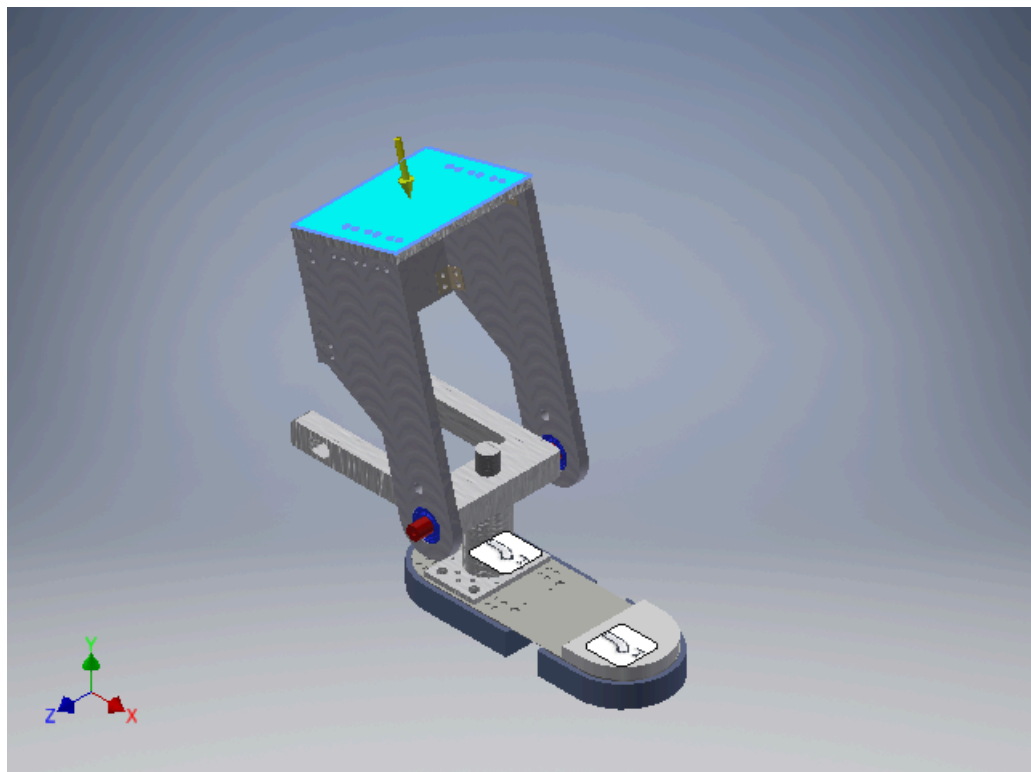


Figure 7-7: stress analysis for assembly in heel strike phase.

Results

Reaction Force and Moment on Constraints

Constraint Name	Reaction Force		Reaction Moment	
	Magnitude	Component (X,Y,Z)	Magnitude	Component (X,Y,Z)
Fixed Constraint:1	1000 N	-258.819 N	87.8095 N m	-0.415541 N m
		965.926 N		-0.111745 N m
		0 N		-87.8085 N m

Result Summary

Name	Minimum	Maximum
Volume	1607740 mm ³	
Mass	4.24829 kg	
Von Mises Stress	0 MPa	97.3773 MPa
1st Principal Stress	-8.94761 MPa	68.656 MPa
3rd Principal Stress	-70.3691 MPa	10.3659 MPa
Displacement	0 mm	0.272611 mm
Safety Factor	2.12575 ul	15 ul
Stress XX	-52.1832 MPa	55.0694 MPa
Stress XY	-31.1024 MPa	53.7959 MPa
Stress XZ	-29.4109 MPa	23.9871 MPa

Stress YY	-61.9535 MPa	62.0903 MPa
Stress YZ	-25.6039 MPa	32.6221 MPa
Stress ZZ	-19.1001 MPa	21.3374 MPa
X Displacement	-0.246031 mm	0.0313534 mm
Y Displacement	-0.17034 mm	0.0161345 mm
Z Displacement	-0.0153956 mm	0.0197517 mm
Equivalent Strain	0 ul	0.000769092 ul
1st Principal Strain	-0.00000194484 ul	0.000671026 ul
3rd Principal Strain	-0.000694082 ul	0.0000055334 ul
Strain XX	-0.000307814 ul	0.000274956 ul
Strain XY	-0.000247556 ul	0.000333022 ul
Strain XZ	-0.000545557 ul	0.000444949 ul
Strain YY	-0.000316405 ul	0.000336698 ul
Strain YZ	-0.00047494 ul	0.000605123 ul
Strain ZZ	-0.000194301 ul	0.000211114 ul
Contact Pressure	0 MPa	100.127 MPa
Contact Pressure X	-66.874 MPa	65.7183 MPa
Contact Pressure Y	-88.0293 MPa	56.4569 MPa

Contact Pressure Z	-18.7354 MPa	33.9305 MPa
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Figures

Von Mises Stress

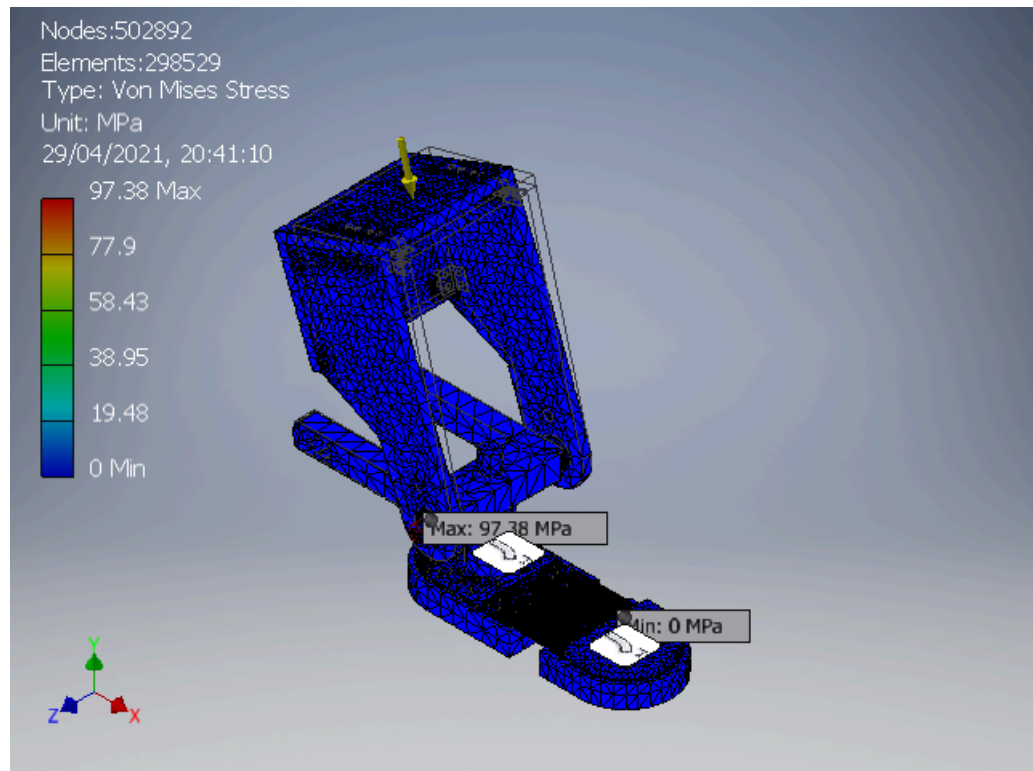


Figure 7-8: Von Mises stresses for assembly in heel strike phase.

Displacement

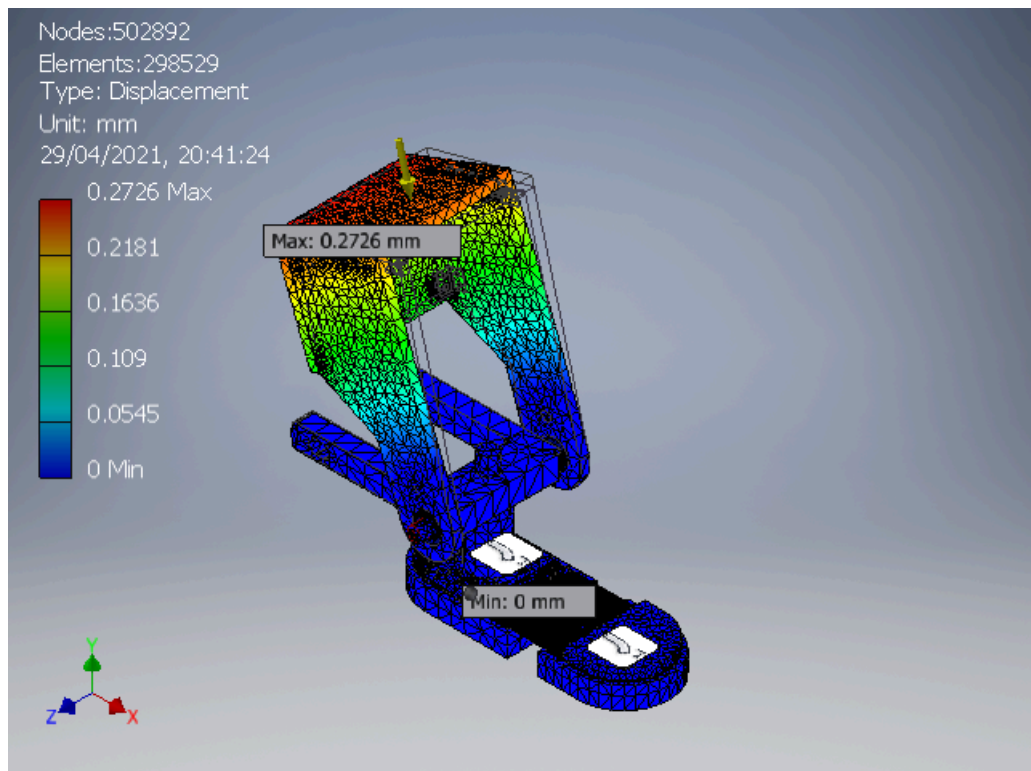


Figure 7-9: max. displacement for assembly in heel strike phase.

Safety Factor

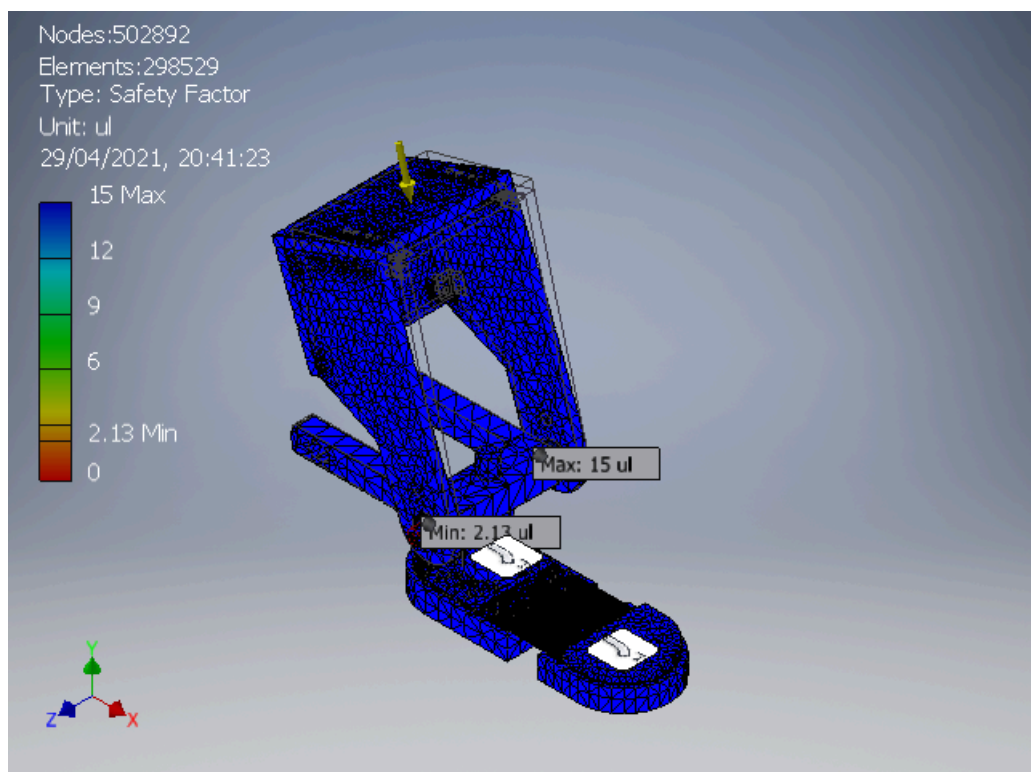


Figure 7-10: max. and min. safety factor for assembly in heel strike phase.

Equivalent Strain

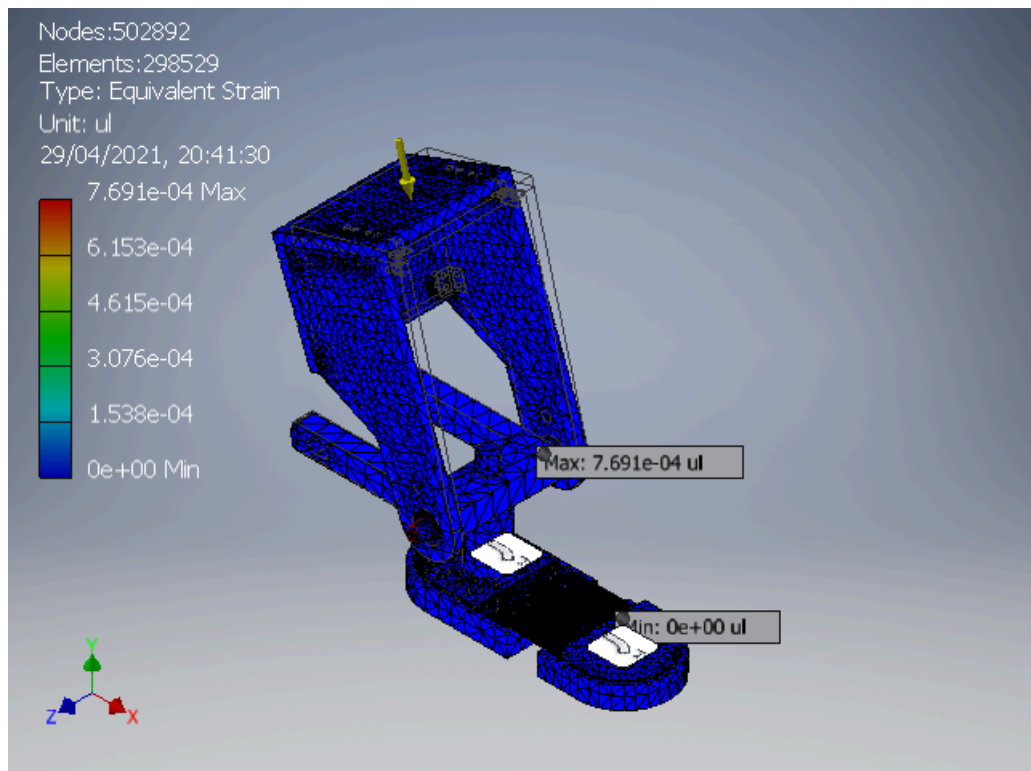


Figure 7-11: max. equivalent strain for assembly in heel strike phase.

Contact Pressure

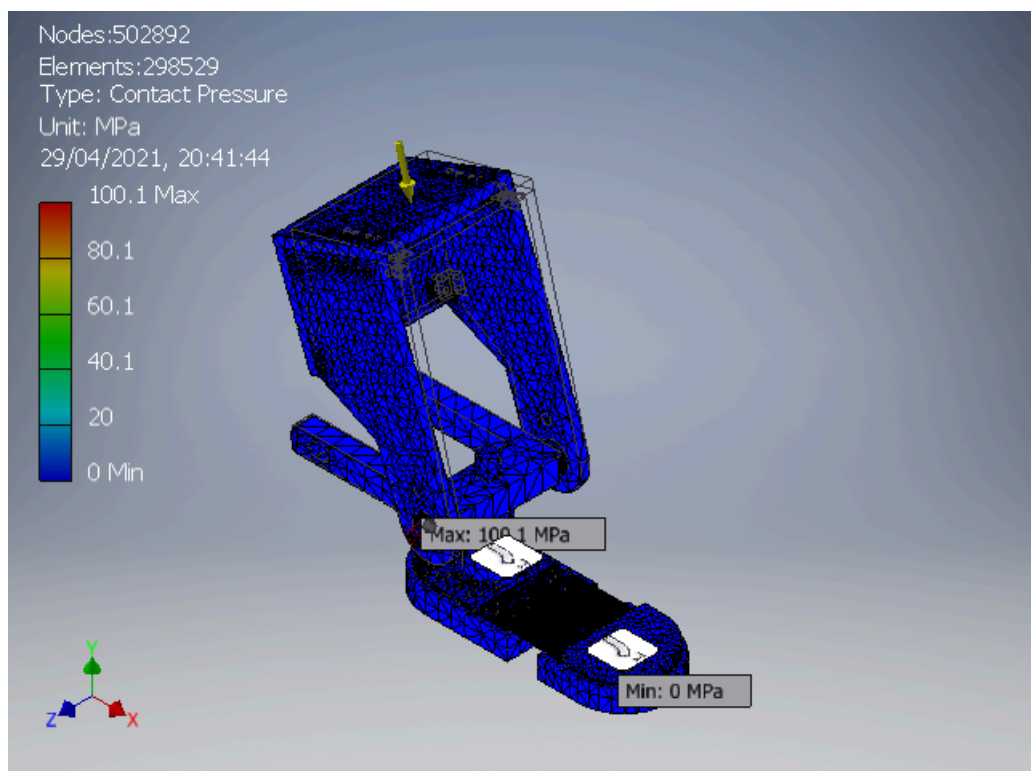


Figure 7-12: max. contact pressure for assembly in heel strike phase.

3. Toe off phase:

Stress Analysis Report



General objective and settings:

Design Objective	Single Point
Study Type	Static Analysis
Last Modification Date	29/04/2021, 20:48
Detect and Eliminate Rigid Body Modes	No
Separate Stresses Across Contact Surfaces	No
Motion Loads Analysis	No

Mesh settings:

Avg. Element Size (fraction of model diameter)	0.1
Min. Element Size (fraction of avg. size)	0.2
Grading Factor	1.5
Max. Turn Angle	60 deg
Create Curved Mesh Elements	No
Use part based measure for Assembly mesh	Yes

Material(s)

Name	Aluminum 7075	
General	Mass Density	2.81 g/cm ³
	Yield Strength	480 MPa
	Ultimate Tensile Strength	550 MPa
Stress	Young's Modulus	71.7 GPa
	Poisson's Ratio	0.33 ul
	Shear Modulus	26.9549 GPa
Part Name(s)	Support side Holes Updated.ipt Support side Holes Updated.ipt Grab Plate3 (5mm).ipt support back.ipt foot shaft.ipt Lower Link updated.ipt Lower link bolt.ipt Lower link bolt.ipt	

Name	Steel	
General	Mass Density	7.85 g/cm ³
	Yield Strength	207 MPa
	Ultimate Tensile Strength	345 MPa
Stress	Young's Modulus	210 GPa
	Poisson's Ratio	0.3 ul

	Shear Modulus	80.7692 GPa
Part Name(s)	L shape 3.ipt L shape 3.ipt L shape 3.ipt L shape 3.ipt L shape 3.ipt L shape 3.ipt L shape 3.ipt L shape 3.ipt L shape 3.ipt L shape 3.ipt User Library-15mm 6002 BEARING.ipt User Library-15mm 6002 BEARING.ipt	

Name	Stainless Steel	
General	Mass Density	8 g/cm ³
	Yield Strength	250 MPa
	Ultimate Tensile Strength	540 MPa
Stress	Young's Modulus	193 GPa
	Poisson's Ratio	0.3 ul
	Shear Modulus	74.2308 GPa
Part Name(s)	u foot.ipt fore foot.ipt spring sheet.ipt u foot 2.ipt fore foot 2.ipt	

Name	Aluminum 6061	
General	Mass Density	2.7 g/cm ³
	Yield Strength	275 MPa
	Ultimate Tensile Strength	310 MPa
Stress	Young's Modulus	68.9 GPa
	Poisson's Ratio	0.33 ul
	Shear Modulus	25.9023 GPa
Part Name(s)	load cell.ipt load cell.ipt load cell.ipt load cell.ipt	

Name	Rubber, Silicone	
General	Mass Density	1.25 g/cm ³
	Yield Strength	10.34 MPa
	Ultimate Tensile Strength	6.5 MPa
Stress	Young's Modulus	0.003 GPa
	Poisson's Ratio	0.49 ul
	Shear Modulus	0.00100671 GPa
Part Name(s)	lower fore foot.ipt lower heel.ipt	

Name	ABS Plastic	
General	Mass Density	1.06 g/cm ³
	Yield Strength	20 MPa
	Ultimate Tensile Strength	29.6 MPa
Stress	Young's Modulus	2.24 GPa
	Poisson's Ratio	0.38 ul
	Shear Modulus	0.811594 GPa
Part Name(s)	Hx711 box.ipt	

Name	Steel, Mild	
General	Mass Density	7.85 g/cm ³
	Yield Strength	207 MPa
	Ultimate Tensile Strength	345 MPa
Stress	Young's Modulus	220 GPa
	Poisson's Ratio	0.275 ul
	Shear Modulus	86.2745 GPa
Part Name(s)	ISO 2491 A A 6 x 4 x 22	

Operating conditions

Force:1

Load Type	Force
Magnitude	1000.000 N
Vector X	-258.819 N
Vector Y	-965.926 N
Vector Z	-0.000 N

Selected Face(s)

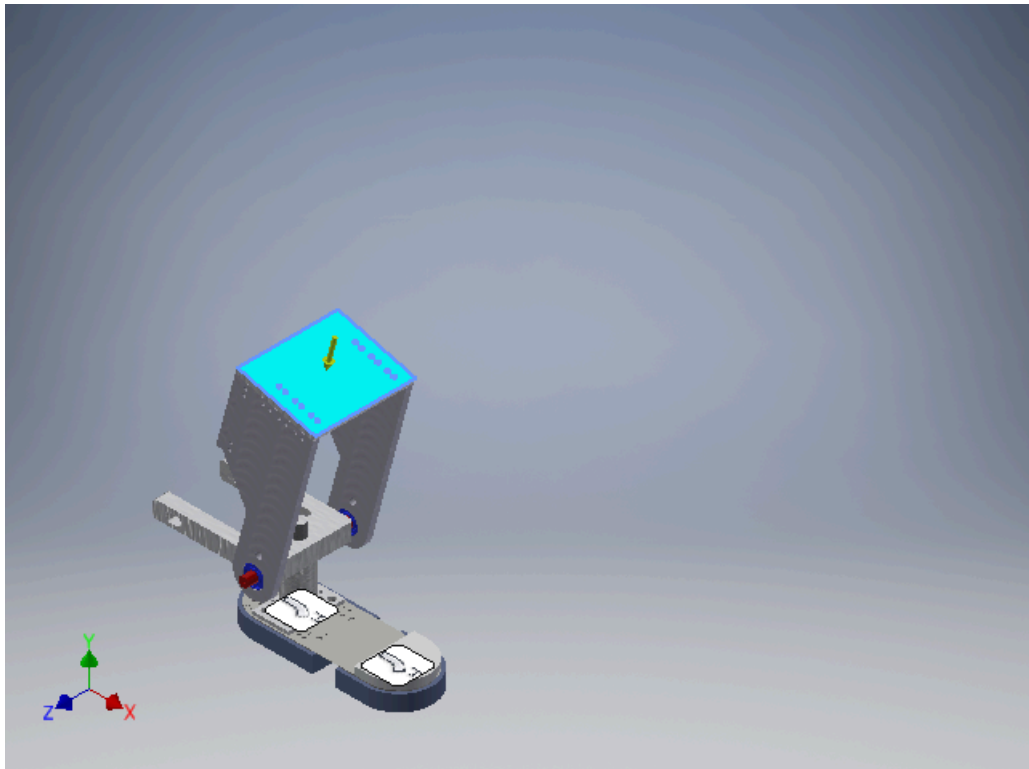


Figure 7-13: stress analysis for assembly in heel strike phase.

Contacts (Bonded)

Results

Reaction Force and Moment on Constraints

Constraint Name	Reaction Force		Reaction Moment	
	Magnitude	Component (X,Y,Z)	Magnitude	Component (X,Y,Z)
Fixed Constraint:1	1000 N	258.819 N	88.8501 N m	-0.374765 N m
		965.926 N		0.119544 N m
		0 N		-88.8492 N m

Figures

Von Mises Stress

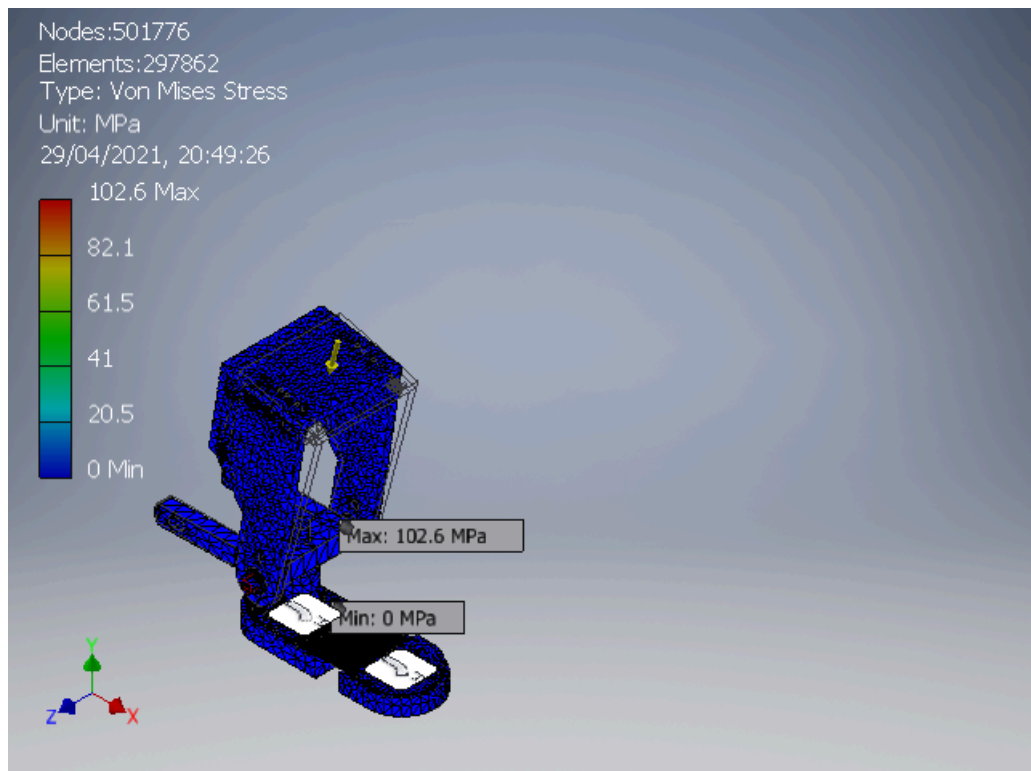


Figure 7-14: Von Mises stress for assembly in toe off phase.

Displacement

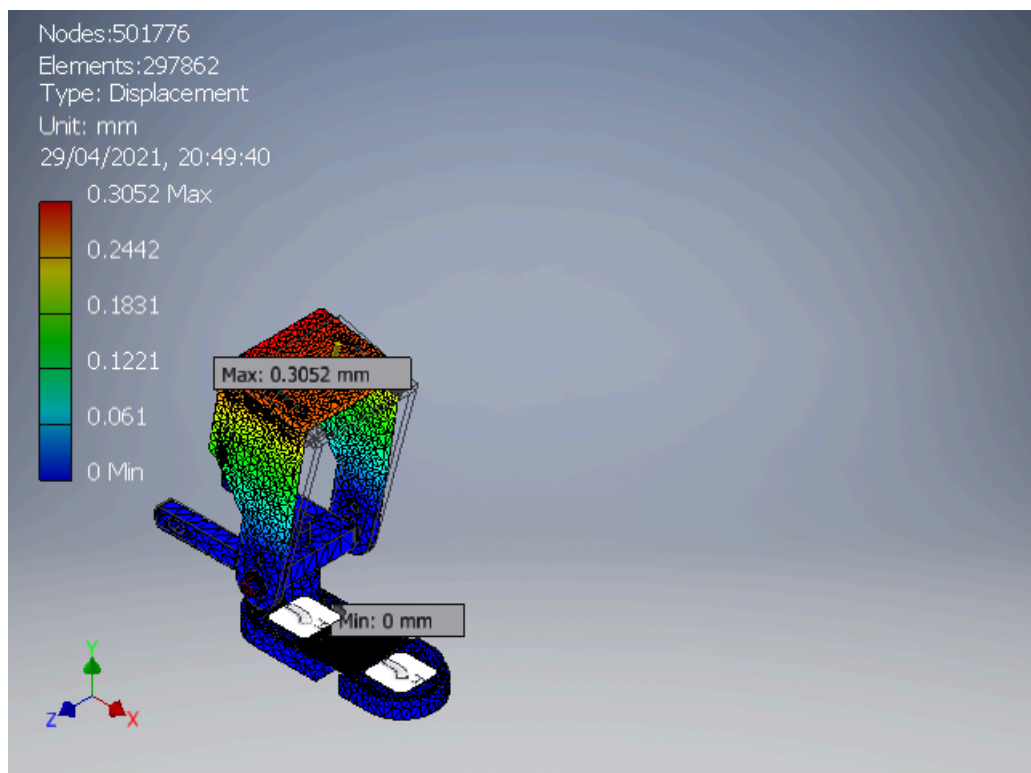


Figure 7-15: max. displacement for assembly in toe off phase.

Safety Factor

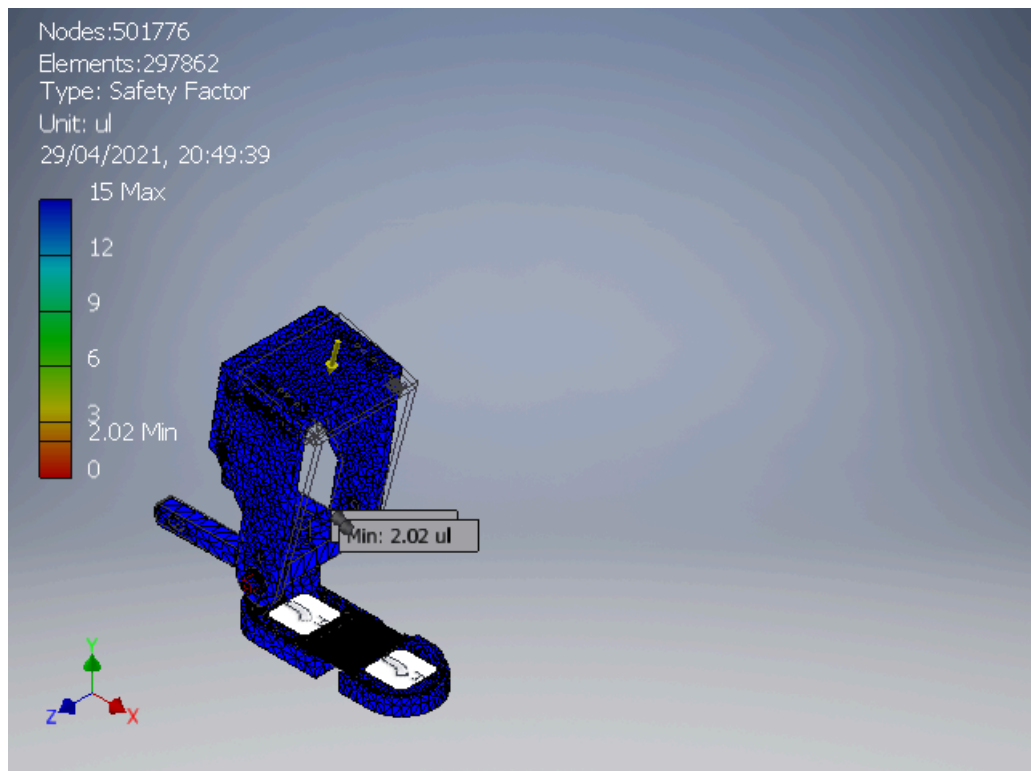


Figure 7-16: max. and min. safety factor for assembly in toe off phase.

Equivalent Strain

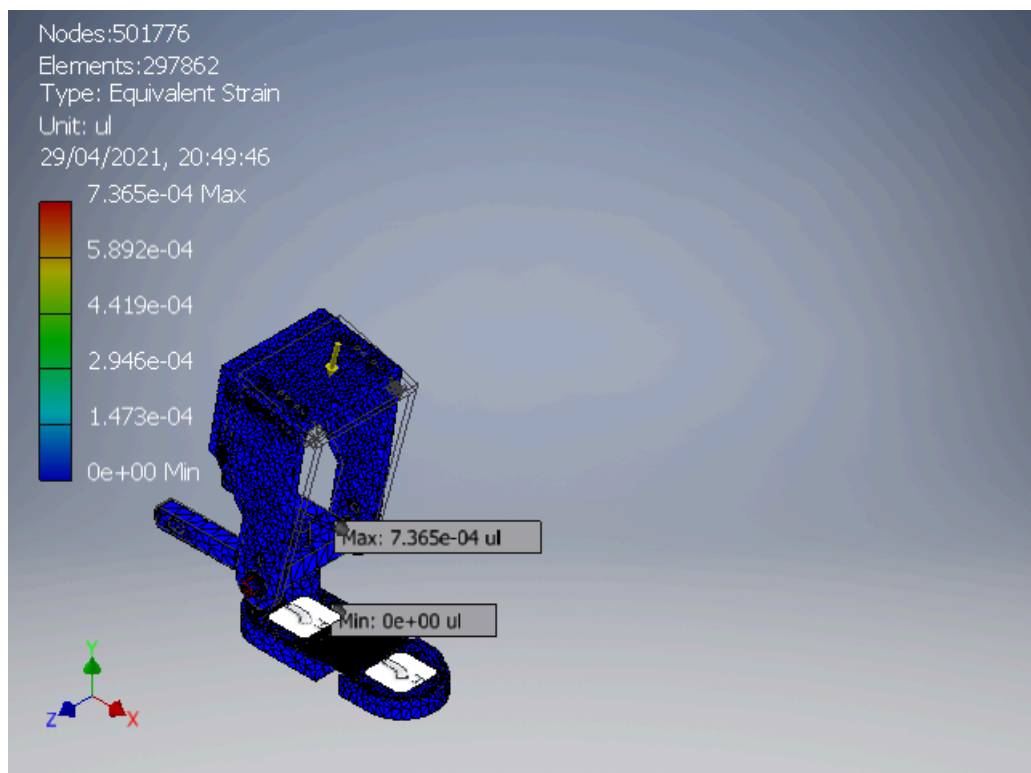


Figure 7-17: max. equivalent strain for assembly in toe off phase.

Contact Pressure

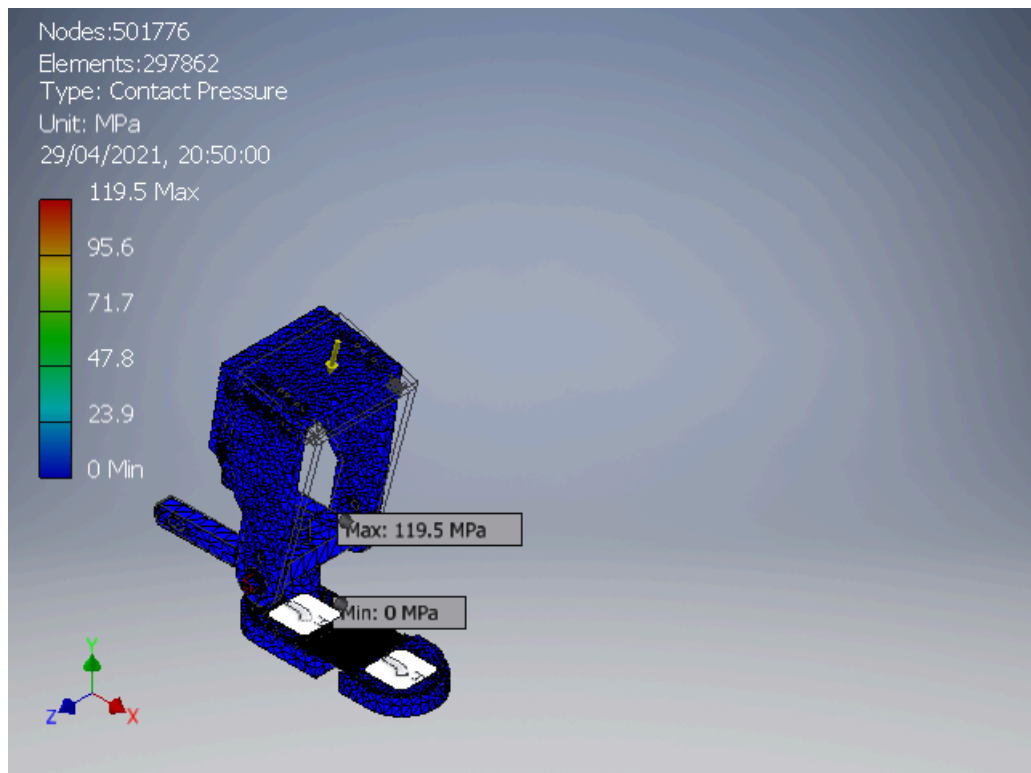


Figure 7-18: max. contact pressure

4. Maxon RE 40 Motor parameters

Nominal voltage	12 V
Load speed	6920 rpm
Load current	241 mA
Nominal speed	6380 rpm
Nominal torque (max. continuous torque)	94.9 mNm
Nominal current (max. continuous current)	6 A
Peak torque	1720 mNm
Peak current	105 A
Max. efficiency	88 %
CHARACTERISTICS	
Nominal resistance	0.115 Ω
Nominal inductance	0.024 mH
Torque constant	16.4 mNm/A
Speed constant	581 rpm/V
Speed / torque gradient	4.05 rpm/mNm
Mechanical time constant	5.89 ms
Motor inertia	139 gcm ²

HEAT THERMAL DATA

Thermal resistance housing-ambient	4.65 K/W
Thermal resistance winding-housing	1.93 K/W
Thermal time constant winding	41.9 s
Thermal time constant motor	809 s
Ambient temperature	-30...+100 °C
Max. winding temperature	+155 °C

MECHANICAL DATA

Bearing type	ball bearings
Max. speed	12000 rpm
Backlash	0.05 - 0.15 mm
Backlash	0.025 mm
Max. axial load (dynamic)	5.6 N
Max. force for press fits (static)	110 N
Max. force (static, shaft supported)	1200 N
Max. radial load	28 N, 5 mm from flange

OTHER SPECIFICATIONS

Number of pole pairs	1
Number of commutator segments	13
Number of autoclave cycles	0
PRODUCT	
Weight	480 g

Table 7- 1: Maxon RE 40 motor parameters.[10]

5. EC 40 BL motor parameters.

VALUES AT NOMINAL VOLTAGE

Nominal voltage	42 V
No load speed	10400 rpm
No load current	258 mA
Nominal speed	9280 rpm
Nominal torque (max. continuous torque)	124 mNm
Nominal current (max. continuous current)	3.44 A
Stall torque	1280 mNm
Stall current	33.5 A
Max. efficiency	84 %

CHARACTERISTICS

Terminal resistance	1.25 Ω
Terminal inductance	0.319 mH
Torque constant	38.2 mNm/A
Speed constant	250 rpm/V
Speed / torque gradient	8.2 rpm/mNm
Mechanical time constant	7.3 ms
Rotor inertia	85 gcm ²

THERMAL DATA

Thermal resistance housing-ambient	3.2 K/W
Thermal resistance winding-housing	1.2 K/W
Thermal time constant winding	16.7 s
Thermal time constant motor	1050 s
Ambient temperature	-20...+125 °C
Max. winding temperature	+125 °C

MECHANICAL DATA

Bearing type	ball bearings
--------------	---------------

Max. speed	18000 rpm
Axial play	0 - 0.14 mm
Max. axial load (dynamic)	10 N
Max. force for press fits (static)	170 N
(static, shaft supported)	5000 N
Max. radial load	70 N, 5 mm from flange
OTHER SPECIFICATIONS	
Number of pole pairs	1
Number of phases	3
Number of autoclave cycles	0
PRODUCT	
Weight	390 g

Table 7- 2: EC 40 BL motor parameters.[10]

REFERENCES

- [1] “Ankle range of motion and injuries - correct footwear fitting.” <https://profeet.co.uk/ankle-range-motion-injuries/> (accessed Jan. 08, 2021).
- [2] “Seattle Foot prosthetic device introduced in 1985. - HistoryLink.org.” <https://www.historylink.org/File/2104> (accessed Jan. 08, 2021).
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